Unit 1

Functions and Relations

1.1 Sets and Numbers

Introduction to Sets

Definition: Set
A **set** is a collection of objects. If an object is contained in the collection, we say it is an **element** of the set.

Notation:

- We use capital letters (A, B, C, etc.) to represent sets.
- We typically use lower-case letters (x, y, z, etc.) to represent objects.
- To say that an object x is an element of a set A, we write $x \in A$.
- To say that an object x is not an element of a set A, we write $x \notin A$.

Definition: Roster Notation
The simplest way to represent a set is **Roster Notation**, in which we list the elements of the set inside curly braces, separated by commas.

Example. $A = \{1, 2, 3\}$ means that A is the set containing the numbers 1, 2, and 3.

If there are a lot of elements in a set, we can write "..." to indicate a pattern continues.

Example. $B = \{1, 2, ..., 20\}$ means that B is the set containing the _____ numbers _____ through _____ .

Example. $C = \{0, 2, 4, ..., 100\}$ means that C is the set containing the _____ numbers _____ through _____ .

We can also indicate that a pattern continues forever.

Example. $D = \{1, 3, 5, ...\}$ means the set of all _____ .

Example. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{-3, -1, 1, 3\}$, and $C = \{3, 5, 7, 9, 11, ...\}$. Fill in the blanks below with $\in$ or $\notin$ as appropriate.

-1 _____ A	2 _____ A	3 _____ A
-1 _____ B	2 _____ B	3 _____ B
-1 _____ C	2 _____ C	3 _____ C

Standard Number Sets

There are certain sets of numbers that we talk about regularly in algebra.

Name	Symbol		Description
Natural Numbers	$\mathbb{N}$		The counting numbers 1, 2, 3, ...
Integers	$\mathbb{Z}$		The natural numbers, their opposites -1, -2, -3, ..., and 0. Numbers with no fractional part.
Rational Numbers	$\mathbb{Q}$		The numbers that can be represented as fractions $\frac{p}{q}$ where p and q are integers, $q \neq 0$.
Real Numbers	$\mathbb{R}$		The rational numbers, together with irrational numbers to "fill in" the number line.

Note. We will sometimes refer to the **positive integers** to mean the natural numbers, and the **nonnegative integers** to mean the natural numbers plus zero.

Note. The same rational number can be written in many forms, e.g. $\frac{3}{5} = \frac{6}{10} = \frac{-9}{-15}$ etc. These are all the same rational number.

Example. For each number below, determine which of the standard number sets contains it.

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	Notes
6					
$-\frac{16}{4}$					
$\frac{3}{5}$					
-0.78					
$0.\overline{3}$					
$\sqrt{2}$					

How can we recognize rational and irrational numbers?

- All _____ are rational.
- Any numbers written as a _____ is rational.
- Every _____ or _____ decimal is rational.
- All other decimals are irrational.
- Square roots of positive integers are either _____ or irrational.

Terminating and Repeating Decimals are Rational

Theorem
Every fraction can be represented as a terminating or repeating decimal.

We prove this by long division. Here, we will show two examples, but this works in general.

$\frac{5}{8}$ as a terminating decimal.

$$8 \overline{)\ 5.000}$$

$\frac{19}{22}$ as a repeating decimal.

$$22 \overline{)\ 19.000}$$

Theorem
Every terminating decimal can be written as a fraction.

We see an example, but this works in general. 0.385 can be written as a fraction.

Denominator is Numerator is

$0.385 =$

Theorem
Every repeating decimal can be written as a fraction.

We see an example again. $0.\overline{87}$ can be written as a fraction.

Square Roots are Typically Irrational

Theorem
The square root of a positive integer is either a positive integer or an irrational number.

Proof. We'll argue that $\sqrt{2}$ is irrational. The same argument works for any prime number; a little more detail is needed for numbers that aren't prime.

- Suppose $\sqrt{2}$ were rational. Then it could be written as a fraction $\frac{p}{q}$ in lowest terms. That means that p and q have no _____ .
- $\sqrt{2} = \dfrac{p}{q}$ means $2 = \dfrac{p^2}{q^2}$. But $2 = \dfrac{2}{1} = \dfrac{2q^2}{q^2}$, so $p^2 = 2q^2$.
- That means _____ is a factor of p^2. Since _____ is prime, it must be a prime factor of p. Say $p = 2r$.
- Then $p^2 = (2r)^2 = 2^2 r^2$.
- Going back to the fraction, we have $2 = \dfrac{p^2}{q^2} = \dfrac{\quad}{q^2} = \dfrac{\quad}{q^2} = \dfrac{2q^2}{q^2}$.
- Now we see that $2r^2 = q^2$, so _____ must be a factor of q^2 and thus q.
- So 2 is a _____ of p and q.
- But at the beginning we picked p and q to have no _____ !
- That means we can't pick values p and q like that, so $\sqrt{2}$ can't be written as a fraction—it is _____ . $\square$

Set Builder Notation

We have an alternate way of describing the elements of a set.

Definition: Set Builder Notation
Set Builder Notation is a method of describing a set in terms of a generic element, followed by a description of rules the elements must follow. Set Builder Notation looks like this: $$\{\text{generic element} \mid \text{rules}\}$$

Example. Let $A = \{2n \mid n \in \mathbb{Z}\}$

- List three elements of A:
- How else could we describe A?

Example. Let $B = \left\{\frac{n}{10} \mid n \in \mathbb{N}\right\}$

- List three elements of B:
- Is $\frac{3}{5} \in B$?
- Are there any integers in B?

Example. Let $C = \left\{\frac{p}{q} \mid p \text{ is even and } q \text{ is odd}\right\}$

- List three elements of D:
- Are there any integers in C?

Properties of Real Numbers

To do math to real numbers, we use two operations: **addition** and **multiplication**. These operations have certain properties.

Commutative Property of Addition For any real numbers a and b, $a + b = b + a$	**Commutative Property of Multiplication** For any real numbers a and b, $ab = ba$
Associative Property of Addition For any real numbers a, b, and c, $(a + b) + c = a + (b + c)$	**Associative Property of Multiplication** For any real numbers a, b, and c, $(ab)c = a(bc)$
Distributive Property For any real numbers a, b, and c, $a(b + c) = ab + ac \qquad (b + c)a = ba + ca$	

You might think of **subtraction** and **division** as operations, but we meet them in these next properties.

Additive Identity Property There is an additive identity 0 such that, for all real numbers a, $a + 0 = 0 + a = a$	**Multiplicative Identity Property** There is a multiplicative identity 1 such that, for all real numbers a, $1a = a \cdot 1 = a$
Additive Inverse Property Every real number a has an additive inverse $-a$ such that $a + (-a) = -a + a = 0$	**Multiplicative Inverse Property** Every real number $a \neq 0$ has a multiplicative inverse $\frac{1}{a}$ such that $a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1$

We define subtraction and division using the inverses:

- $a - b =$ _____
- $a \div b =$ _____

Simplifying Expressions: Basic Moves

Recall:

Definition

- An **expression** is a sequence of numbers, variables, and symbols that can be resolved to represent a number.
- To **evaluate** an expression is to determine what number it represents.
- Two expressions are said to be **equivalent** if they represent the same number, no matter what any variables represent. To **rewrite** an expression is to write an equivalent expression in a different form.
- To **simplify** an expression is to write an equivalent expression in some standard "simplest" from.
- The **terms** of an expression are parts of an expression combined using *addition*. The **factors** of an expression are parts of an expression combined using *multiplication*.

There are a number of common steps we use to simplify expressions.

- Cancel common factors in the numerator and denominator of a fraction.
- Use the commutative property to rearrange terms and factors.
- Multiply numbers within terms.
- Use the distributive property to eliminate parentheses around addition (or _____).
- "Combine Like Terms" (using the distributive property).
- Use properties of exponents to combine factors.

Example. Simplify $3(x+4)-5(x-3)$.

Step	Why?

Properties of Equality

Definition: Equation

An **equation** is a statement that two expressions are equal. It is of the form

$$\text{expression} = \text{expression}$$

The equation is *true* if the two expressions represent the same number.

- A **solution** to an equation is a value(s) of the variable(s) in the equation that makes the equation true.
- The **solution set** of an equation is the set of all solutions of the equation.
- Two equations are **equivalent** if they have the same solution set.
- To **solve** an equation means to find its solution set.

We use equations in several ways.

- To show equivalent expressions.

  Example: _____

- To define a variable symbol.

  Example: _____

- To describe an unknown quantity.

  Example: _____

The following properties of equality allow us to get equivalent equations.

Additive Property of Equality	**Multiplicative Property of Equality**
For any expressions X and Y and any real number c, the equations $$X = Y \quad \text{and} \quad X + c = Y + c$$ are equivalent.	For any expressions X and Y and any real number $c \neq 0$, the equations $$X = Y \quad \text{and} \quad c(X) = c(Y)$$ are equivalent.

Symmetric Property of Equality

For any expressions X and Y, the equations

$$X = Y \quad \text{and} \quad Y = X$$

are equivalent.

Solving Equations: Basic Moves

The properties of equality give us two basic moves that we can use to solve equations.

- We can _____ the same number to each side of an equation.

  We can also _____ (this is just adding the opposite).

- We can _____ by the same number on each side of an equation.

  We can also _____ (this is just multiplying by the inverse).

Note. We can _____ any number, so it is safe to _____ variable expressions. However, we must be careful not to _____ by zero, so we don't _____ by variable expressions at this stage.

We choose basic moves in order to isolate the variable on one side.

Example. Solve $2x - 3 = 7$

Step	Why?

Example. Solve $5x - 7 = 2x + 5$

Step	Why?

Example. Solve $\frac{2}{3}x - \frac{3}{5} = \frac{1}{6}x + \frac{4}{5}$

Step	Why?

Unit 2

Polynomial and Rational Functions

2.1 Monomials and Polynomials

Review: Exponents

Definition

We represent repeated multiplication using **positive integer exponents**. The expression b^p is defined as

$$b^p = \underbrace{b \cdot b \cdots b}_{p \text{ copies}}$$

The number b in this expression is called the **base**, and the number p is called the **exponent**.

Note. The exponent is written as a _____ , _____ and _____ the main line of text.

Example. Consider the expression 5^3:

- The base is _____ and the exponent is _____ .
- Rewrite 5^3 as repeated multiplication: _____
- Find 5^3 by hand: _____

Note. We can add exponent _____ to any expression without changing the value, that is, for any x, $x =$ _____.

Exponent Rules (Product and Quotient of Powers)

We have certain rules that allow us to combine expressions with exponents.

Theorem: Product of Powers Rule

For any real number x and positive integers m and n,

$$x^m x^n = x^{m+n}$$

$x^m x^n =$

Theorem: Quotient of Powers Rule

For any real number x and positive integers $m \neq n$,

$$\text{If } m > n,\ \frac{x^m}{x^n} = x^{m-n} \qquad \text{If } m < n,\ \frac{x^m}{x^n} = \frac{1}{x^{n-m}}$$

If $m > n$, $\dfrac{x^m}{x^n} =$

If $m < n$, $\dfrac{x^m}{x^n} =$

Exponent Rules (Power of a Product, Quotient, and Power)

Theorem: Power of a Power Rule

For any real number x and any positive integers n and p,

$$(x^n)^p = x^{np}$$

$(x^n)^p =$

Theorem: Power of a Product Rule

For any real numbers x and y and any positive integer n,

$$(xy)^n = x^n y^n$$

$(xy)^n =$

Theorem: Power of a Quotient Rule

For any real numbers x and $y \neq 0$ and any positive integer n,

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$$

$\left(\frac{x}{y}\right)^n =$

Exponent Rules (Negative Exponents and Examples)

We define the remaining integer exponents to be consistent with the product and quotient rules.

Definition: Exponent Zero

For any real number $x \neq 0$, we define

$$x^0 = 1$$

- This is consistent with the product rule:

  $x^0 x^n = x^{0+n} = \underline{\qquad} = \underline{\qquad} \cdot x^n$

- This is consistent with the quotient rule (if we allow $m = n$):

  $\dfrac{x^n}{x^n} = x^{n-n} = \underline{\qquad}$, but $\dfrac{x^n}{x^n} = \underline{\qquad}$

Definition: Negative Integer Exponents

For any real number $x \neq 0$ and positive integer n,

$$x^{-n} = \frac{1}{x^n}$$

This lets us simplify the statement of the quotient rule:

Theorem: Quotient of Powers Rule

For any real number x and integers m and n,

$$\frac{x^m}{x^n} = x^{m-n} = \frac{1}{x^{n-m}}$$

We end with some examples of applying these rules.

Example. Simplify. Assume $x \neq 0$.

$x^3 x^4 =$

$\dfrac{x^7}{x^3} =$

$\dfrac{x^5}{x^8} =$

$(x^3)^5 =$

$(x^{-3})^2 =$

$(4x)^3 =$

$\left(\dfrac{x}{3}\right)^4 =$

$\left(\dfrac{x^{-2}}{y^3}\right)^{-5} =$

Monomials

Definition: Monomial

A **monomial** is a combination of numbers and variables, put together with *multiplication only*.

In simplest form, we combine all the numeric factors into a single **coefficient**, and write repeated variable factors using a single positive integer exponent.

The **degree** of a monomial is the total number of variable factors, that is, the sum of all the exponents.

Example. For each expression, determine whether or not it is a monomial. If it is, write it in simplest form.

a. $3x \cdot 5xy \cdot (-6yy)$

b. $\frac{2}{3}x^3yxy^5$

c. $\dfrac{5y^3}{x}$

d. $3x^{-4}y^3$

e. 7

Example. For each monomial below, determine the coefficient and degree.

a. $4x^2$

Coefficient: _____ Degree: _____

b. $-\frac{5}{3}x^3y^5$

Coefficient: _____ Degree: _____

c. $7.4xyzw$

Coefficient: _____ Degree: _____

Polynomials

Definition: Polynomial
A **polynomial** (or **polynomial expression**) is an expression given as a sum of one or more monomials. The monomials are called the **terms** of the polynomial.

Example. For each expression below, determine whether or not it is a polynomial. If it is not, explain why not.

a. $3x^2 + 5x - 7$

b. $x - 3\sqrt{x} + 2$

c. $4x^{-2} - 3$

d. $2x^3y$

e. 5

Describing Polynomials (Part 1)

Definition

- The **degree of a polynomial** is the highest degree of any term.
- The **leading term** is the term with highest degree.
- The **leading coefficient** is the coefficient of the leading term.

If a polynomial has more than one term of the highest degree, it does not have a leading term.

Example. For each polynomial below, identify the leading term, leading coefficient, and degree.

a. $3x^3 + 2x^2 + 5x + 3$ L. term: _____ L. coeff: _____ Deg: _____

b. $x + \frac{2}{3}x^3 - 7x^5$ L. term: _____ L. coeff: _____ Deg: _____

c. $3x^2y - 4xy^2 + 1.3x^2y^2$ L. term: _____ L. coeff: _____ Deg: _____

d. $x^2 - 5xy + 6y^2$ L. term: _____ L. coeff: _____ Deg: _____

Definition

- A **polynomial in one variable** is one in which only one variable appears.
- A **homogeneous** polynomial is one in which every term has equal degree.

Example. For each polynomial below, identify it as a polynomial in one variable, a homogeneous polynomial, or neither. Also determine the degree of the polynomial.

a. $x^4 - x^2 + 2x + 1$ One variable | Homogeneous | Neither Degree: _____

b. $x^2 - 5xy - 6y^2$ One variable | Homogeneous | Neither Degree: _____

c. $6xy - 10x + 9y$ One variable | Homogeneous | Neither Degree: _____

Definition

- A polynomial is written in **descending order** if the terms are ordered from highest to lowest degree.
- A polynomial is written in **ascending order** if the terms are ordered from lowest to highest degree.

Example. Write $5x - 3x^2 - 2x^3 + 6$ in descending order.

_____

Example. Write $5y + 15 - 6y^5 + 2y^7$ in ascending order.

_____

Describing Polynomials (Part 2)

Definition

We can name polynomials by their degree. (These words are *adjectives*.)

Degree	Name	Example
0	Constant	7
1	Linear	$5x - 3$
2	Quadratic	$x^2 - 3x + 2$
3	Cubic	$x^3 - 5$

For higher degrees, we just say "of degree n."

Definition

We can name polynomials by their number of terms. (These words are *nouns*.)

Terms	Name	Examples	
1	Monomial	x^3	$2x$
2	Binomial	$x - 3$	$x^3 - 8$
3	Trinomial	$x^2 - 3x + 2$	$x^5 - x^3 + x$

For more terms, we just say "A polynomial with n terms."

Example. Describe each polynomial.

a. $x^3 - 7$ _____

b. $x^4 + x^5 - 3$ _____

c. y^2 _____

d. $2x^2 - 5xy + 2y^2$ _____

e. $x + y + z + w$ _____

f. $2x - 5$ _____

g. $x^3 - 5x^2 + 2x - 10$ _____

h. $x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + y^4$ _____

Polynomial Functions

Definition: Polynomial Function
A function is a polynomial function if its expression is a polynomial.

Example. Let $f(x) = 2x^3 - 5x^2 + 4x + 7$. Evaluate:

a. $f(3) =$ _____

b. $f(0) =$ _____

c. $f(-2) =$ _____

Polynomials give an insight into the Order of Operations.

Definition: Order of Operations
In expressions with more than one operation, they should be completed in the following order: 1. Exponents 2. Multiplication (and division) 3. Addition (and subtraction) To violate this order, use grouping symbols (like parentheses).

The reason that this order is chosen is to make polynomials work as expected. To evaluate $f(-2)$, we:

1. found each power of -2
2. multiplied by its coefficient
3. then added up all the terms

This is why _____ come before _____ in order of operations: we want $-x^2$ to mean _____ the number x, then _____ .

Example. Evaluate each of the following.

$2^2 =$ _____ $\qquad -2^2 =$ _____ $\qquad (-2)^2 =$ _____

2.2 Polynomial Arithmetic

Adding Polynomials

Adding polynomials is an application of the associative and commutative property.

Example. Add $(2x^2 - 5x + 6) + (3x^2 - 11x - 5)$, justifying each step.

Step	Why?

We will typically not write all of these steps. If you remember that _____ signs go with their term, you may be able to directly write down the sum.

Example. Add $(3x^4 - 5x^3 + 2x + 3) + (3x^3 - 7x^2 + x - 11)$

Negating a Polynomial

It is often necessary to take the opposite of a polynomial. Recall that we can understand writing a negative sign in front of parentheses as multiplying by _____ .

Example. Simplify $-(2x^4 - 5x^3 + 6x^2 + 7x - 3)$, justifying each step.

Step	Why?

We will usually not write all of these details, either.

Example. Simplify $-(-3x^2 + 4xy - 6y^2 + 2y)$

We can think of this move as "distribute the negative sign."

Subtracting Polynomials

To subtract polynomials, we remember that subtraction is _____ the _____ .

Example. Subtract $(3x^2 - 5x + 3) - (2x^2 + 6x - 5)$, justifying each step.

Step	Why?

Typically, the only step we write out is the one where we "distribute" the negative sign.

Example. Subtract $(5x^2 - 4xy - 6y^2) - (3y^2 + 3xy - 4y^2)$

We can also think of this process as "subtracting like terms."

Example. Subtract $(5x^2 - 6x + 3) - (2x^2 + 3x - 3)$

Write:	Think:

Multiplying a Monomial and a Polynomial

Multiplying a monomial and a polynomial is an application of the _____ property, together with exponent rules.

Example. Multiply $3x^2y(4x^2 - 6xy + 3y^2)$, justifying each step.

Step	Why?

We will typically write fewer steps here, as well.

Example. Multiply $-2x(-x^2 + 7x - 3)$

Multiplying Polynomials: Introduction

To multiply two polynomials (with more than one term), we still apply the _____ property, but we apply it _____ .

Example. Multiply $(x^2 - 3x + 2)(2x - 1)$, justifying each step.

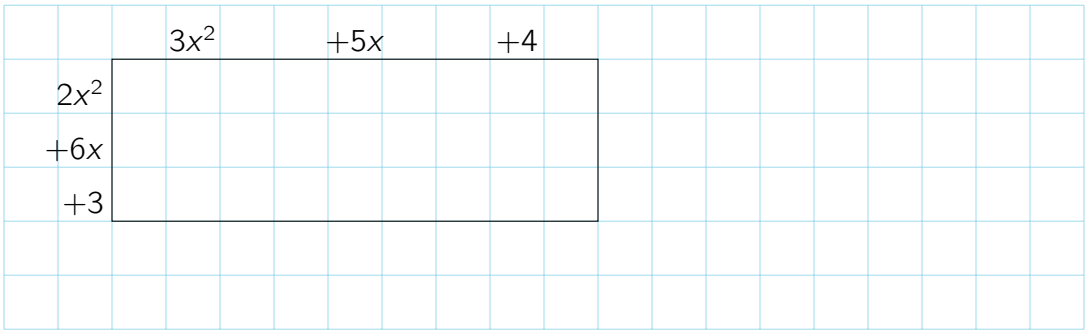
Step	Why?

The result is to multiply every _____ in the first polynomial by every _____ in the second. In the next two videos, we'll see two ways to arrange this.

Multiplying Polynomials: Rectangle Method

We can think of multiplication as describing the _____ of a rectangle.

Example. Multiply $(3x^2 + 5x + 4)(2x^2 + 6x + 3)$ by interpreting the polynomials as describing the sides of a rectangle.



Example. Multiply $(4x + 3y - 6)(3x - 5y + 2)$.

Multiplying Polynomials: FOIL Method

When we multiply two _____ , there are _____ possible products.

Example. Multiply $(A + B)(C + D)$ as a rectangle.

The _____ possible products are:

F Multiply the **F**irst terms of each binomial
O Multiply the **O**uter pair of the product
I Multiply the **I**nner pair of the product
L Multiply the **L**ast terms of each binomial

Using this as a shortcut to multiply binomials is called the **FOIL method**. Remember, we can only use this to multiply two _____ .

Example. Multiply $(2x - 3)(x + 5)$ as a rectangle and using the FOIL method.

When we multiply two _____ binomials, we know we will combine the middle terms. With practice, we may skip writing this step.

Example. Multiply $(2x - 3)(3x - 2)$ using the FOIL method.

Multiplying Three Polynomials

If we wish to multiply three or more polynomials, we do it the same we do with numbers: multiply the _____ , then multiply _____ .

Example. Multiply $(x-4)(x+2)(x-1)$.

First we multiply _____ using the _____ .

Now we must multiply _____ .

$(x-4)(x+2)(x-1) =$ _____

Powers of Polynomials

We take powers of polynomials by _____ .

Example. Find $(2x - 3y)^2$

Example. Find $(x + 3)^3$

Example. Find $(x^2 - 2x + 2)^2$

2.3 Factoring: Basic Methods

Factoring Numbers Review

Let's review the notion of factoring integers.

Definition

- A **factor** (noun) of an integer n is an integer k such that $n = kd$ for some integer d.
- To **factor** (verb) an integer is to determine its factors.
- A **prime number** p is an integer whose only factors are $\pm p$ or ± 1.
- The **prime factorization** of an integer as a product of prime numbers.
- We can find the prime factorization of an integer by making a **factor tree**.

Example. Make a factor tree to find the prime factorization of the number 18.

Theorem: Fundamental Theorem of Arithmetic

Any integer can be written uniquely as a product of prime numbers.

Note. For purposes of "uniqueness," we ignore negative signs and order; so $10 = 2 \cdot 5 = 5 \cdot 2 = (-2) \cdot (-5)$ would be considered "the same" factorization.

Introduction to Factoring Polynomials

When we consider factoring polynomials, we will consider polynomials with *integer* coefficients.

Definition
• A **factor** of an integer n is a number k such that $n = kd$ for some integer d. • A **factor** of a monomial M is a monomial A such that $M = AB$ for some monomial B. • A **factor** of a polynomial P is a polynomial Q such that $P = QS$ for some polynomial S.

Note that these definitions are very similar!

Example.

As an integer, 5 is a factor of 30 because $30 = 5 \cdot$ _____.

As a monomial, $3x^2$ is a factor of $9x^3y$ because $9x^3y = (3x^2)($_____$)$.

As a polynomial, $4x+3$ is a factor of $4x^2+3x$ because $4x^2+3x = ($_____$)(4x+3)$.

When we factor (verb) a polynomial, our goal is to rewrite it in a factored form.

Definition
An expression is in "factored form" if it is written as a product of simpler expressions, that is, if the last step to evaluate it would be _____ .

Example. For each expression below, determine whether it is in a factored form.

a. $3x^2y$ _____

b. $(2x-3)(5x+2)$ _____

c. $3x(x+5)-7(x+5)$ _____

d. $4x^2+6x-3$ _____

e. $x^2(3x^2-4x-4)$ _____

f. $3x(5-3x)+4$ _____

Note that _____ are automatically in a factored form.

Greatest Common Factor

Definition: Greatest Common Factor (integers)
A **common factor** of two (or more) integers is a number that is a factor of both (or all). The **greatest common factor** (or **GCF**) of two (or more) integers is their largest (positive) common factor.

Example. List all (positive) factors of the integers 8 and 20. Then, give their GCF.

- Factors of 8 _____
- Factors of 20 _____

The largest number on both lists is _____ , so this is the greatest common factor of 8 and 20.

We can also use a calculator or computer to find the GCF of two numbers.

Definition: Greatest Common Factor (monomials)
A **common factor** of two (or more) monomials is a monomial that is a factor of both (or all). The **greatest common factor** (or **GCF**) of two (or more) monomial is their common factor that has: • the largest possible coefficient • the highest possible degree

To find the GCF of monomials, we find:

- the GCF of the coefficients
- the lowest power of each variable that appears in the monomials

If we take the prime factorization of each coefficient, we can treat prime numbers like variables. Or, we can use a calculator to find the GCF of the coefficients.

Example. Find the GCF of $3x^4y^3$, $6x^3y^5$, and $9x^2y^7$.

- For the coefficient: _____
- For x: _____
- For y: _____

The GCF of $3x^4y^3$, $6x^3y^5$, and $9x^2y^7$ is _____ .

Factoring Out the GCF

The first step in factoring any polynomial is to find any monomial factor. We call this **factoring out the GCF**. Let's see an example, then summarize the process.

Example. Factor out the GCF from the polynomial $6x^5y^2 - 18x^4y^3 + 12y^4$

Procedure: Factoring Out the GCF

1. Find the GCF of the polynomial's terms.
2. Divide each term by the GCF to determine the remaining factor, using properties of exponents.
3. Write the original polynomial as the GCF multiplied by the polynomial made up of the remaining factors.
4. To check, note that:
   (a) The remaining factors have no common factors.
   (b) When you multiply, you get the original polynomial.

A Negative GCF

It is more convenient to work with polynomials that have a _____ leading term. If the leading term is _____ , we can fix this by factoring out a negative GCF.

Example. Factor out the GCF from $-5x^5 + 20x^4 - 25x^3$ so that the leading term of the polynomial factor is positive.

Sometimes, we simply factor out -1.

Example. Factor out the GCF from $-3x^2 - 7x + 9$ so that the leading term of the polynomial factor is positive.

Factoring by Grouping

Sometimes, when we multiply two binomials, we get a polynomial in four terms. To factor such a polynomial, we want to get the two binomials back.

Example. Multiply $(2x-3)(3y+1)$ by applying the distributive property twice.

$(2x-3)(3y+4) =$

Note that in this example, there are no _____ to combine.

If we want to get back to the binomial factors, we can apply this process backward!

Example. Factor $6xy - 9y + 8x - 12$ as the product of two binomials.

Step	Why?

Procedure: Factor by Grouping

To factor a polynomial with four terms by grouping:

1. Use the associative property of addition to "group" the terms into two pairs.
2. For each pair of terms, factor out the GCF.
3. Note that the remaining binomial factors for each pair are the same.
4. Factor out the remaining binomial factors to write the polynomial as a product of two binomials.
5. To check, multiply the two binomials, and confirm you get the original polynomial back.

Note that it is important that, when you create the groups, the operation combining them is *addition*.

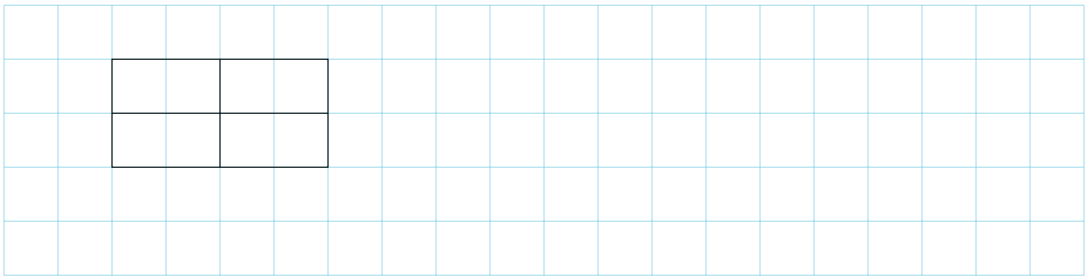
Example. Factor $2y^3 - 3y^2 - 10y + 15$ by grouping.

$2y^2 - 3y^2 - 10y + 15 =$

Factoring by Grouping Rectangle Method

We can also perform factoring by grouping by running the rectangle method of multiplication backward. This can make it easier to keep track of signs.

Example. Factor $p^2 + 3ps - 2pq - 6qs$ by grouping, using the rectangle method.



Example. Factor $2x^5 - x^3 + 10x^2 - 5$ by grouping, using the rectangle method.

2.4 Factoring: Trinomials and Special Forms

Factoring Quadratic Trinomials: the AC Method

When we multiply two linear binomials (in one variable or homogeneous in the same two variables), the result is a quadratic trinomial.

Example. Multiply $(2x-3)(x+2)$, using the rectangle method and the FOIL shortcut.

When we multiply, we must _____ . When we factor, we must reverse this, i.e. "_____" the middle terms.

Observing the multiplication, we see that the middle terms (to be combined) have the same _____ as the first and last terms. This gives us the following technique.

Procedure: The AC Method

To factor a quadratic trinomial:

1. Multiply the coefficients of the first and last terms.
2. Find two numbers that have the same product, and whose sum is the coefficient of the middle term.
3. Rewrite the middle term as a sum of two terms using these coefficients.
4. The polynomial now has four terms. Factor by grouping.

If we write a quadratic trinomial as $ax^2 + bx + c$ (or $ax^2 + bxy + cy^2$), the two numbers we seek in step 2 have product ac and sum b.

Example. Factor $2x^2 + x - 6$ using the AC Method.

Example. Factor $6x^2 - 11xy + 3y^2$ using the AC Method.

AC Method with the Calculator

When using the AC Method, we often need to find all factors of a large number with many factors.

Example. Use the AC Method to factor $6x^2 + 23xy + 20y^2$ using the AC Method.

We must find two numbers whose product is _____ and whose sum is _____ . This requires we find all factors of _____ !

1	×	120	= 120		1	+	120	=	121
2	×	60	= 120		2	+	60	=	62
3	×	40	= 120		3	+	40	=	43
4	×	30	= 120		4	+	30	=	34
5	×	24	= 120		5	+	24	=	29
6	×	20	= 120		6	+	20	=	26
8	×	15	= 120		8	+	15	=	23
10	×	12	= 120		10	+	12	=	22

This is a lot of factors! Whenever we want to do highly repetative arithmetic, we want a _____ to do it for us. The procedure here uses the TI-84 calculator, but it can be done with many other calculators or with a computer.

Procedure: Factors for the AC Method on the TI-84

1. Press [Y=] to access the function entry screen.
2. Enter `##/X` as Y_1 and `##/X+X` as Y_2, where `##` is the desired product.
3. Press [2ND] [GRAPH] to access the "Table" feature.
4. The `X` and Y_1 columns show the two factors, and the Y_2 column shows their sum. Use the up and down arrow keys to scroll through the table until the desired sum is found.

Example. Factor $12x^2 - 29x - 60$ using the AC Method, using a calculator to help split the middle term.

Quadratic Trinomials with Leading Coefficient 1

In general, when we use the AC Method, the numbers used to split the middle term do not appear directly in the factors. However, look what happens when the leading coefficient is 1.

Example. Factor using the AC Method.

$x^2 - 5x + 6 =$

$x^2 + 4xy - 12y^2 =$

We notice that the numbers that appear in the factors are _____ as the ones we found in the AC Method. This gives us a shortcut, as long as the leading coefficient (in at least one variable) is 1.

Example. Determine whether the shortcut for the AC Method can be used. Factor using the shortcut if possible.

$2x^2 + 5x - 3$

$x^2 + 3x - 18$

Prime Quadratic Polynomials

Recall the definition of a prime number.

Definition: Prime Number
An integer p is **prime** if its only factors are ± 1 and $\pm p$.

The concept is that a prime number "cannot be factored." We have a parallel definition for polynomials.

Definition: Prime Polynomial
A polynomial with integer coefficients is said to be **prime** if it cannot be written as a product of polynomials with integer coefficients of lower degree or with smaller coefficients.

We can always factor out ± 1 from any polynomial; it is prime if we can't factor out anything else. How can we recognize a prime polynomial?

Assuming no greatest common factor (except ± 1), the following polynomials are prime.

- Any linear polynomial.
- Any quadratic polynomial with four terms for which grouping fails.
- Any quadratic trinomial for which the AC Method fails.

Example. Identify each polynomial as prime or not prime. If it is not prime, factor.

$x^2 - 4xy + 3x - 12y$

$x^2 - 4xy + 5y^2$

$2x^2 - 7x + 6$

$xy - 5x - 3y + 7$

Square and Cube Numbers

In factoring, it can be useful to recognize certain numbers as "perfect squares" and "perfect cubes," that is, powers of integers.

x	1	2	3	4	5	6	7	8	9
x^2	1	4	9	16	25	36	49	64	81
x^3	1	8	27	64	125	216	343	512	729

If you spend a lot of time factoring, you may memorize these lists. If you don't have a list like this memorized, you can use a calculator or computer to make one.

Procedure: List Squares and Cubes on the TI-84

1. Press [Y=] to access the function entry screen.
2. Enter $\mathtt{X}^2$ as $\mathtt{Y}_1$ and $\mathtt{X}^3$ as $\mathtt{Y}_2$.
3. Press [2ND] [GRAPH] to access the "Table" feature.
4. The $\mathtt{Y}_1$ column shows perfect squares, and the $\mathtt{Y}_2$ column shows perfect cubes. Use the up and down arrow keys to scroll through the table to see if a particular number appears.

Example. Use the table above or a table generated on the calculator to identify each of the following numbers as a perfect square, a perfect cube, both, or neither.

1. 20 Perfect square Perfect cube Neither
2. 49 Perfect square Perfect cube Neither
3. 125 Perfect square Perfect cube Neither
4. 64 Perfect square Perfect cube Neither
5. 400 Perfect square Perfect cube Neither
6. 200 Perfect square Perfect cube Neither

Special Forms: Perfect Squares

To help us to factor, there are certain special products that it is worthwhile to memorize. The first form we memorize is the perfect square of a binomial.

To see these forms, find each product using the FOIL shortcut.

$(A+B)^2 =$

$(A-B)^2 =$

Recognizing a Perfect Square

A trinomial is a **perfect square trinomial** when:

- Two terms are positive perfect square monomials (_____ a square number, and _____ powers of variables) $+A^2$ and $+B^2$.
- The third term is $+2AB$ or $-2AB$.

Example. For each polynomial below:

a. Determine whether two terms are perfect squares.
b. Determine whether the trinomial is a perfect square.
c. If it is, factor it as such.

$4x^4 - 12x^2 + 9$

a.

b.

c.

$x^2 + 8xy + 16y^2$

a.

b.

c.

$x^2 - 5xy + 4y^2$

a.

b.

c.

$x^2 - 6x - 9$

a.

b.

c.

Special Forms: Difference of Squares

Example. Factor $x^2 - 9$ using the AC Method.

- To apply the AC Method, we need to add the term _____ .

We can generalize this to a form called **difference of squares** by multiplying:

$(A - B)(A + B) =$

Recognizing a Difference of Squares
A binomial is a **difference of squares** when: • Each term is a perfect square monomial (ignoring signs). • One is positive, and the other negative.

Example. For each polynomial below:

a. Determine whether it is a difference of squares.
b. If it is, factor it as such.

$9x^2 - 16$

a.

b.

$x^4y^2 - 36z^6$

a.

b.

$z^4 + 81$

a.

b.

$x^3 - y^2$

a.

b.

A Sum of Squares is Prime

What happens in the difference of squares form if we replace B by $-C$?

- $A^2 - B^2 = A^2 - (_____)^2 =$ _____
- $(A-B)(A+B) = (A-(_____))(A+(_____)) =$ _____

This is the _____ form!

Example. Try to factor $x^2 + 9$ using the AC Method. Show that it is prime.

Note that $x^2 + 9 = x^2 +$ $\qquad$ $+ 9$.

1	$\times$	9	$=$	9		1	$+$	9	$=$										
3	$\times$	3	$=$	9		3	$+$	3	$=$										
-1	$\times$	-9	$=$	9		-1	$+$	-9	$=$										
-3	$\times$	-3	$=$	9		-3	$+$	-3	$=$										

Note that any two numbers whose product is $+9$ must have the same _____ , so their sum can never be _____ .

Theorem
As long as we are limited to polynomails with real number coefficients, any polynomial of the form $A^2 + B^2$ cannot be factored as a product of two binomials.

Special Forms: Difference of Cubes

Let's see what happens when we multiply $(A-B)(A^2+AB+B^2)$ using the rectangle method.

Notice that all the _____ cancel out! This gives us a special form called a **difference of cubes**.

Recognizing a Difference of Cubes
A binomial is a **difference of cubes** when: • Each term is a perfect cube monomials (_____ a cubed number, and powers of variables multiples of _____). • One is positive, the other negative.

Example. Factor each difference of cubes.

$x^3 - 27 =$

$8x^6y^3 - 27 =$

Note. We can generalize this for *any* powers (not just 2 and 3) but we will only consider 2 and 3 for this class.

Special Forms: Sum of Cubes

What happens in the difference of cubes form if we replace B by $-C$?

- $A^3 - B^3 = A^3 - (_____)^3 =$ _____
- $(A - B)(A^2 + AB + B^2) = (A - (_____))(A^2 + A(_____) + (_____)^2)$

  $=$ _____

This is a _____ form, the _____ **of cubes**.

Recognizing a Sum of Cubes
A binomial is a **sum of cubes** when: • Each term is a perfect cube monomials (_____ a cubed number, and powers of variables multiples of _____). • Both terms are positive.

Note. If both terms are _____ , we can factor out _____ to get a sum of cubes.

Example. Factor each sum of cubes.

$8x^3 + 1 =$

$x^6 + y^6 =$

Note. We can generalize this for any _____ powers (not just 3). For _____ powers, the sum is prime.

Using Special Forms

We now have five special forms of polynomials, that we can use to help us factor.

Perfect Square Trinomials	
$(A+B)^2 = A^2 + 2AB + B^2$	$(A-B)^2 = A^2 - 2AB + B^2$

Difference of Squares
$A^2 - B^2 = (A-B)(A+B)$

Sum and Difference of Cubes	
$A^3 - B^3 = (A-B)(A^2 + AB + B^2)$	$A^3 + B^3 = (A+B)(A^2 - AB + B^2)$

How do we use this? We should be in the habit, when factoring, of being on the lookout for these special forms.

- When factoring a trinomial, two _____ terms should "jump out" at us, and we should check for a perfect square.
- When factoring a _____ , we can *only* factor if we have one of the special forms!

Example. For each binomial, determine whether it is a special form, and if so, factor it.

$x^4 - 1$

$8y^6 + 343$

$5x^4 - 1$

$25y^2 + 16$

$x^3 - 25$

$x^6 - 64$

Note. If a binomial is *both* a difference of squares and of cubes, we'll factor it first as a difference of _____ .

2.5 Factoring Fully and Polynomial Equations

Factoring Fully

Definition: Factoring Fully

A polynomial is **fully factored** if:

- The greatest common factor, if any, has been factored out.
- All polynomial factors are prime.

Example. Fully factor the polynomial $x^4y^2 - 16y^2$

1. Factor out the GCF

2. Factor the polynomial factor

3. Factor each new polynomial factor

4. Put it all together

Factoring Trinomials

We know how to factor quadratic trinomials. When they have higher degree, we may be able to pretend they're quadratic.

Example. Fully factor $10x^4 - 25x^2 - 60$.

1. Factor out the GCF

2. Factor the polynomial factor

3. Factor each new polynomial factor

4. Put it all together

Disguised Difference of Squares

Sometimes, we have a difference of squares in which one or both squares is a _____ .

Example. Fully factor the polynomial $x^4 - 10x^3 + 25x^2 - 36$.

1. Factor out the GCF
2. Factor the polynomial factor
3. Factor each new polynomial factor
4. Put it all together

We can check for this whenever we have 3 terms that are _____ .

Example. Fully factor the polynomial $x^4 - 4x^2 + 12x - 9$.

1. Factor out the GCF
2. Factor the polynomial factor
3. Factor each new polynomial factor
4. Put it all together

Multiple Iterations

We must continue factoring until all remaining factors are prime.

Example. Fully factor $5x^6 + 15x^5 - 5x^2 - 15x$.

1. Factor out the GCF

2. Factor the polynomial factor

3. Factor each new polynomial factor

4. Put it all together

Factor Any Polynomial

Follow these steps to factor any polynomial.

1. Factor out the GCF.

2. The next step depends on the number of terms:

   Two Terms Can only factor if it's a special form.

   (a) For degree 2 or 3, anything other than $A^2 - B^2$, $A^3 - B^3$, or $A^3 + B^3$ is prime.

   (b) For degree 4+, may need extra special forms; if this fails, turn to advanced techniques.

   Three Terms Try these things.

   (a) If it's quadratic, use the AC Method. If this fails, it's prime.

   (b) If it's higher degree, try to pretend it's quadratic and use the AC Method. If this fails, turn to advanced techniques.

   Four Terms Try these things.

   (a) Factor by grouping. If this fails and it's quadratic, can't factor as product of binomials.

   (b) If grouping fails, try the difference of squares trick.

   (c) If both fail, turn to advanced techniques.

   Five+ Terms Turn to advanced techniques.

3. After factoring, return to step 2 with each factor.

2.6 Solving Polynomial Equations

Polynomial Equations

Definition: Polynomial Equation
An equation is a **polynomial equation** if the expressions on either side are polynomials, at least one of which is degree 2 or higher.

Example. $x^2 - 5x - 14 = 0$ is a polynomial equation. Determine whether each value below is a solution to this equation:

a. $x = 7$ _____

b. $x = 2$ _____

c. $x = -2$ _____

Example. $(x - 5)(x + 3) = 9$ is a polynomial equation. Determine whether each value below is a solution to the equation:

a. $x = 6$ _____

b. $x = 3$ _____

c. $x = -4$ _____

Note that polynomial equations typically have _____ solution.

Polynomial Equations in Factored Form

Zero Property of Multiplication
For any real numbers a and b, if $ab = 0$ then $a = 0$ or $b = 0$

We can use this fact to solve polynomial equations.

Example. Use the Zero Property of Multiplication to solve the polynomial equation

$$(2x - 1)(x + 3)(x - 5) = 0$$

Since the _____ is _____ , one of the _____ is _____ .

We can check that these really are solutions to the original equation:

Note. We will solve *many* equations of the form $ax + b = 0$ as we continue to work with polynomials! The solution is always $x = -\frac{b}{a}$.

Polynomial Equations in Standard Form

Definition: Standard Form
A polynomial equation is said to be in **standard form** if it has an unfactored polynomial on one side, and zero on the other.

Given a polynomial equation in standard form, we first factor the polynomial.

Example. Solve $6x^4 + 14x^3 - 40x^2 = 0$.

Factor $6x^4 + 14x^3 - 40x^2$

Solve:

Check each solution

Polynomial Equations in General Forms

Procedure: Solving a Polynomial Equation
1. Write the equation in standard form. This may involve doing multiplication and/or adding or subtracting on both sides. 2. Once the polynomial is in standard form, factor the polynomial side. 3. Use the Zero Property of Multiplication to solve.

Note. If the polynomial is given in the form factored form $= 0$, there is no need to multiply out, then factor again. However, if it is in the form factored form $= c$ for some $c \neq 0$, it *is* necessary to multiply out, subtract c from both sides, then factor again. The eventual factors will be different.

Example. Solve $(x-3)(x-5) = 8$

1. Write in Standard Form

2. Factor:

3. Set each factor equal to zero

Check each solution

Example. Solve $x^3 - 4x^2 + 8 = x^3 - 5x^2 + 3x + 6$

1.

2.

3.

Check:

Solve Any Equation

Solving polynomial equations uses a whole new technique! (New steps in red.)

Solve Any Equation

Basic moves

- Simplify the expressions on either side.
- Add or subtract the same expression on both sides of the equation.
- Multiply or divide by the same expression on both sides of the equation.
  - We must be careful not to multiply or divide by zero.

How does the variable appear?

- Once, inside a function $\rightarrow$ Isolate, then eliminate the function
  - If the function is not one-to-one, there may be many solutions.
  - If the function's range is not all real numbers, there may be no solutions solutions.
- Many times, as a polynomial $\rightarrow$ Write in standard form, factor, use the zero property.

2.7 Function Algebra

Function Algebra

We have learned one way to combine functions: _____ . We can also combine them the way we do numbers.

Definition: Sum, Difference, Product, Quotient of Functions

For any functions f and g:

- Their **sum** $f+g$ is given by $(f+g)(x) = f(x) + g(x)$
- Their **difference** $f-g$ is given by $(f-g)(x) = f(x) - g(x)$
- Their **product** $f \cdot g$ is given by $(f \cdot g)(x) = f(x) \cdot g(x)$
- Their **quotient** $\frac{f}{g}$ is given by $\left(\frac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}$

Note. We may write fg rather than $f \cdot g$. It is important to distinguish between $f \cdot g$ and $f \circ g$ (the composition).

Example. Let $f(x) = 3x^2 - 5x + 2$, $g(x) = 2\,|x-5|$. Find each of the following:

$f(3) =$

$g(3) =$

$(f+g)(3) =$

$(f-g)(3) =$

$(f \cdot g)(3) =$

$\left(\frac{f}{g}\right)(3) =$

Constructing new functions in this way is referred to as "function algebra."

Domains of New Functions

For x to be in the domain of the sum, difference, product, or quotient of two functions f and g, we need to be able to do three things:

- Evaluate $f(x)$
- Evaluate $g(x)$
- Perform the appropriate operation (addition, subtraction, multiplication, or division).

We can add, subtract, or multiply _____ two numbers, but we can't divide by _____ .

Domain of Sum, Difference, Product, and Quotient
For any two functions f and g, $$\begin{aligned} \text{Dom}(f+g) &= (\text{Dom}\, f) \cap (\text{Dom}\, g) \\ \text{Dom}(f-g) &= (\text{Dom}\, f) \cap (\text{Dom}\, g) \\ \text{Dom}(f \cdot g) &= (\text{Dom}\, f) \cap (\text{Dom}\, g) \end{aligned}$$ $$\text{Dom}\left(\frac{f}{g}\right) = (\text{Dom}\, f) \cap (\text{Dom}\, g) \cap \{x \mid g(x) \neq 0\}$$

Example. Suppose $\text{Dom}\, f = (-\infty, 3]$ and $\text{Dom}\, g = (-6, 15)$. Find each domain:

$\text{Dom}\,(f+g) =$

$\text{Dom}\,(f-g) =$

$\text{Dom}\,(f \cdot g) =$

Suppose also the solution set of $g(x) = 0$ is $\{-2, 1\}$. Find the domain:

$\text{Dom}\left(\frac{f}{g}\right) =$

Examples from Pairs

Example. Define the following functions:

$$\begin{aligned} f &= \{(-8,\ \ 8),\ (-6,-3),\ (3,\ \ 0),\ (4,\ \ 3),\ (9,-3)\} \\ g &= \{(-8,-6),\ (-4,\ \ 5),\ (3,-7),\ (7,-3),\ (9,\ \ 0)\} \\ h &= \{(-8,\ \ 3),\ (\ \ 0,-9),\ (1,-8),\ (3,-3),\ (9,\ \ 7)\} \end{aligned}$$

Find each of the following.

a. $(f+g)(3)$ _____

b. $(h-g)(1)$ _____

c. $(f \cdot g)(9)$ _____

d. $\left(\frac{g}{h}\right)(-8)$ _____

e. $\left(\frac{f}{h}\right)(3)$ _____

Example. Define the following functions:

$$\begin{aligned} f &= \{(-8,\ \ 8),\ (-6,-3),\ (3,\ \ 0),\ (4,\ \ 3),\ (9,-3)\} \\ g &= \{(-8,-6),\ (-4,\ \ 5),\ (3,-7),\ (7,-3),\ (9,\ \ 0)\} \\ h &= \{(-8,\ \ 3),\ (\ \ 0,-9),\ (1,-8),\ (3,-3),\ (9,\ \ 7)\} \end{aligned}$$

Find each of the following.

a. $\operatorname{Dom} f$ _____

b. $\operatorname{Dom} g$ _____

c. $\operatorname{Dom} h$ _____

d. $\operatorname{Dom}(f+g)$ _____

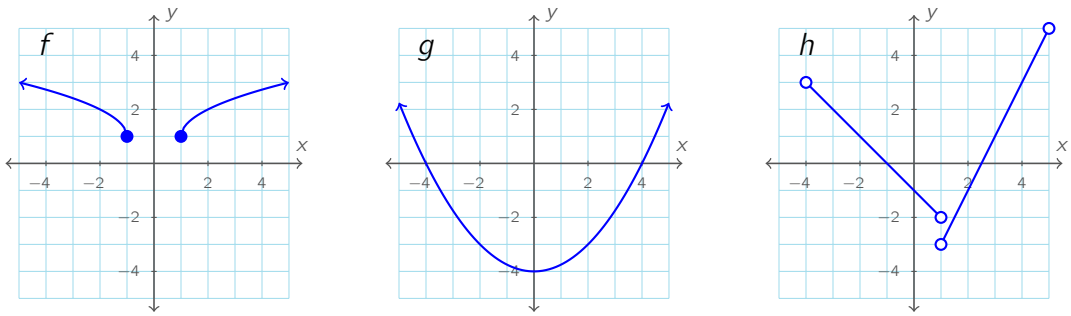
e. $\operatorname{Dom}(f-h)$ _____

f. $\operatorname{Dom}(g \cdot h)$ _____

g. $\operatorname{Dom}\left(\frac{h}{g}\right)$ _____

Examples from Graphs

Example. The functions f, g, and h are graphed below.



Find each of the following:

a. $(f+g)(2)$ _____

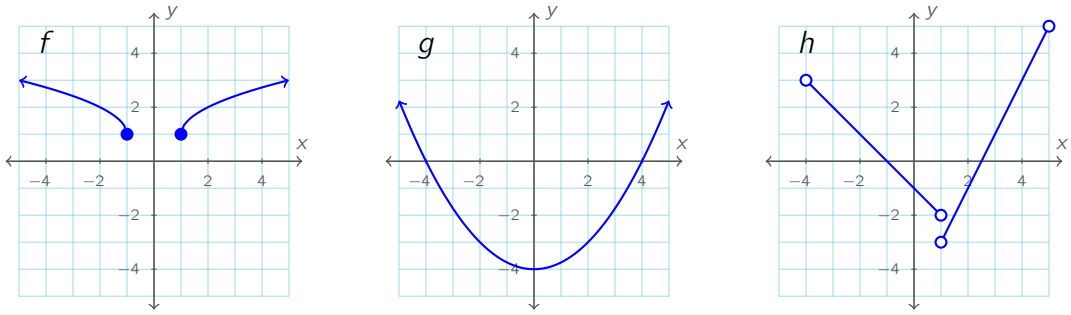
b. $(g-h)(4)$ _____

c. $(f \cdot h)(1)$ _____

d. $\left(\frac{f}{h}\right)(-2)$ _____

e. $\left(\frac{h}{g}\right)(4)$ _____

Example. The functions f, g, and h are graphed below.



Find each of the following:

a. Dom f _____

b. Dom g _____

c. Dom h _____

d. Dom$(f+h)$ _____

e. Dom $\left(\frac{g}{f}\right)$ _____

Examples from Equations

Example. Define the following functions.

$f(x) = 2x - 3$ $\quad$ $g(x) = x^2 - 3x - 10$ $\quad$ $h(x) = x^2 - 9$

Find each of the following.

a. $(f + g)(2)$ _____

b. $(f - h)(-1)$ _____

c. $(g \cdot h)(0)$ _____

d. $\left(\frac{h}{g}\right)(5)$ _____

e. $\left(\frac{f}{g}\right)(-4)$ _____

Example. Define the following functions.

$f(x) = 2x - 3$ $\quad$ $g(x) = x^2 - 3x - 10$ $\quad$ $h(x) = x^2 - 9$

Find expressions for the following. Simplify as much as possible.

a. $(f + g)(x) =$

b. $(h - g)(x) =$

c. $(g \cdot h)(x) =$

Write an expression for the following. Do not simplify.

d. $\left(\dfrac{g}{h}\right)(x) =$

Polynomials and Function Algebra

Theorem
Polynomials are exactly those functions that can be built from linear and constant functions by addition, subtraction, and multiplication.

Example. Show how $P(x) = 2x^2 + 2x - 7$ can be built from linear and constant functions by addition, subtraction, and multiplication.

Note. There are many other ways we could break this function apart.

Theorem
The domain of any polynomial function is _____ . In notation, if P is a polynomial function, $$\text{Dom } P = ____ = _____$$

Studying the arithmetic of polynomial expressions, we discovered the following fact about polynomial functions.

Theorem: Closure of Polynomials
The set of polynomial functions is **closed** under addition, subtraction, and multiplication. That is, if P and Q are polynomial functions, $P + Q$, $P - Q$, and $P \cdot Q$ are also polynomial functions.

The same is not true for _____ .

2.8 Rational Functions

Rational Expressions

Definition: Rational Expression
A **rational expression** is a fraction whose numerator and denominator are both polynomials.

Note. Any polynomial is a rational expression with denominator _____ . This is like the relationship between _____ and _____ .

Example. Determine whether each of the following are rational expressions.

a. $\dfrac{3x^2 - 5x + 7}{2x^2 + 4x - 11}$

b. $\dfrac{1}{x^2 - 5x - 2}$

c. $4x^5 - 3$

d. $\dfrac{3x^2}{2y}$

e. $\dfrac{2x + 5}{\sqrt{3x - 7}}$

f. $\dfrac{|x^2 + 5x + 4|}{2x - 7}$

g. $\dfrac{x^2 - x^{1.5} + 3x}{x^{2.5} - 3}$

Review: Simplifying Fractions

Before we manipulate rational expressions, we must remember how we manipulate fractions. Recall the following definitions:

Definition

- Two fractions are **equivalent** if they represent the same number.
- A fraction is in **lowest terms** if its numerator and denominator are as small as possible and it includes at most one negative sign.
- To **simplify** a fraction is to write an equivalent in lowest terms.

We can simplify a fraction by "canceling" any common factors of the numerator and denominator.

Example. Simplify the following fractions.

a. $\dfrac{12}{16} =$

b. $\dfrac{-15}{-10} =$

c. $\dfrac{18}{-21} =$

When we deal with larger numbers, this process is easier if we factor our numerator and denominator into primes.

Example. Simplify the fraction $\dfrac{252}{540}$ by factoring the numerator and denominator into primes.

Simplifying Ratios of Monomials

To simplify a rational expression in which the numerator and denominator are both monomials, we use the properties of exponents, specifically:

Theorem: Quotient of Powers Rule

For any real number x and positive integers $m \neq n$,

$$\text{If } m > n,\ \frac{x^m}{x^n} = x^{m-n} \qquad \text{If } m < n,\ \frac{x^m}{x^n} = \frac{1}{x^{n-m}}$$

We can apply this to each variable.

Example. Simplify.

a. $\dfrac{x^2y^3z}{x^3yz^5} =$

b. $\dfrac{3a^2b^5c}{9a^2c^4d^2} =$

While they aren't actually monomials, we can also do this with expressions that include negative exponents.

Example. Simplify. Give your answer without using negative exponents.

$\dfrac{x^{-3}y^2z^5}{x^2y^{-4}z^3} =$

Note that if a variable appears only once with a negative exponent, we can write it with a positive exponent by "moving" it between the numerator and denominator.

Example. Simplify. Give your answer without using negative exponents.

$\dfrac{4a^{-3}b}{5b^{-2}c^{-1}} =$

Simplifying Ratios of Polynomials

Simplifying with monomials was easy, because they are automatically factored. With polynomials, just like with numbers, we must factor before we can simplify a rational expression.

Example. Simplify $\dfrac{2x^3 - 7x^2 + 6x}{2x^3 - 16}$.

To do this, we first need to fully factor the numerator and denominator.

$2x^3 - 7x^2 + 6x =$

$2x^3 - 16 =$

Then, we rewrite with the numerator and denominator in factored form, and cancel any common factors.

$$\frac{2x^3 - 7x^2 + 6x}{2x^3 - 16} =$$

To finish, we may or may not multiply any remaining factors. Often, we will leave our answer in factored form.

Example. Simplify $\dfrac{2x^3 + 3x^2 - 18x - 27}{4x^3 - 12x^2 - 9x + 27}$.

Term Order in Factors

Sometimes, factors appear to be common, but the order of terms will be different.

Example. Simplify $\dfrac{16 + 12x - 4x^2 - 3x^3}{9x^3 - 18x^2 - 16x + 32}$.

To do this, we first need to fully factor the numerator and denominator.

$16 + 12x - 4x^2 - 3x^3 =$

$9x^3 - 18x^2 - 16x + 32 =$

Then, we rewrite with the numerator and denominator in factored form.

$$\frac{16 + 12x - 4x^2 - 3x^3}{9x^3 - 18x^2 - 16x + 32} =$$

We see two pairs of factors that are not *quite* the same.

- _____ and _____

  _____

- _____ and _____

  _____

Using this to rewrite:

$$\frac{16 + 12x - 4x^2 - 3x^3}{9x^3 - 18x^2 - 16x + 32} =$$

Rational Functions

Definition: Rational Function

We can define a rational function in two ways.

- A rational function is one whose expression is a rational expression.
- A rational function is one that can be found by dividing two polynomial functions, that is, if P and Q are polynomial functions, $R = \dfrac{P}{Q}$ is a rational function.

Example. Let $R(x) = \dfrac{2x^2 - 7x + 5}{3x^2 - 7x + 10}$. Evaluate each of the following, if defined.

a. $R(4) =$

b. $R(-2) =$

c. $R(0) =$

As we study the arithmetic of rational expressions, we will discover the following fact.

Theorem: Closure of Rational Functions

The set of rational functions is **closed** under addition, subtraction, multiplication, and division. That is, if R and S are rational functions, $R + S$, $R - S$, $R \cdot S$, and $\dfrac{R}{S}$ are also rational functions.

Domain of a Rational Function

Recall that we get a rational function by dividing two polynomial functions, and that the domain of a polynomial function is all real numbers. This tells us how to find the domain of any rational function.

Theorem: Domain of a Rational Function
The domain of a rational function $R(x) = \dfrac{P(x)}{Q(x)}$ is $\{x \mid Q(x) \neq 0\}$.

We can also see this directly: we can plug any number we like into a polynomial, but we can't _____ by _____ .

Example. Find the domain of the rational function $f(x) = \dfrac{3x-5}{2x+1}$.

To do this we must solve

Dom f =

Example. Find the domain of the rational function $g(x) = \dfrac{2x-3}{x^2-7x+12}$.

To do this we must solve

Dom g =

Simplification and Domain Restrictions

When we simplify the expression for a rational function, *the domain stays the same*. Sometimes, we must note a domain restriction.

Example. Let $h(x) = \dfrac{3x^2 - 2x - 8}{x^2 + 4x - 12}$.

a. Find the domain of h.

b. Simplify the expression for h.

c. Write the simplified form of h, noting domain restrictions.

2.9 Arithmetic of Rational Expressions

Review: Multiplying and Dividing Fractions

Multiplying and dividing fractions is straightforward.

Procedure: Multiplying Fractions
To multiply fractions, multiply the numerators, then multiply the denominators.

Example. Multiply $\frac{3}{8} \cdot \frac{10}{9}$. Simplify your answer as much as possible.

$$\frac{3}{8} \cdot \frac{10}{9} =$$

Note. When we simplify, we must _____ the numerator and denominator. We didn't need to multiply _____ or _____ ! In fact, we could have just factored _____ .

Procedure: Dividing Fractions
To divide fractions, rewrite as multiplication by the reciprocal. That is, for any fractions $\frac{a}{b}$ and $\frac{c}{d} \neq 0$, $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$.

Example. Divide $\frac{7}{9} \div \frac{14}{15}$. Simplify your answer as much as possible.

$$\frac{7}{9} \div \frac{14}{15} =$$

If we want to include integers, we write them with denominator _____ .

Example. Find each of the following.

a. $\frac{5}{6} \cdot 9 =$

b. $\frac{4}{5} \div 8 =$

Multiplying Rational Expressions

We multiply rational expressions the same way we do fractions.

Example. Multiply. Simplify your answer as much as possible.

$$\frac{5x^2 - 25x}{x^2 - 3x + 2} \cdot \frac{x^3 - 8}{x^2 - 25} =$$

$$\frac{5x^2 - 25x}{x^2 - 3x + 2} \cdot \frac{x^3 - 8}{x^2 - 25} =$$

We can also multiply a rational expression by a polynomial.

Example. Multiply. Simplify your answer as much as possible.

$$\frac{3x + 5}{x^3 + 27} \cdot (3x^3 - 9x^2 + 27x) =$$

$$\frac{3x + 5}{x^3 + 27} \cdot (3x^3 - 9x^2 + 27x) =$$

Dividing Rational Expressions

We divide rational expressions the same way we do fractions.

Example. Divide. Simplify your answer as much as possible.

$$\frac{3x^4y - 6x^3y - 9x^2y}{5x^2 - 20} \div \frac{3x^5y - 27x^3y}{5x^2 + 30x + 40} =$$

$$\frac{3x^4y - 6x^3y - 9x^2y}{5x^2 - 20} \div \frac{3x^5y - 27x^3y}{5x^2 + 30x + 40} =$$

We can divide polynomials in this way. Sometimes, the result is itself a polynomial.

Example. Divide. Simplify your answer as much as possible.

$$(4x^3 + 16x^2 - 9x - 36) \div (2x^2 + 5x - 12) =$$

$$(4x^3 + 16x^2 - 9x - 36) \div (2x^2 + 5x - 12) =$$

Review: Adding and Subtracting Fractions

We can add fractions with the same _____ .

Procedure: Adding or Subtracting Fractions with Like Denominator
If fractions have the same denominator, we can add or subtract them by adding or subtracting the numerators.

Example. Add or subtract. Simplify your answer as much as possible.

a. $\frac{5}{9} + \frac{7}{9} =$

b. $\frac{9}{8} - \frac{3}{8} =$

Otherwise, we have to _____ them to have the same denominator.

Procedure: Adding or Subtracting Fractions
1. Find the least common multiple of the denominators (LCD). 2. Write equivalent fractions with that denominator. 3. Add the new fractions (that have the same denominator). 4. Simplify, if necessary.

There are many methods we can use to find a common denominator with numbers. In this example, we'll use the method we'll use for rational expressions.

Example. Add $\frac{5}{6} + \frac{7}{10}$.

1. Find the LCD by looking at factors.

   $6 =$ $10 =$

   LCD $=$

2. Rewrite the fractions with the LCD.

   $\frac{5}{6} =$ $\frac{7}{10} =$

3. Add.

   $\frac{5}{6} + \frac{7}{10} =$

4. Simplify if necessary.

Adding and Subtracting with Like Denominators

As with fractions, we can add and subtract rational expressions with the same _____ by adding or subtracting the _____ .

Example. Add or subtract. Simplify your answer as much as possible.

a. $\dfrac{3x}{5y} + \dfrac{4x}{5y} =$

b. $\dfrac{11x^2}{4x+1} - \dfrac{7}{4x+1} =$

c. $\dfrac{x^2 - 7x + 2}{6x^2 - x - 15} + \dfrac{x^2 + 2x - 14}{6x^2 - x - 15} =$

d. $\dfrac{x^2 + 6x + 9}{x^4 - 81} - \dfrac{12x}{x^4 - 81} =$

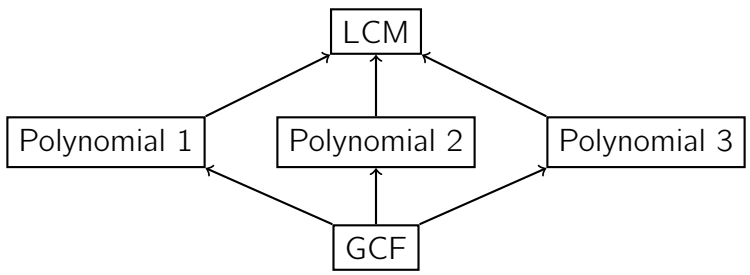
Least Common Multiple of Polynomials

To find the LCD for rational expressions, we must be able to find the least common multiple (LCM) of two (or more) polynomials.

Procedure: Finding Least Common Multiple for Polynomials

1. Factor all denominators.
2. Build the LCM:
   - Must have all factors that appear in any denominator.
   - If factors appear with different powers, take the highest power.

Note. The LCM and the GCF (greatest common factor) are found in similar ways, but the GCF must be a factor of each polynomial, and each polynomial must be a factor of the LCM.



Example. Find the least common multiple of $3x^2y$, $2x^5z$, and $9xy^3$.

LCM =

Example. Find the least common multiple of the three polynomials below. Leave the answer in factored form.

$6x^2 - 9x - 6 =$

$3x^4 - 12x^3 + 12x^2 =$

$2x^4 + x^3 - 8x^2 - 4x =$

LCM =

Adding and Subtracting Rational Expressions

To add or subtract rational expressions with different denominators, we follow the same steps as we do to add or subtract fractions of numbers:

Procedure: Adding or Subtracting Fractions
1. Find the least common multiple of the denominators (LCD). 2. Write equivalent fractions with that denominator. 3. Add the new fractions (that have the same denominator). 4. Simplify, if necessary.

Example. Add $\dfrac{x+4}{2x^2+x-6} + \dfrac{3x+1}{2x^2-13x+15}$. Simplify your answer as much as possible.

1. Find the LCD

2. Rewrite

3. Add

4. Simplify

2.10 Solving Rational Equations

Solutions to Rational Equations

Definition: Rational Equation
A **rational equation** is an equation in which both sides are rational expressions, at least one of which has a variable in the denominator.

Example. $\dfrac{3x-1}{x-2} = 4$ is a rational equation. Determine whether each value below is a solution to the equation:

a. $x = 0$ _____

b. $x = 3$ _____

c. $x = 7$ _____

Example. $\dfrac{2x-1}{x-3} = \dfrac{3x+5}{x+1}$ is a polynomial equation. Determine whether each value below is a solution to the equation:

a. $x = -2$ _____

b. $x = 0$ _____

c. $x = 2$ _____

d. $x = 7$ _____

Note that rational equations typically have _____ solution.

Clearing Denominators

We can _____ both sides of an equation by the LCD to "clear denominators."

Example. Solve $\frac{2}{3}x - \frac{4}{5} = \frac{3}{5}x + \frac{8}{15}$ by first clearing denominators.

a. Find the LCD

b. Multiply both sides by

c. Solve the new equation

d. Check

We can do the same thing when the denominators contain variables.

Example. Solve $\dfrac{x^2 - 10}{x^2} = \dfrac{3}{x}$.

a. Find the LCD

b. Multiply both sides by

c. Solve the new equation

d. Check

Extraneous Solutions

When we clear denominators in a rational equation, we multiply both sides by a _____ expression, which we've avoided so far. Why?

Example. Solve $\dfrac{3x+2}{x^2-4x-12} = \dfrac{2x+1}{x^2-2x-8}$ by clearing denominators.

a. Find the LCD

$x^2 - 4x - 12 =$ $\qquad$ $x^2 - 2x - 8 =$

LCD =

b. Multiply both sides by

c. Solve the new equation

d. Check

Wait—what happened?

If $x =$ _____, _____ $=$ _____, so we accidentally multiplied both sides by _____ . $x =$ _____ is an **extraneous solution**.

Definition: Extraneous Solution
An **extraneous solution** to an equation is one which appears through the solving process, but was not an actual solution to the equation. One reason this may occur is accidentally multiplying both sides of the equation by _____ .

Flagging Extraneous Solutions

There are two ways we can control extraneous solutions.

1. When we know one of our steps might have caused an extraneous solution, we can check our solutions in the original equation.
2. When we perform a step that might cause an extraneous solution, we can "flag" the forbidden values.

Let's see an example with the second method.

Example. Solve $\dfrac{8x-10}{x^2-3x+2} - \dfrac{x+29}{x^2+3x-4} = \dfrac{8x+2}{x^2+2x-8}$ by clearing denominators.

a. Find the LCD

$x^2 - 3x + 2 =$ $\qquad$ $x^2 + 3x - 4 =$

$x^2 + 2x - 8 =$ $\qquad$ LCD $=$

b. Multiply both sides by

$x \neq$

c. Solve the new equation

d. Check

Solve Any Equation

Solving rational equation doesn't introduce a new *technique*, but we discover a new thing that can go wrong! (New complications in red.)

Solve Any Equation

Basic moves

- Simplify the expressions on either side.
- Add or subtract the same expression on both sides of the equation.
- Multiply or divide by the same expression on both sides of the equation.
  - We must be careful not to multiply or divide by zero.
  - If we accidentally multiply by zero, we introduce **extraneous solutions**.
  - If we multiplied by a variable expression, we must check.

How does the variable appear?

- Once, inside a function $\rightarrow$ Isolate, then eliminate the function
  - If the function is not one-to-one, there may be many solutions.
  - If the function's range is not all real numbers, there may be no solutions solutions.
- Many times, as a polynomial $\rightarrow$ Write in standard form, factor, use the zero property.

Unit 3

Power Functions and Quadratic Equations

3.1 Power Functions

Power Functions

For each possible exponent p, we have a family of **power functions**.

Definition: Power Function
A **power function** is one that can be written in the form $$f(x) = a(bx - h)^p + k$$ where a, b, h, and k are any real numbers. This is a family of functions with base function $$f(x) = x^p$$ Each exponent p gives a different family.

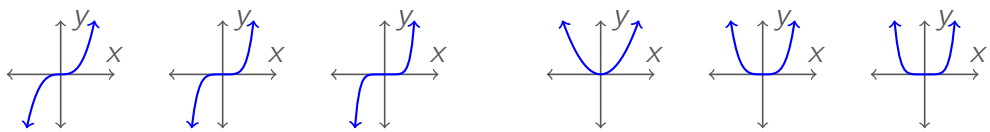
In practice, we will usually exclude exponents _____ and _____ :

- If $p =$ _____, the base function is $f(x) =$ _____, which is a _____ function.
- If $p =$ _____, the base function is $f(x) =$ _____, which is a _____ function.

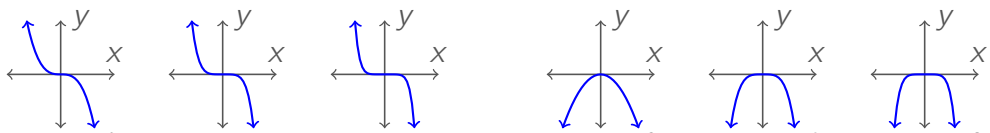
To begin, we will limit our discussion to _____ exponents.

Graphs of Power Functions

Let's consider the graphs of some power functions. (Note the unlabeled axes: we're most interested in the general shape here.)



$y = x^3$ $y = x^5$ $y = x^7$ $y = x^2$ $y = x^4$ $y = x^6$



$y = -x^3$ $y = -x^5$ $y = -x^7$ $y = -x^2$ $y = -x^4$ $y = -x^6$

We observe the following:

- _____ exponents give an _____ or _____ shaped graph.
- _____ exponents give a _____ or _____ shaped graph.
- Each graph is flat near (___, ___). A _____ exponent is flatter.
- When x is far away from _____ , the graph _____ . This is stronger for _____ exponents.
- If we add a negative sign, the effect is to _____ .

Domain and Range of Power Functions

We can understand the domain and range of a power function by looking at its graph, or by considering the operations.

Domain and Range of a Power Function - Positive Odd Integers

If $p \geq 3$ is a positive odd integer, the graph of $f(x) = a(bx - h)^p + k$ looks like one of the following.

We see that:

- The domain and range are _____
- This function is _____

We can also see this numerically: the function is defined for any _____ , and it can have any _____ or _____ number as output.

Domain and Range of a Power Function - Positive Even Integers

If $p \geq 2$ is a positive even integer, the graph of $f(x) = a(bx - h)^p + k$ looks like one of the following.

It's useful to think about this numerically: the function is defined for any _____ , but _____ can't be _____ .

- The domain is _____
- The range is _____ if $a > 0$, _____ if $a < 0$
- This function is _____

Example. Give the domain and range of each function.

a. $f(x) = 2(x-3)^4 + 1$ Dom $f =$ _____ Rng $f =$ _____

b. $g(x) = -3(x+1)^2 + 5$ Dom $g =$ _____ Rng $g =$ _____

c. $h(x) = 2(x-4)^7 - 5$ Dom $h =$ _____ Rng $h =$ _____

Odd Radicals

Since odd power functions are _____ , they have an _____ function. We need new notation to describe this function.

Definition: Radical Function (odd index)

For a positive odd integer $p \geq 3$, the ***p*th root function**

$$f(x) = \sqrt[p]{x}$$

is defined to be the function such that $f^{-1}(x) = x^p$.

In other words, $\sqrt[p]{x}$ is the unique real number such that $\left(\sqrt[p]{x}\right)^p = x$.

In this notation, the symbol $\sqrt{\ }$ is called a **radical**. The number p is called the **index**, and the number x is called the **argument**.

Example. Find each root by rewriting in terms of a power and using trial and error.

a. $\sqrt[3]{27}$

b. $\sqrt[5]{32}$

c. $\sqrt[5]{-3125}$

Often, the root is not an integer. (In fact, roots are frequently _____ .) In this case, we have several options.

- Leave the root as-is.
- Simplify the root (we will discuss this later).
- Get a decimal approximation using a calculator.
- Get a rough approximation using trial and error.

Example. Use trial and error to determine what two consecutive integers each root is between.

a. $\sqrt[3]{45}$ $\qquad < \sqrt[3]{45} <$

b. $\sqrt[5]{-18}$ $\qquad < \sqrt[5]{-18} <$

Domain and Range of a Radical Function (odd index)

If $f(x) = \sqrt[p]{x}$ and $f^{-1}(x) = x^p$, we know that:

- $\operatorname{Dom} f^{-1} =$ _____ , so $\operatorname{Rng} f =$ _____
- $\operatorname{Rng} f^{-1} =$ _____ , so $\operatorname{Dom} f =$ _____

Even Radicals

Even power functions are *not* one-to-one, but we would like a way to reverse them, so we define a _____ inverse, as follows.

Definition: Radical Function (even index)

For a positive even integer $p \geq 2$, the ***p*th root function**

$$f(x) = \sqrt[p]{x}$$

is defined to be the function such that

$$f^{-1}(x) = x^p, \ x \geq 0$$

In other words, $\sqrt[p]{x}$ is the unique *nonnegative* real number such that $\left(\sqrt[p]{x}\right)^p = x$.

Note. If we want the *negative* number that, raised to the pth power, gives us x, we can simply write $-\sqrt[p]{x}$.

The domain and range of even roots is more interesting.

Domain and Range of a Radical Function (even index)

If $f(x) = \sqrt[p]{x}$ and $f^{-1}(x) = x^p, \ x \geq 0$, we know that:

- $\operatorname{Dom} f^{-1} =$ _____, so $\operatorname{Rng} f =$ _____
- $\operatorname{Rng} f^{-1} =$ _____, so $\operatorname{Dom} f =$ _____

Example. Find the domain and range of the radical function.

$f(x) = 3\sqrt[6]{6-x} + 1$

$\operatorname{Dom} f =$ $\operatorname{Rng} f =$

Example. Find each root by rewriting in terms of a power and using trial and error.

a. $\sqrt[4]{16}$

b. $\sqrt[2]{25}$

c. $\sqrt[4]{-81}$

Like odd roots, even roots are frequently _____ .

Special Notation for Square Roots

We take the root of index _____ often enough that it has special notation.

Definition: Square Root

The **square root** is the root of index 2. We write

$$f(x) = \sqrt{x}$$

to mean the function such that

$$f^{-1}(x) = x^2, \ x \geq 0$$

Note that there is no index shown for this radical. When no index is shown, the index is _____ .

Note. Historically, the square root predates the other roots. It represents "the side of a square with the given area." The other radical functions generalize this to higher powers (with less geometric reality).

Example. Find each root by rewriting in terms of a power and using trial and error.

a. $\sqrt{16}$

b. $\sqrt{25}$

c. $\sqrt{-81}$

d. $\sqrt{400}$

e. $\sqrt{-49}$

Roots on the Calculator

When we're working with an irrational root in a concrete application, we often want a _____ approximation of the value. We'll demonstrate this on the TI-84, but any scientific or graphing calculator will have the appropriate functions.

Procedure: Taking Square Roots on the TI-84

1. Press [2ND] [x^2] to access the square root function.
2. Enter the number whose root you want to take.
3. Press [▶] to move the cursor outside the square root.

If your calculator has older firmware, or you have a TI-83, you may need to hit [)] to end the root, instead.

Example. Find a decimal approximation of each square root. Round your answer to three decimal places.

a. $\sqrt{18} \approx$

b. $\sqrt{94} \approx$

c. $\sqrt{289} \approx$

d. $\sqrt{-20} \approx$

For roots with other indexes, we need to find the appropriate command in the `MATH` menu.

Procedure: Taking Arbitrary Roots on the TI-84

1. Enter the index.
2. Press [MATH] to access the math menu.
3. Use the [▲] and [▼] arrows to find `5:`$\sqrt[x]{\ }$ and press [ENTER], or press [5].
4. Proceed as for the square root.

You can also select `4:`$\sqrt[3]{\ }$`(` or press [4] for the third root.

Example. Find a decimal approximation of each root. Round your answer to three decimal places.

a. $\sqrt[3]{18} \approx$

b. $\sqrt[4]{21} \approx$

c. $\sqrt[4]{81} \approx$

d. $\sqrt[5]{-200} \approx$

e. $\sqrt[7]{995} \approx$

f. $\sqrt[8]{-215} \approx$

3.2 Root Properties

The Odd Root Property

We can use radicals to find the solutions of equations, by rewriting power functions in terms of radical functions.

Definition: Odd Root Property

For any expression X, and real number c, and odd integer p, the equations

$$X^p = c \quad \text{and} \quad X = \sqrt[p]{c}$$

are equivalent.

Note. This is a case of a general move. If f is a one-to-one function and c is a number in the range of f, the equations $f(X) = c$ and $X = f^{-1}(c)$ are equivalent.

Procedure: Solving Equations with the Odd Root Property

1. Isolate the power expression.
2. Apply the odd root property to rewrite.
3. Continue solving.

Example. Solve each equation below using the Odd Root Property.

$x^3 = 125$

Check:

$(2x+1)^5 = 32$

Check:

$4(x-3)^7 + 8 = 4$

Check:

We can solve some equations of this type as polynomial equations.

Example. Solve $x^3 = 125$ as a polynomial equation by factoring.

The Even Root Property

With even powers, using roots to solve equations is more complicated, because even power functions are not one-to-one.

Definition: Even Root Property

For any expression X, and real number $c \geq 0$, and even integer p, the equation

$$X^p = c$$

has the same solutions as the two equations

$$X = \sqrt[p]{c} \quad \text{or} \quad X = -\sqrt[p]{c}$$

If $c < 0$, $X^p = c$ has no solution.

Note. To simplify our notation, we will sometimes write $X = \pm\sqrt[p]{c}$ as a shortcut of $X = \sqrt[p]{c}$ or $X = -\sqrt[p]{c}$.

Procedure: Solving Equations with the Even Root Property

1. Isolate the power expression.
2. Apply the even root property to rewrite, if possible.
3. Continue solving.

Example. Solve each equation below using the Even Root Property.

$$x^4 = 16$$

Check:

$$(x+3)^6 = -1$$

$$2(2x+1)^2 = 50$$

Check:

We can solve some equations of this type as polynomial equations.

Example. Solve $x^4 = 16$ as a polynomial equation by factoring.

Solve Any Equation

We don't have any new techniques here: this is the same move we used with the absolute value, just made more formal (see in red below).

Solve Any Equation

Basic moves

- Simplify the expressions on either side.
- Add or subtract the same expression on both sides of the equation.
- Multiply or divide by the same expression on both sides of the equation.
  - We must be careful not to multiply or divide by zero.
  - If we accidentally multiply by zero, we introduce extraneous solutions.
  - If we multiplied by a variable expression, we must check.

How does the variable appear?

- Once, inside a function $\rightarrow$ Isolate, then eliminate the function
  - For a one-to-one function f and c in the range of f, _____ and _____ are equivalent.
  - If the function is not one-to-one, there may be many solutions: we cut up the _____ of f to have a _____ inverse.
  - If the function's range is not all real numbers, there may be no solutions solutions.
- Many times, as a polynomial $\rightarrow$ Write in standard form, factor, use the zero property.

Simplifying Roots of Numbers

Often, when we solve an equation using the root properties, we are left with an irrational root in our solution.

Example. Solve $x^3 = 54$ using the appropriate root property.

We know we can get a decimal approximation for this number. We can also **simplify** it. Recall:

Definition
To **simplify** an expression is to write an equivalent expression in some standard "simplest" from.

To simplify a radical expression, we use the following property of roots (which is the same as a property of powers).

Theorem: Root of a Product Rule
For any real numbers x and y and any positive integer p where all the roots below are real numbers, $$\sqrt[p]{xy} = \sqrt[p]{x} \cdot \sqrt[p]{y}$$

To simplify a root, then, is to take any possible integer roots.

Definition
For any real number x and integer index p, the radical expression $\sqrt[p]{x}$ is said to be **simplified** if it is written in the form $a\sqrt[p]{b}$ where $a^p b = x$ and b is as small as possible.

To accomplish this, we need to find the largest possible factor of x that is a perfect pth power.

Example. Simplify $\sqrt[3]{54}$.

$1^3 =$	$54 \div$	$=$	$\sqrt[3]{54} =$
$2^3 =$	$54 \div$	$=$	
$3^3 =$	$54 \div$	$=$	
$4^3 =$			

Using a Calculator to Simplify

Sometimes, when we simplify a root, we need to search many possible factors.

Example. Simplify $\sqrt{108}$.

$1^2 = 1$	$108 \div 1 = 108$	$\sqrt{108} =$
$2^2 = 4$	$108 \div 4 = 27$	
$3^2 = 9$	$108 \div 9 = 12$	
$4^2 = 16$	$108 \div 16 = 6.75$	
$5^2 = 25$	$108 \div 25 = 4.32$	
$6^2 = 36$	$108 \div 36 = 3$	
$7^2 = 49$	$108 \div 49 \approx 2.204$	
$8^2 = 64$	$108 \div 64 = 1.6875$	

We can use a computer to do the repetitive arithmetic. The procedure here uses the TI-84 calculator, but it can be done with many other calculators or with a computer.

Procedure: Simplifying Roots with the TI-84

1. Press [Y=] to access the function entry screen.
2. Enter $\mathtt{X}^{\mathtt{p}}$ as $\mathtt{Y}_1$ and $\mathtt{\#\#/X}^{\mathtt{p}}$ as $\mathtt{Y}_2$, where ## is the number whose root you want to simplify, and p is the index.
3. Press [2ND] [GRAPH] to access the "Table" feature.
4. The $\mathtt{Y}_1$ column shows the perfect *p*th powers, the X column shows the *p*th root, and the $\mathtt{Y}_2$ column shows the remaining factor. You are looking for the largest possible X which gives an integer in the $\mathtt{Y}_2$ column. You know you're done searching when the number in the $\mathtt{Y}_2$ is 2 or less.

Example. Simplify $\sqrt[3]{112}$.

Rationalizing the Denominator (basic)

When we take a root of a fraction, we end up with a radical in the denominator according to the following property of roots, which corresponds to a property of exponents.

Theorem: Root of a Quotient Rule

For any real numbers x and $y \neq 0$ and any positive integer p where all the roots below are real numbers,

$$\sqrt[p]{\frac{x}{y}} = \frac{\sqrt[p]{x}}{\sqrt[p]{y}}$$

Example. Simplify $\sqrt[3]{\frac{10}{27}}$.

$\sqrt[3]{\frac{10}{27}} =$

We consider a quotient of radical expressions to be simplified only when there are no radicals in the denominator (for reasons that will become clear later). If we have a fraction with one term in the denominator, we use the following procedure.

Procedure: Rationalizing the Denominator (basic)

To rationalize the denominator of a fraction $\frac{X}{\sqrt[p]{a}}$:

1. Find the smallest number b such that ab is a perfect pth power. (A table on a calculator may be helpful here.)
2. Multiply the numerator and denominator by $\sqrt[p]{b}$.
3. Simplify the numerator and denominator, combining radicals if possible.

Example. Simplify $\sqrt[4]{\frac{7}{8}}$.

$\sqrt[4]{\frac{7}{8}} =$

Trying 4th powers...

Simplifying Solutions to Power Equations

When we solve equations using the root properties, we often get solutions that have very specific forms.

Example. Solve $(6x - 12)^4 = 162$

It seems like we should be able to simplify this more! Can we "cancel" that 3?

Definition
If we have an expression in the form $$\frac{a + b\sqrt[p]{c}}{d}$$ it is considered to be in its **simplest form** if the radical expression $b\sqrt[p]{c}$ is simplified, the denominator is rationalized, and a, b, and d have no factors in common.

To cancel any factors shared by a, b, and d, we treat this like a rational expression.

Example. Simplify $\dfrac{12 \pm 3\sqrt[4]{2}}{6}$.

$$\frac{12 \pm 3\sqrt[4]{2}}{6} =$$

Note. We *cannot* cancel a factor that appears in only one term of the numerator, just like with rational expressions.

Example. Solve $3(2x - 6)^2 + 14 = 230$. Simplify your answer as much as possible.

3.3 Quadratic Functions

Quadratic Functions

Definition: Quadratic Function
A **quadratic function** is a polynomial function of degree 2 (that is, a polynomial function in which the polynomial is quadratic).

Example. Determine whether each function is quadratic.

a. $f(x) = x^2 - 5x + 3$

b. $g(x) = x^2 - 7$

c. $h(x) = (x - 5)^2 + 2$

d. $k(x) = \sqrt{4x + 3}$

e. $p(x) = x^3 - 4x^2 + 2x - 1$

f. $q(x) = (2x - 5)^4 - 1$

Solving Quadratic Equations by Factoring

Definition: Quadratic Equation
A **quadratic equation** is one in which one or both sides is a quadratic polynomial (and the degree two terms are not equal).

We know how to solve quadratic equations as _____ equations.

Example. Solve by factoring.

$x^2 - 7x - 18 = 0$

Check:

In the previous example, the equation was given in _____ form. The next example shows a quadratic equation given in a different from.

Example. Solve by factoring.

$(x - 4)^2 = 9$

Check:

Solving Quadratic Equations by the Square Root Property

If a quadratic equation is given in the appropriate form, we can also solve it using the _____ root property. (We'll say "square root property" here because we're interested specifically in index _____ .)

Example. Solve using the square root property.

$(x-4)^2 = 9$

Check:

We can solve quadratic equations using the square root property that could not be solved by factoring.

Example. The equation $(x+1)^2 = 6$ cannot be solved by factoring.

$(x+1)^2 = 6$

However, it can be solved using the square root property.

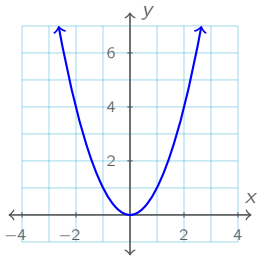
$(x+1)^2 = 6$

Check:

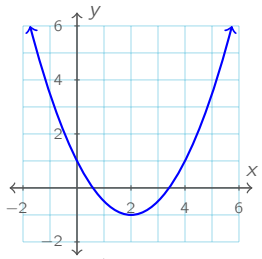
In order to use the square root property to solve, we have to write our equation in the form of a _____ equation rather than a _____ equation. Is this always possible?

Graphs of Quadratic Functions

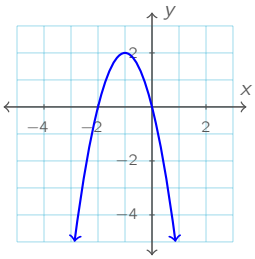
Let's look at the graphs of some quadratic functions.



$f(x) = x^2$



$g(x) = \frac{1}{2}x^2 - 2x + 1$



$h(x) = -2x^2 - 4x$

We see that these graphs all have the same _____ .

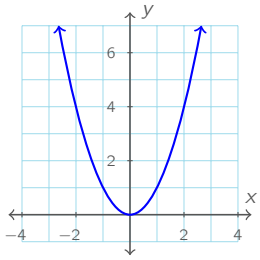
Definition: Parabola

A **parabola** is the geometric shape of the curve formed by the graph of a quadratic function. The shape has the following features.

- A highest or lowest point, called the **vertex**.
- Reflective **symmetry** across a vertical line through the vertex, the **axis of symmetry**.

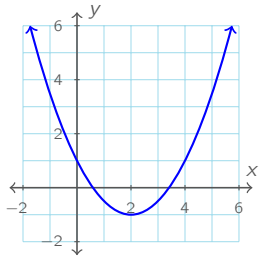
The symmetry means that points at the same distance to either side of the axis of symmetry will have the same _____ coordinate.

Example. For each of the following parabolas, identify the vertex and axis of symmetry. Label the vertex with a dot and the axis of symmetry with a dashed line.



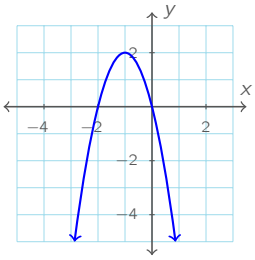
Vertex: (___, ___)

AoS: _____



Vertex: (___, ___)

AoS: _____



Vertex: (___, ___)

AoS: _____

Graphing Quadratic Functions in Standard Form

Definition

A quadratic function written as

$$f(x) = ax^2 + bx + c$$

where $a \neq 0$, b, and c are real numbers is said to be written in **standard form**.

To graph a quadratic function given in standard form, we will use the following procedure.

Procedure: Graphing a Quadratic Function in Standard Form

1. Create a table of values showing (integer) pairs x and $y = f(x)$. Include enough x values that y values both increase and decrease.
2. If possible, find two pairs with the same y-coordinate. The axis of symmetry lies halfway between their x-coordinates.
3. Find the coordinates of the vertex, if it doesn't have integer coordinates.
4. Plot the vertex, along with at least two points on each side of the axis of symmetry.
5. Draw a parabola passing through the two plotted points.

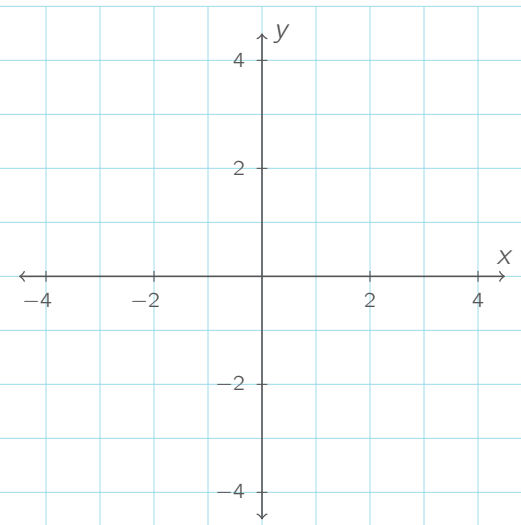
This process will require a certain amount of trial and error to find "appropriate" points! (We will have a more efficient method once we know more about quadratic functions.)

Example. Fill in the table below. Then, sketch the graph of $f(x) = x^2 - 4x + 3$.

x	$y = f(x)$
-4	
-3	
-2	
-1	
0	
1	
2	
3	
4	

AoS:

Vertex:



The x- and y-intercepts of a Quadratic Function

It can be helpful, in planning a graph, to find its x- and y-intercepts. Recall the following definitions.

Definition: Intercepts
The ***x*-intercept(s)** of a graph is(are) the point(s) at which the graph intersects the x-axis. The ***y*-intercept(s)** of a graph is(are) the point(s) at which the graph intersects the $y-$axis.

The graph of a function only has one _____ -intercept, but might have more than one _____ -intercept.

- To find the y-intercept, plug in _____ $= 0$ and _____ .
- To find the x-intercept, set _____ $= 0$ and _____ .

Example. Find the x- and $y-$ intercepts of the graph of $g(x) = x^2 - x - 6$.

Set $x = 0$

y-intercept:

Set $g(x) = 0$

x-intercepts:

We can use this information to help us sketch the graph. Note that the y-intercepts have the same y-coordinate, so the axis of symmetry is halfway between them.

Axis of Symmetry:

Vertex:

x	$y = g(x)$

3.4 Completing the Square

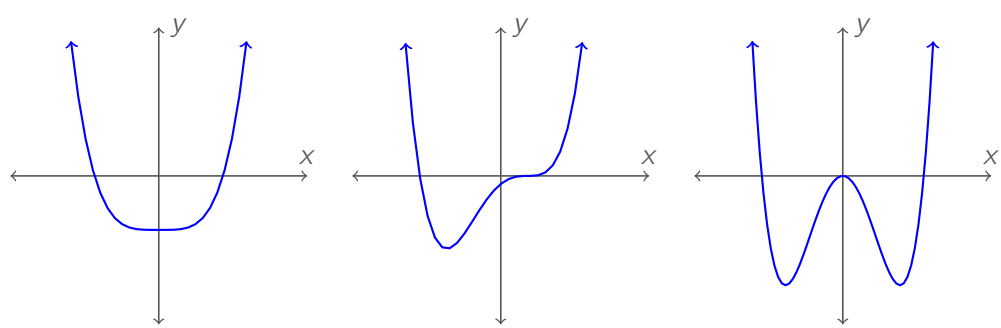
All Quadratic Functions are Power Functions

We saw before that all graphs of quadratic functions have the same shape, a _____ . This suggests a family of functions!

Claim: All Quadratic Functions are Power Functions
All quadratic functions can be written in the form $$f(x) = a(x-h)^2 + k$$ for real numbers a, h, and k. This form is called the **vertex form** of the parabola.

We will prove this claim by developing a procedure that will let us write any quadratic function in this form. However, note that this isn't true of all polynomials.

Example. The graphs of three polynomial functions of degree 4 are shown below. (Note the unlabeled axes; we are interested in the general shape of the graph.)



$f(x) = x^4 - 2$ $\qquad$ $g(x) = x^4 - 6x^2 + 8x - 3$ $\qquad$ $h(x) = x^4 - 9x^2$

We can see that these have very different _____ , so these functions to *not* form a family!

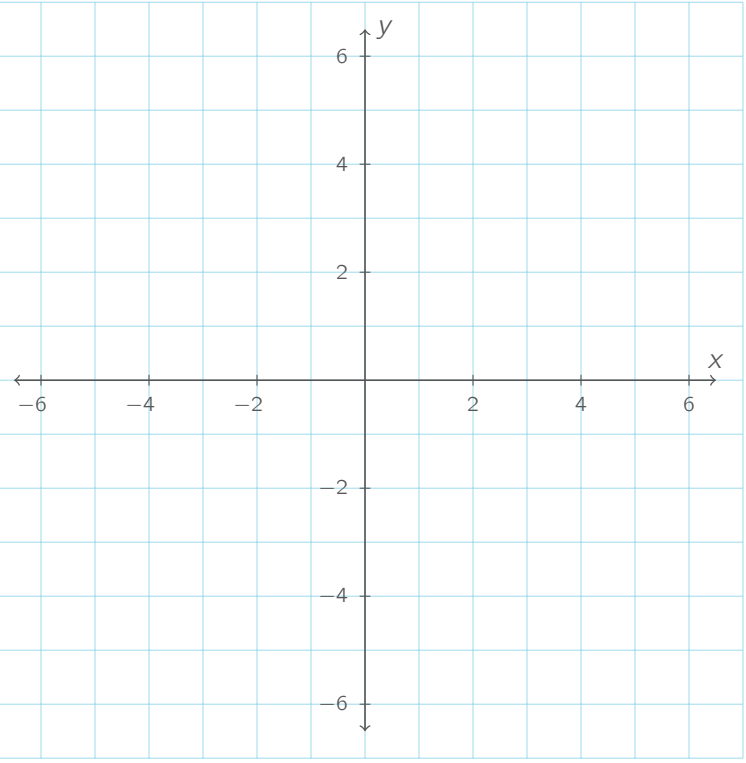
Vertex Form to Standard Form

Given a quadratic function in vertex form, we can write it in standard form just by simplifying.

Example. Write the function in standard form and graph.

$f(x) = -\frac{1}{2}(x-2)^2 + 5$

x	$y = f(x)$
-6	
-4	
-2	
0	
2	
4	
6	



We notice that the vertex is at $(___, ___)$. We can see this pair in the vertex form.

Graphing a Quadratic Function in Vertex Form

If a quadratic function is written in vertex form as

$$f(x) = a(x-h)^2 + k$$

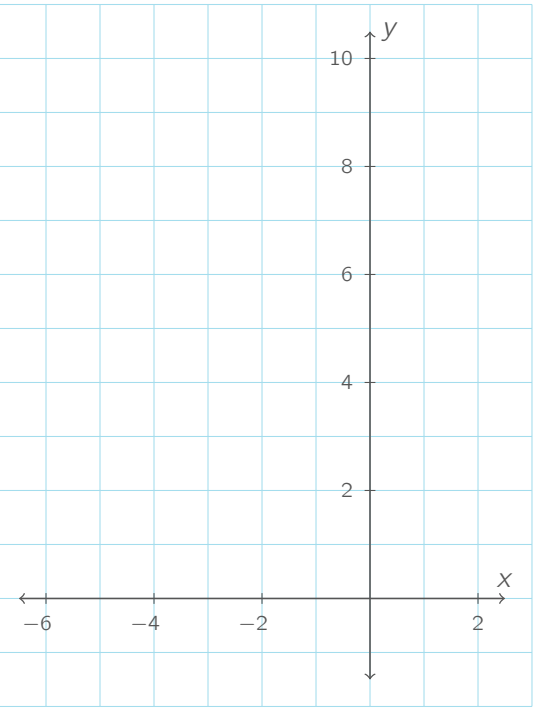
- Its vertex is at (h, k).
- If $a > 0$, the parabola opens upward. If $a < 0$, the parabola opens downward.
- For any number d, the points where $x = h + d$ and $x = h - d$ have the same y-coordinate (because of _____).

To graph a parabola, we need the vertex and two points to either side. When the function is given in vertex form, this is easy to find.

Example. Sketch the graph of the function $h(x) = 2(x+3)^2 + 1$.

Vertex:

x	$y = h(x)$



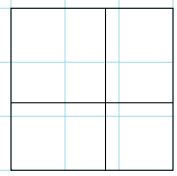
Completing the Square

Claim: We can "complete" a square
For any expression $x^2 + bx$, we can find a constant c such that $x^2 + bx + c$ is a perfect square trinomial.

Thinking of multiplication as representing the area of a rectangle, we can think of the constant c as the "missing piece" of a square.

Example. Find the constant c such that $x^2 + 6x + c$ is a perfect square, and write it as such.

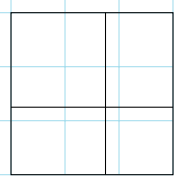
$x^2 + 6x + \quad = (x + \quad)^2$



Sometimes, fractions appear in the constant.

Example. Complete the square.

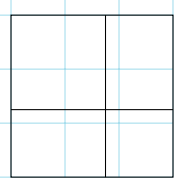
$x^2 - 5x + \quad = (x + \quad)^2$



We can do this even when the number b is already a fraction.

Example. Complete the square.

$x^2 + \frac{5}{3}x + \quad = (x + \quad)^2$



Solving Quadratic Equations by Completing the Square

We can use the "completing the square" procedure to write quadratic equations in a form in which we can solve using the square root property.

Procedure: Solve by Completing the Square
1. By adding, subtracting, and multiplying or dividing by numbers, write the equation in the form $x^2 + bx = \#$. 2. Complete the square on the left by adding the same constant to both sides. 3. Write the equation in the form $(x + \#)^2 = \#$. 4. Solve using the square root property.

Example. Solve $x^2 + 8x + 7 = 0$ by completing the square.

Example. Solve $2x^2 - 12x - 5 = 0$ by completing the square.

Writing a Quadratic Function in Vertex Form

We can also complete the square to write a quadratic function in vertex form. However, in this case we are rewriting an _____ , so we can't do things to "both sides." To do this, we will need a new simplifying move:

> We can add and subtract the same number to an expression without changing its value. That, is, for any expression X and real number c,
>
> $$X = X + c - c$$
>
> for all values of any variables in X.

Writing vertex form is easier for quadratic functions whose leading coefficient is 1.

Example. Write $f(x) = x^2 + 12x - 7$ in vertex form. Give the coordinates of the vertex of its graph.

Vertex:

We can still write vertex form when the leading coefficient is not 1, but the algebra requires a little care.

Example. Write $g(x) = -2x^2 + 8x - 5$ in vertex form. Give the coordinates of the vertex of its graph.

Group the variable terms:

Factor out the coefficient:

Complete the square:

Distribute:

Combine the constants:

Vertex:

3.5 The Quadratic Formula

The Quadratic Formula

Completing the square will always get us the solutions to a quadratic equation (if they exist), but it's tedious. We can complete the square "in general" and get a formula for the solutions.

Start in standard form	$ax^2 + bx + c = 0$
Complete the square	
Square root property	
Simplify the root	
Finish solving	

The Quadratic Formula

For a quadratic equation $ax^2 + bx + c = 0$, the solutions (if any) are given by the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Using the Quadratic Formula

Procedure: Solving a Quadratic Equation using the Quadratic Formula
1. Write the equation in standard form $ax^2 + bx + c = 0$ 2. Identify the values of a, b, and c. 3. Plug into the formula. 4. If needed, simplify or get a decimal approximation for the solutions.

Example. Use the quadratic formula to solve $5x^2 - 8x + 4 = 10$. Give exact solutions, simplified as much as possible.

The Discriminant

The quadratic formula "normally" gives 2 solutions, but it may give 1 or 0, depending on the equation.

Example. Use the quadratic formula to solve each quadratic equation.

$x^2 + 4x + 3 = 0$

$x^2 + 4x + 4 = 0$

$x^2 + 4x + 7 = 0$

The number of solutions depends on the number inside the square root. This number has a name.

Definition

Discriminant The **discriminant** of a quadratic equation $ax^2 + bx + c = 0$ is the number

$$D = b^2 - 4ac$$

whose square root appears in the quadratic formula. The quadratic equation has:

- Two real solutions if $D > 0$
- One real solution if $D = 0$
- No real solutions if $D < 0$

If D is a perfect square, the solutions are rational. Otherwise, they are irrational.

Example. For each quadratic equation below, use the discriminant to determine how many real solutions the equation has.

$3x^2 - 11x - 5 = 0$

$t^2 + 5t + 25 = 0$

$x^2 + 6x + 9 = 0$

Quadratic Formula with Zeros

Sometimes, $b = 0$ or $c = 0$. In these cases, another method will be easier, but we *can* use the quadratic formula.

Example. Use the quadratic formula to solve $3x^2 - 25 = 0$

We could more easily solve this using the _____ .

Example. Use the quadratic formula to solve $4x^2 - 11x = 0$

We could more easily solve this by _____ .

Quadratic Formula with Unpleasant Numbers

Quadratic equations often come up in real-life situations that give unpleasant numbers. We should work carefully when using them in the quadratic formula. *When getting a decimal approximation, it's important not to round until the end.*

Example. The height in meters of a thrown ball t seconds after it is thrown is given by the function

$$h(t) = -4.9t^2 + 17.3t + 0.8$$

Use the quadratic formula to determine at what time(s) the ball is 5 meters off the ground. Give your answer as a decimal approximation, rounded to 2 decimal places.

Check:

Formulas on the Calculator

When we plug unpleasant numbers into a formula on the calculator, it's very easy to mis-click. We can take advantage of the calculator's ability to store numbers to avoid re-typing values. Every graphing calculator will be able to do this, and some scientific calculators have the ability to store multiple values like this, too.

Procedure: Using Stored Values to Apply the Quadratic Formula

1. First, we will store our values of a, b, and c.
   (a) Enter the value of a, press [STO], then press [ALPHA] [MATH] to enter the letter `A`.
   (b) Repeat for b (pressing [ALPHA] [APPS] for the letter `B`)
   (c) Repeat for c (pressing [ALPHA] [PRGM] for the letter `C`)
2. Next, we'll compute the discriminant. Enter `B`^{2}`-4AC`, then press [STO] [ALPHA] [x^{-1}] to store this value as `D`.
3. If $D \geq 0$, enter `(-B+√D)/(2A)` to find the first solution, and `(-B-√D)/(2A)` to find the second. Round appropriately.

Example. A company determines that its profit, in millions of dollars, from selling x million widgets is given by the function

$$P(x) = -0.06x^2 + 3.8x$$

Use the quadratic formula to determine how many widgets they should sell to make a profit of \$50.1 million. Round your answers to one decimal place.

Check:

Quadratic Equations: Choosing a Method

We have three methods we can use to solve a quadratic equation:

1. Factoring
2. The Square Root Property
3. The Quadratic Formula

When we encounter a quadratic equation, it can be hard to decide which method to use. This is a general approach.

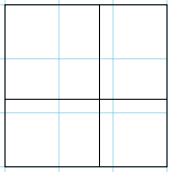
- Use the square root property if the equation is already set up for that.
  - No _____ term (only _____ or _____)
- Otherwise, write the equation in standard form.
- Factor if it seems easy.
  - _____
  - _____
- If factoring fails but the numbers are nice you might complete the square.
  - Completing the square is easy when _____
  - _____
- If factoring seems hard, or if it fails and completing the square seems hard, use the quadratic formula.

3.6 The Vertex Formula

The Vertex Formula

Like solving quadratic equations, we can go through the process of writing in vertex form just once, and record the resulting formula.

Start in standard form	$f(x) = ax^2 + bx + c$
Factor out a	
Complete the square	
Distribute	
Simplify	
Compare vertex form	$f(x) = a(x-h)^2 + k$



The Vertex Formula

For a quadratic function $f(x) = ax^2 + bx + c$, coordinates of the vertex of the graph are given by

$$(h, k) = \left(\frac{-b}{2a}, \frac{-b^2 + 4ac}{4a}\right)$$

Note. If you prefer, you can just remember $h = \frac{-b}{2a}$ and $f(h) = k$.

Vertex Formula for Vertex Form

Instead of manipulating the equation, we can use the vertex formula to write a function in vertex form, for example, in order to graph it.

Example. Use the vertex formula to rewrite $f(x) = \frac{1}{2}x^2 - 4x + 3$ in vertex form. Then, use the vertex form to sketch the graph of the equation.

Vertex as Maximum or Minimum

The vertex gives a maximum or minimum value of the quadratic function (depending on whether it's at the top or bottom of the graph).

- If the graph opens up ($a > 0$) the vertex gives the minimum value of the function.
- If the graph opens down ($a < 0$) the vertex gives the maximum value of the function.

Example. Let $f(x) = 0.003x^2 - 0.15x + 17$.

a. Find the vertex of the graph of x.

b. Does f have a maximum value or a minimum value?

c. What is the maximum/minimum value? For what value of x is it obtained?

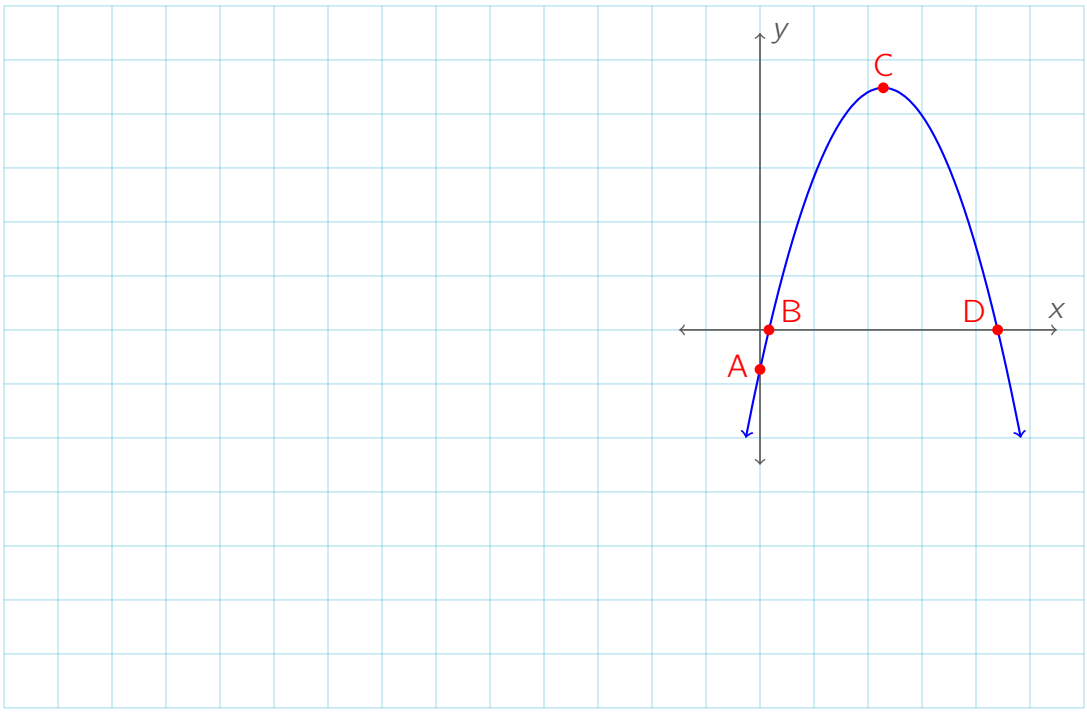
This is especially useful in applications.

Example. A model rocket is fired into the air, and its height in feet above the ground t seconds after it was launched is given by the function $H(t) = -16t^2 + 352t$. Determine the maximum height the rocket reaches, and how many seconds it takes to reach this height.

Identifying Points on Graphs

Often, we will have a computer-generated graph, especially for a function with unpleasant numbers, and we still want to identify the key points on the graph. The quadratic formula and vertex formula can be useful for this purpose.

Example. The graph shows the function $P(x) = -0.03x^2 + 13.7x - 219$. Find the coordinates of the points shown. Give the coordinates rounded to the nearest integer.



3.7 Radical Equations

Equations with One Radical: Basic

Radical functions are all one-to-one, with power functions as their inverses. So if we want to solve an equation with one radical, we have a simple process.

Procedure: Solving Equations with One Radical
1. Isolate the radical. 2. Rewrite in terms of the corresponding power function. 3. Finish solving.

Example. Solve each radical equation.

$$\sqrt{x} = 5$$

Check:

$$2\sqrt[4]{x} - 3 = 1$$

Check:

$$\sqrt[3]{3x - 1} = -4$$

Check:

$$2\sqrt{x + 7} - 4 = 8$$

Check:

Radical Equations with No Solution

An equation involving a function might have no solution due to the limited _____ of the function.

- If n is _____ , the range of $\sqrt[n]{x}$ is $\mathbb{R}$
- If n is _____ , the range of $\sqrt[n]{x}$ is $\{y|$ _____ $\}$

So the equation $\sqrt[n]{X} = c$ has no solution if n is _____ and c is _____ .

Example. Solve each radical equation, if it has solutions.

$$\sqrt[3]{x} = 5$$

$$\sqrt[4]{x} = 2$$

$$\sqrt[5]{x} = -3$$

$$\sqrt{x} = -7$$

As usual, we need to check this *after* isolating the radical.

Example. Solve each radical equation, if it has a solution.

$$-2\sqrt{3x-1} + 5 = -3$$

Check:

$$3\sqrt{4x+1} + 7 = 4$$

Check:

Equations with One Radical: Advanced

Because power functions are _____ functions, we can solve radical equations which also have occurrences of the variable outside the radical.

Note. In this example, we use an _____ root, so we don't have any worries about the range.

Example. Solve $\sqrt[3]{x^2 + 12x + 19} = x + 1$.

Check:

Extraneous Solutions to Radical Equations

Often, when we solve equations with an even root, something happens.

Example. Solve $\sqrt{x+5} = x - 1$

Check:

What happened? When $x =$ _____, the right-hand side of this equation was _____ .

Extraneous Solutions to Radical Equations
If we take an _____ power of both sides of an equation, this can introduce an **extraneous solution** if one side of the equation is _____ . This happens because the inverses—the _____ power functions—are not _____ .

As we could for rational equations, we can handle the possibility of extraneous solutions in two ways:

- Flag values that might result in extraneous solution.
  - In this example, the only valid solutions have the right-hand side positive, so _____ ≥ 0
- Check solutions in the original equation.

For radical equations, it's usually easier to check.

Equations with Two Radicals

We can also solve equations with two radicals.

Procedure: Solving an Equation with Two Radicals
1. Isolate and eliminate one radical. 2. Simplify as much as possible. If we've worked correctly, there should now be only one radical in the equation. 3. Isolate and eliminate the remaining radical. 4. Check for extraneous solutions, if necessary.

In theory, we could solve in this way with any number of radicals; but the algebra quickly becomes unweildy.

Example. Solve $\sqrt{2x+1} - \sqrt{x+4} = 1$.

Eliminate $\sqrt{2x+1}$

Simplify

Eliminate $\sqrt{x+4}$

Finish solving

Check

Note. Since we took an _____ power to eliminate the roots, we *must* check for extraneous solutions.

Solve Any Equation

In working with radical equations, we have two new issues in solving equations. New moves in red, below.

Solve Any Equation

Basic moves

- Simplify the expressions on either side.
- Add or subtract the same expression on both sides of the equation.
- Multiply or divide by the same expression on both sides of the equation.
  - We must be careful not to multiply or divide by zero.
  - If we accidentally multiply by zero, we introduce extraneous solutions.
  - If we multiplied by a variable expression, we must check.

How does the variable appear?

- Once, inside a function $\rightarrow$ Isolate, then eliminate the function
  - For a one-to-one function f and c in the range of f, $f(X) = c$ and $X = f^{-1}(c)$ are equivalent.
  - If the function is not one-to-one, there may be many solutions: we cut up the domain of f to have a partial inverse.
  - If the function's range is not all real numbers, there may be no solutions solutions.
  - If the *inverse* is not one-to-one, we may introduce **extraneous solutions**, and must check.
- Many times, inside a function $\rightarrow$ Isolate, then eliminate one instance of the function.
- Many times, as a polynomial $\rightarrow$ Write in standard form, factor, use the zero property.

Inverses with Odd Powers

Since odd power functions are one-to-one and odd roots are their inverses, we can write the inverses of odd power functions in general.

Example. Let $f(x) = 2(x-3)^3 + 6$. Find f^{-1}.

Swap x and y

Isolate the 3rd power

Rewrite using a radical function

Finish solving for y

Rewrite in function notation

Since we know that an odd power function is also the inverse of an odd root, we can also write the inverse of an odd radical function with the same logic.

Example. Let $g(x) = \frac{1}{2}\sqrt[3]{5x+1} - 1$. Find g^{-1}.

Swap x and y

Isolate the radical

Rewrite as a power function

Finish solving for y

Rewrite in function notation

Inverses with Even Powers

When we want to take the inverse of an even root, we need to be careful of the _____ of the resulting function.

Example. Let $f(x) = \sqrt{x+3} + 4$. Find f^{-1}.

Swap x and y

Isolate the root

Consider the domain

Rewrite as a power function

Finish solving for y

Rewrite in function notation

We can't take an inverse of an even power function unless the _____ is restricted. If it is, we need to be careful of whether we want the _____ or _____ root, so the range of the inverse matches the domain restriction. (We can't use both if we want to get a function!)

Example. Let $g(x) = (2-x)^4 - 7,\ x \geq 2$. Find g^{-1}.

Swap x and y

Isolate the 4th power

Consider the range

Rewrite as a radical function

Finish solving for y

Rewrite in function notation

3.8 Rational Exponents

Defining Rational Exponents: 1/q

So far, we have defined all _____ exponents.

Definition

For positive integers n and any real number x,

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ copies}}$$

For any real number $x \neq 0$,

$$x^0 = 1$$

For any negative integer $-n$ and real number $x \neq 0$,

$$x^{-n} = \frac{1}{x^n}$$

Now we'll define $x^{1/q}$ for rational numbers $\frac{1}{q}$. We want this definition to be consistent with the power of a power rule.

Power of a Power Rule (integers)

For any real number x for which all expressions are defined and any integers n and p,

$$(x^n)^p = x^{np}$$

We define $x^{1/q}$ to be consistent with this definition. In particular:

$$\left(x^{1/q}\right)^q = _____$$

Since $\left(x^{1/q}\right)^q =$ _____, it must be that $x^{1/q} =$ _____.

Definition

For any positive integer q and any real number x for which $\sqrt[q]{x}$ is defined,

$$x^{1/q} = \sqrt[q]{x}$$

Defining Rational Exponents: p/q

We have now defined the following exponents:

Definition

For positive integers n and any real number x,

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ copies}}$$

For any real number $x \neq 0$,

$$x^0 = 1$$

For any negative integer $-n$ and real number $x \neq 0$,

$$x^{-n} = \frac{1}{x^n}$$

For any positive integer q and any real number x for which $\sqrt[q]{x}$ is defined,

$$x^{1/q} = \sqrt[q]{x}$$

We define rational exponents $\frac{p}{q}$ to be consistent with the power of a power rule.

Note that $\frac{p}{q} =$

$x^{p/q} =$

$x^{p/q} =$

Definition

For any integers p and q with $q > 0$, and any real number x for which $\sqrt[q]{x}$ is defined,

$$x^{p/q} = \sqrt[q]{x^p} = \left(\sqrt[q]{x}\right)^p$$

Rational Power Functions Domain and Range

We made the following definitions for rational exponents.

Definition

For any integers p and q with $q > 0$, and any real number x for which $\sqrt[q]{x}$ is defined,

$$x^{1/q} = \sqrt[q]{x} \qquad x^{p/q} = \sqrt[q]{x^p} = \left(\sqrt[q]{x}\right)^p$$

The function $f(x) = x^{p/q} = \left(\sqrt[q]{x}\right)^p$ can be thought of as a composition of a radical function and a power function, and we can use this fact to figure out its domain and range.

Let $f(x) = x^{p/q} = \left(\sqrt[q]{x}\right)^p$, where the fraction $\frac{p}{q}$ is written in lowest terms.

- If p and q are positive odd integers:
  - Dom f = _____
  - Rng f = _____
- If p is a positive even integer and q is a positive odd integer:
  - Dom f = _____
  - Rng f = _____
- If p is a positive odd integer and q is a positive even integer:
  - Dom f = _____
  - Rng f = _____
- If p is a negative integer, then _____ is not in Dom f and Rng f.

We don't consider the case where p and q are both _____ , since in this case $\frac{p}{q}$ is not in lowest terms.

Example. Give the domain and range of each function.

$f(x) = x^{2/3}$

Dom f =

Rng f =

$g(x) = x^{3/5}$

Dom g =

Rng g =

$h(x) = x^{7/2}$

Dom h =

Rng h =

$k(x) = x^{-4/5}$

Dom k =

Rng k =

Evaluating with Rational Exponents

When we evaluate a rational power, we apply the definition. In this case, we will usually use $x^{p/q} = \left(\sqrt[q]{x}\right)^p$.

Example. Evaluate each of the following, if it is defined.

$27^{2/3} =$

$4^{-3/2} =$

$(-32)^{4/5} =$

$(-8)^{-1/3} =$

$(-81)^{3/4} =$

If we take an irrational root, we will usually use $x^{p/q} = \sqrt[q]{x^p}$ to write an expression without exponents.

Example. Write each of the following as a simplified radical, including rationalizing any denominators.

$2^{2/3} =$

$5^{-1/2} =$

$(8)^{4/5} =$

$(-4)^{-2/3} =$

Simplifying with Rational Exponents

When we rewrite a rational power of a variable expression, we will usually use $x^{p/q} = \sqrt[q]{x^p}$.

Example. Rewrite each expression using radicals and positive integer exponents.

$x^{2/3} =$

$x^{-3/2} =$

$x^{-3/5} =$

We can apply the usual rules of exponents to rational exponents.

Example. Simplify each of the following.

$\left(x^{2/3}y^{3/4}\right)^{3/5} =$

$x^{1/3}x^{1/5}x^{1/2} =$

$\left(\dfrac{x^{1/2}y^{2/3}}{x^{1/3}y^{3/4}}\right)^{-4/5} =$

3.9 Equations with Rational Exponents

Solving Equations with Rational Powers

We can generalize the way that we solve power equations to solve equations involving rational exponents, using the following facts.

> If X is an expression that represents a positive real number,
>
> $$\left(X^{p/q}\right)^{q/p} = X$$
>
> for all values of the variables in X.

We know this is true because, when all the expressions involved are defined, and we only take even powers of positive numbers,

Note that here, we're restricting ourselves to _____ numbers. This works for all real numbers when p and q are both _____ , but we need to worry about signs if p or q is _____ .

> If X and Y represent positive real numbers, and $\frac{p}{q}$ is a rational number written in lowest terms the equations
>
> $$X = Y \quad \text{and} \quad X^{p/q} = Y^{p/q}$$
>
> are equivalent.

> If p and q are both odd numbers, the above facts hold even if X and Y are negative.

Example. Use the facts above to find all positive real solutions to each equation below.

$x^{2/3} = 9$

$3x^{3/4} = 81$

$(x+1)^{3/5} = 8$

Rational Powers with Even Numerator

When p is even, $x^{p/q} = \left(\sqrt[q]{x}\right)^p$ contains an even power; in eliminating it, we use the _____ .

> When p is an even integer, q is an odd integer, and X is any expression, the equation
>
> $$X^{p/q} = c$$
>
> is equivalent to the two equations
>
> $$X = c^{q/p} \quad \text{and} \quad X = -c^{q/p}$$
>
> when $c \geq 0$, and has no solution when $c < 0$.

Example. Solve each equation.

$$2x^{2/3} + 3 = 21$$

Check:

$$(3x+1)^{4/5} = 16$$

Check:

Rational Powers with Even Denominator

Recall that if q is even, $x^{p/q}$, has domain and range _____ . That means that:

- $X^{p/q} = c$ has _____ when $c < 0$.
- If an apparent solution to $X^{p/q} = c$ makes X negative, this solution is _____ .

Example. Solve each equation, or conclude that it has no solution.

$$3x^{3/2} - 9 = -3$$

Check:

$$(2x-1)^{3/4} = -8$$

Summary of Rules for Equations with Rational Powers

Let $\frac{p}{q}$ be a fraction written in lowest terms (so p and q have no common factors).

- If p, q are both odd, $X^{p/q} = c$ is equivalent to $X = c^{q/p}$.
- If p or q are even, $X^{p/q} = c$ has no solutions if $c < 0$.
- If p is even, q is odd, and $c \geq 0$, $X^{p/q} = c$ is equivalent to $X = c^{q/p}$ or $X = -c^{q/p}$. There is no risk of extraneous solutions.
- If p is odd, q is even, and $c \geq 0$, solutions to $X^{p/q} = c$ are also solutions to $X = c^{q/p}$, but we must check for extraneous solutions.

There isn't a case where p and q are both even, since then they would have common factor _____ .

Pretend Polynomials with Rational Exponents

We can have expressions that look like polynomials, except that some of the exponents are rational.

Example. The following expressions have the form of polynomials, except one or more exponents are not positive integers.

- $x^{2/3} - 3x^{1/3} - 4$
- $x^{5/3} - x^{2/3} - 20x^{-1/3}$
- $2x^{-2} - 7x^{-1} - 15$
- $4x - 8\sqrt{x} + 3 =$ _____

Equations with expressions like these can be solved just like polynomial equations (usually by _____), except with an extra final step.

Example. Solve each equation by treating it as a polynomial equation.

$$x^{2/3} - 3x^{1/3} - 4 = 0$$

$$x^{5/3} - x^{2/3} - 20x^{-1/3} = 0$$

$$2x^{-2} - 7x^{-1} - 15 = 0$$

$$4x - 8x^{1/2} + 3 = 0$$

Check:

Solve Any Equation

Old moves that are especially important are in blue. New moves in red, below.

Solve Any Equation

Basic moves

- Simplify the expressions on either side.
- Add or subtract the same expression on both sides of the equation.
- Multiply or divide by the same expression on both sides of the equation.
  - We must be careful not to multiply or divide by zero.
  - If we accidentally multiply by zero, we introduce extraneous solutions.
  - If we multiplied by a variable expression, we must check.

How does the variable appear?

- Once, inside a function $\rightarrow$ Isolate, then eliminate the function
  - For a one-to-one function f and c in the range of f, $f(X) = c$ and $X = f^{-1}(c)$ are equivalent.
  - If the function is not one-to-one, there may be many solutions: we cut up the domain of f to have a partial inverse.
  - If the function's range is not all real numbers, there may be no solutions solutions.
  - If the *inverse* is not one-to-one, we may introduce extraneous solutions, and must check.
- Many times, inside a function
  - If we can, pretend it's a polynomial.
  - If all else fails, isolate then eliminate one instance of the function.
- Many times, as a polynomial $\rightarrow$ Write in standard form, factor, use the zero property.

3.10 Simplifying Radical Expressions

Roots Canceling Powers

So far we have simplified roots of numbers.

Example. Simplify $\sqrt{80}$.

$\sqrt{80} =$

We can do this because we can actually take the square root of _____ . When we work with variables, there is a complication.

Example. Evaluate $\sqrt{x^2}$ for each of the values of x shown.

$x = 3$	$\sqrt{3^2} =$
$x = -3$	$\sqrt{(-3)^2} =$
$x = 5$	$\sqrt{5^2} =$
$x = -5$	$\sqrt{(-5)^2} =$

We see that when we take $\sqrt{x^2}$ we get _____ .

Example. Evaluate $\sqrt[3]{x^3}$ for each of the values of x shown.

$x = 3$	$\sqrt[3]{3^3} =$
$x = -3$	$\sqrt[3]{(-3)^3} =$
$x = 5$	$\sqrt[3]{5^3} =$
$x = -5$	$\sqrt[3]{(-5)^3} =$

We see that when we take $\sqrt[3]{x^3}$, we just get _____ .

Root of a Power Property

If p an even positive integer, then for any real number x,

$$\sqrt[p]{x^p} = |x|$$

If p is an odd positive integer, then for any real number x,

$$\sqrt[p]{x^p} = x$$

Example. Simplify each of the following.

$\sqrt{x^2} =$ _____ $\sqrt[3]{x^3} =$ _____ $\sqrt[4]{x^4} =$ _____ $\sqrt[5]{x^5} =$ _____

Simplifying Radicals with One Variable

If we have a root of a power of one variable, we can simplify using the Root of a Power Property.

If the power of the variable is a multiple of the index of the root, we can proceed as follows.

Example. Simplify $\sqrt[3]{x^{18}}$.

Note that $18 =$ So $x^{18} =$

$\sqrt[3]{x^{18}} =$

We (do | do not) need an absolute value because _____ is _____ .

Example. Simplify $\sqrt[4]{x^{20}}$.

Note that $20 =$ So $x^{20} =$

$\sqrt[4]{x^{20}} =$

We (do | do not) need an absolute value because _____ is _____ .

If the power of the root is *not* a multiple of the index, we must use division with remainder.

Example. Simplify $\sqrt[5]{x^{17}}$.

$17 \div 5 =$ $17 =$ $x^{17} =$

$\sqrt[5]{x^{17}} =$

We (do | do not) need an absolute value because _____ is _____ .

Example. Simplify $\sqrt[6]{x^{17}}$.

$17 \div 6 =$ $17 =$ $x^{17} =$

$\sqrt[6]{x^{17}} =$

We (do | do not) need an absolute value because _____ is _____ .

Note. If the power of the varaible is _____ than the index, we can't simplify further.

Simplifying Radicals with Multiple Variables

If we have a radical expression with multiple variables (and possibly a coefficient), we can just treat each factor separately.

Example. Simplify $\sqrt[3]{40x^7y^6z^{11}}$

$$\sqrt[3]{40} =$$

$$\sqrt[3]{x^7} =$$

$$\sqrt[3]{y^6} =$$

$$\sqrt[3]{z^{11}} =$$

$$\sqrt[3]{40x^7y^6z^{11}} =$$

By factoring them into primes, we can handle coefficients the same way that we do variables. (This can be especially helpful for large coefficients.)

Example. Simplify $\sqrt{972x^5y^3z^8w}$

$$\sqrt{972} =$$

$$\sqrt{x^5} =$$

$$\sqrt{y^3} =$$

$$\sqrt{z^8} =$$

$$\sqrt{w} =$$

$$\sqrt{972x^5y^3z^8w} =$$

Usually, we won't write out each variable separately. Of course, you can write out more detail if it helps you keep track.

Example. Simplify $\sqrt[4]{18x^7y^2z^{16}w^9}$

$$\sqrt[4]{18x^7y^2z^{16}w^9} =$$

Minding the Absolute Value

Often, we can ignore the absolute value when simplifying even roots of even powers.

- We may take an even power of the resulting variable, which is positive anyway.
- The remaining root may only be defined when the variable is positive.

However, it is important to *include* the absolute value *first*, then decide whether it can be ignored.

Example. Simplify each expression. As a last step, decide whether the absolute value must be included.

$\sqrt{x^8} =$

$\sqrt[4]{x^7} =$

$\sqrt[4]{x^6} =$

$\sqrt{x^{10}} =$

Note. When taking an odd root, we *cannot* include an absolute value, since if p is odd, $\sqrt[p]{x^p} \neq |x|$.

Example. Simplify $\sqrt[3]{27x^4}$. Confirm that when $x = -8$, $\sqrt[3]{27x^4} = 3x\sqrt[3]{x}$, and $\sqrt[3]{27x^4} \neq 3\,|x|\,\sqrt[3]{x}$.

$\sqrt[3]{27(-8)^4} =$

$3(-8)\sqrt[3]{-8} =$

$3\,|-8|\,\sqrt[3]{-8} =$

Note. Often, when working with radical expressions, we are in a setting where only positive real numbers are interesting. In this case, we can ignore the absolute values.

Example. Simplify $\sqrt{3x^4y^7}$. *Assume all variables represent positive real numbers.*

$\sqrt{3x^4y^7} =$

Rationalizing Denominators with Variables

Previously, we have rationalized denominators which contained roots of numbers.

Example. Simplify

$$\sqrt{\frac{25}{12}}$$

as much as possible, including rationalizing the denominator.

$$\sqrt{\frac{25}{12}} =$$

When working with variable expressions, it is easier to find a denominator whose root we can take.

Example. Simplify

$$\frac{3x^3}{\sqrt[4]{y^9}}$$

by rationalizing the denominator. Assume all variables represent positive real numbers.

$$\frac{3x^3}{\sqrt[4]{y^9}} =$$

If we factor coefficients into primes, we can handle them the same way that we do variables.

Example. Simplify

$$\sqrt[3]{\frac{8x}{25y}}$$

as much as possible, including rationalizing the denominator. Assume all variables represent positive real numbers.

$$\sqrt[3]{\frac{8x}{25y}} =$$

Note. We can also "rationalize the numerator" using a similar procedure.

Rationalizing Denominators with Multiple Terms

If there is more than one term in the denominator, it can be more complicated to rationalize the denominator.

Example. Consider the fraction $\dfrac{7}{3-\sqrt{5}}$. If we multiply the numerator and denominator by $\sqrt{5}$, does this rationalize the denominator?

$$\frac{7}{3-\sqrt{5}} =$$

This doesn't work because _____

To rationalize such denominators, it is valuable to recall the difference of squares special form.

Difference of Squares
$$(x-y)(x+y) = x^2 - y^2$$

This gives us a tool we can use to rationalize denominators with two terms, one or both of which is a square root.

Definition: Conjugates
For positive real numbers x and y, the expressions $$\sqrt{x}-\sqrt{y} \quad \text{and} \quad \sqrt{x}+\sqrt{y}$$ are called **conjugates**. This is useful because $$\left(\sqrt{x}-\sqrt{y}\right)\left(\sqrt{x}+\sqrt{y}\right) = \left(\sqrt{x}\right)^2 - \left(\sqrt{y}\right)^2 = x - y$$ In words, multiplying by the conjugate allows us to eliminate the square roots.

Note. This definition makes sense even when one term doesn't contain a radical, since any positive real number is the square root of its square.

Example. Identify the conjugate of $3-\sqrt{5}$. Then, simplify $\dfrac{7}{3-\sqrt{5}}$ by rationalizing the denominator.

The conjugate of $3-\sqrt{5}$ is

$$\frac{7}{3-\sqrt{5}} =$$

Note. We can make similar moves for higher-order roots using other identities, like the sum or difference of cubes for cube roots.

Multiple Radical Terms

We noted before that when an expression has one term with a radical and one term without, we can't combine those terms. When an expression has multiple radical terms, we can simplify by combining "like radicals." It is useful to first fully simplify each radical term.

Example. Simplify $\sqrt{90} - 3\sqrt{20} + 2\sqrt{\frac{18}{5}} + 4\sqrt{80}$.

$\sqrt{90} =$

$\sqrt{20} =$

$\sqrt{\dfrac{18}{5}} =$

$\sqrt{80} =$

$\sqrt{90} - 3\sqrt{20} + 2\sqrt{\frac{18}{5}} + 4\sqrt{80} =$

We can check that this worked by taking a decimal approximation of the original expression and the simplified expression.

$\sqrt{90} - 3\sqrt{20} + 2\sqrt{\frac{18}{5}} + 4\sqrt{80} \approx$

When terms include both variables and radicals, both must be exactly the same to combine the terms.

Example. Simplify $xy^2\sqrt{20x^5y} + 3x^2y\sqrt{45x^3y^3}$. Assume all variables represent positive real numbers.

$\sqrt{20x^5y} =$

$\sqrt{45x^3y^3} =$

Rewriting with Rational Exponents

We have properties of roots that let us combine and separate roots with the same index:

Properties of radicals
For any values of x and y for which all roots below are real numbers, $$\sqrt[n]{xy} = \sqrt[n]{x}\sqrt[n]{y} \qquad \sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$$

For roots with different indexes, we *could* work out a system similar to the one for exponents, but instead, we can just rewrite using rational exponents and use the exponent rules directly.

Example. Write $\sqrt[3]{x^2y}\sqrt[4]{xy^3}$ as a single radical. Assume all variables represent positive real numbers.

Rewrite each expression with rational exponents.

Use properties of exponents to combine.

Write all exponents with the same denominator.

Rewrite in radical notation.

Simplify the radical.

Roots in Lowest Terms

Sometimes, we want to rewrite a radical by putting a rational exponent in lowest terms. Usually, we will do this because we want to compare two results.

Example. Rewrite $\sqrt[6]{x^4}$ with the smallest possible index. Assume all variables represent positive real numbers.

$\sqrt[6]{x^4} =$

By factoring them into primes, we can also do this with numbers.

Example. Rewrite $\sqrt[4]{25}$ with the smallest possible index.

$\sqrt[4]{25} =$

Roots Within Roots

If we are taking a root of a root, we will use the properties of rational exponents to rewrite.

Example. Rewrite each expression as a single simplified radical. Assume all variables represent positive real numbers.

$\sqrt[3]{\sqrt{x}} =$

$\sqrt{x\sqrt[5]{x^2}} =$

$\sqrt[3]{2x\sqrt{3xy}} =$

Unit 4

Exponentials and Logarithms

4.1 Exponential Functions

We now introduce a new family of functions: the *exponential* function. In an exponential function, we put a variable *in the exponent*. These functions have a structure that is very different from that of the power functions that we saw earlier.

Introduction to Exponential Functions

Definition: Exponential Function
An **exponential function** is a function of the form $$f(x) = b^x$$ where $b > 0$ and $b \neq 1$. The number b is called the **base**.

Why do we require $b > 0$ and $b \neq 1$?

- If $b =$ _____, b^x is undefined when x is _____ or _____ .
- If b is _____ , b^x is undefined when x is a _____ with an even _____ .
- If $b =$ _____, $b^x =$ _____ for _____ of x.
- So b^x is only defined (and interesting!) for _____ when _____ .

Like other functions, we can compose an exponential with _____ functions. The following are exponential functions:

$$f(x) = 2 \cdot 3^{2x-3} + 1 \qquad g(x) = \left(\tfrac{2}{3}\right)^{x+5} - 7 \qquad h(x) = -4^{x-4} + 2$$

Basic Exponential Graphs

What does the graph of an exponential function look like?

Example. Sketch the graph of $f(x) = 2^x$

x	y
-3	
-2	
-1	
0	
1	
2	
3	

We see several things in this shape.

- The domain is _____ .
- The range is _____ .
- We can also describe the end behavior:
  - When $x \to +\infty$, $f(x) \to$ _____.
  - When $x \to -\infty$, $f(x) \to$ _____.
- The dashed red line shows that the graph gets close to _____ but never touches it. This line is called an *asymptote*. This is a _____ asymptote at _____ .

We will see the same basic shape as long as _____ .

Exponentials with Base Less than One

Previously, we saw what the graph of an exponential looks like if _____ . What if _____ ?

Example. Sketch the graph of $f(x) = \left(\frac{1}{2}\right)^x$

x	y
-3	
-2	
-1	
0	
1	
2	
3	

What do we see in this shape?

- The domain is _____ .
- The range is _____ .
- We can also describe the end behavior:
  - When $x \to +\infty$, $f(x) \to$ _____ .
  - When $x \to -\infty$, $f(x) \to$ _____ .
- This graph also has a _____ asymptote at _____ .

The key difference between $b > 1$ and $b < 1$ is the _____ .

Graphing General Exponential Functions

We can also compose an exponential function with _____ functions. The most general form is

$$f(x) = a(b)^{cx-h} + k$$

To graph this, we will be explicit about a strategy we have been using for a while: looking at the input and output of the "base function."

Example. Graph $f(x) = -2(3)^{\frac{1}{2}x-3} + 5$

x	input	output	y

Here are the key features we will use to graph.

- The horizontal asymptote at _____ .
- If _____ is negative, the graph is _____ the asymptote.
- To find two key points, we will use the points where the _____ is _____ , _____ , and _____ ; these will usually give us at least two nice points and one fraction.

Properties of Exponential Functions

We can describe the properties of general exponential functions

$$f(x) = a(b)^{cx-h} + k$$

Domain The domain of an exponential function is always _____ .

Range It's always the case that b^{cx-h} _____. So we can find the range of any exponential function.

$a > 0$	$a < 0$
Range:	Range:

Example. Find the domain and range of $f(x) = 4\left(\frac{1}{2}\right)^{2x-3} + 4$.

Domain:	Range:

Inverse An exponential function is always _____ , which means it has an _____ . However, we don't yet have a name for this function.

Basic Exponential Equations

Because we don't know the _____ of $f(x) = b^x$, solving exponential equations can be challenging at this point. However, we can solve them in a special case.

Example. Solve $5^x = 125$

We can generalize this method to let us solve any exponential equation, as long as we can rewrite each side in the same _____ .

Example. Solve $\left(\frac{1}{5}\right)^{x-5} = 25^{3x-1}$

However, since the _____ of an exponential is not all real numbers, we will sometimes have _____ .

Example. Solve $4^x = -4$.

This equation has no solution, because 4^x _____ for every x.

4.2 The Logarithm Function

Previously, we noted that exponential functions must have an inverse, but it wasn't one of the functions we already know. In this section, we introduce a new function, defined to be the inverse of the exponential.

Introducing the Logarithm

Definition: Logarithm

We define the **logarithm base b of x** (written as $\log_b(x)$) as the function such that if

$$f(x) = b^x$$

then

$$f^{-1}(x) = \log_b(x)$$

The number b is called the **base**, and the number x to which the log is applied is called the **argument**.

Logarithm functions are defined for b _____, b _____, just like _____ functions.

This means that statement involving an exponent can be rewritten in in the form of a statement about a logarithm.

Example. Rewrite the statement $5^3 = 125$ in log form.

Since the original exponential statement is _____ , the log form statement is also _____ .

We can also rewrite a statement about logarithms in exponential form.

Example. Rewrite the statement $\log_3(81) = 4$ in exponential form, and determine whether it's true.

Since the exponential statement is _____ , the log form statement is also _____ .

Like other functions, we can compose a logarithm with something _____ and still have a logarithm function. For example, the following are logarithm functions:

$$f(x) = \log_4(x-2) \qquad g(x) = -2\log_{1/2}(x) + 4 \qquad h(x) = 3\log_2\left(\frac{1}{3}x - 5\right) - 2$$

Note. If the argument is a complicated expression, we write it in parentheses. If the argument contains a single term, the parentheses are optional.

Inverses with Logarithms

We can use the logarithm to take the inverse of any log function.

Example. If $f(x) = 2^{x-3} + 5$, write an expression for $f^{-1}(x)$.

Isolate the exponential

Rewrite in log form

Solve for y

Rewrite in terms of f^{-1}

The same method, in reverse, can be used to find the inverse of any logarithm function.

Example. If $f(x) = \log_7(3x - 5) + 2$, write an expression for $f^{-1}(x)$.

Isolate the log

Rewrite in exponential form

Solve for y

Rewrite in terms of f^{-1}

Basic Properties of Logarithms

Even though the logarithm is only defined as an inverse, we figure out some basic properties.

Domain The domain of $f(x) = \log_b(x)$ is the same as the _____ of $f^{-1}(x) = b^x$, _____ .

Range The range of $f(x) = \log_b(x)$ is the same as the _____ of $f^{-1}(x) = b^x$, _____ .

More generally, the domain is wherever the _____ of the log is _____ , and the range is always _____ .

Example. Find the domain and range of $f(x) = \log_3(2x - 5) + 4$

We can also give some basic values of logarithms. All of these can be found by rewriting in exponential form, but they may be worth memorizing. These are true for any number b where the log is defined: _____ .

Exponential Fact	Logarithm Fact
$b^0 = 1$	$\log_b(1) =$
$b^1 = b$	$\log_b(b) =$
$b^{-1} = \frac{1}{b}$	$\log_b\left(\frac{1}{b}\right) =$
$b^n = b^n$	$\log_b b^n =$

Example. Find each log.

- $\log_4(1) =$ _____
- $\log_4(4) =$ _____
- $\log_4\left(\frac{1}{4}\right) =$ _____
- $\log_4\left(4^{3/2}\right) =$ _____

Basic Log Graphs

When we want to graph a logarithm, because we are graphing an _____ , we start by choosing values for _____ .

Example. Sketch the graph of $f(x) = \log_2(x)$

x	y
	-3
	-2
	-1
	0
	1
	2
	3

What do we see in this graph?

- This looks like the graph of _____ , but _____ across a diagonal line.
- The domain (as we knew) is _____
- The range (as we knew) is _____
- We can describe the end behavior, as well (though notice that one "end" is at _____):
  - When $x \to +\infty$, $f(x) \to$ _____.
  - When $x \to$ _____, $f(x) \to$ _____.
- We see a _____ asymptote, as _____ can get close to _____ but can never _____ it. In this graph, the asymptote is at _____ .

We will see this same general shape as long as _____ .

Logs with Base Less than 1

How does the graph of a logarithm differ when the base is less than 1?

Example. Sketch the graph of $f(x) = \log_{1/2}(x)$

x	y
	-3
	-2
	-1
	0
	1
	2
	3

Like with the _____ graphs, the only change is to the end behavior.

- The domain is still _____ .
- The range is still _____ .
- There is still a _____ asymptote at _____ .
- But now the end behavior is:
  - When $x \to +\infty$, $f(x) \to$ _____.
  - When $x \to$ _____, $f(x) \to$ _____.

Graphing General Logarithms

Graphing a general logarithm is not *difficult*, but it combines two new strategies.

- We want to look at the _____ and _____ of the *base function* instead of the given function.
- Because we are graphing an _____ we want to choose values of the _____ .

Example. Sketch the graph of $2\log_{1/3}(4-x)-3$.

x	input	output	y

We can see the key features of this graph, which will help us graph more easily in the future.

- The vertical asymptote is where _____ equals _____ .
- If the _____ in the _____ is negative (as in this example) the graph is to the _____ of the asymptote.
- To find nice key points to draw our graph, we can start with the _____ -1, 0, and 1, which correspond to _____ of _____ , _____ , and _____ . This will usually give us two nice points and one fraction.

Basic Logarithm Equations

While we will need more powerful tools to solve equations with logarithms in general, may basic equations can be solved by rewriting in exponential form.

Example. Solve $\log_4(x) = -2$

Some trial and error (or taking a root) may be required.

Example. Solve $\log_x(81) = 4$

Remember, since it's the base of a logarithm, x must be _____ .

It can also be useful to use this technique when _____ a logarithm.

Example. Solve $\log_4(32) = x$

The Common Log

Definition: Common Logarithm
When we write the symbol log without a base (that is, "log" rather than "$\log_b$"), it refers to the **Common Logarithm**, which in our context is the logarithm base 10.

Example. Rewrite $\log 100 = 2$ in exponential form, and determine whether it is true.

Since the exponential statement is _____ , the log form statement is also _____ .

One nice thing about the common log is that the calculator has a button for it.

Example. Use your calculator to compute each logarithm. Round to three decimal places if necessary.

- $\log 1000 =$ _____ (Check: _____)
- $\log 0.01 =$ _____ (Check: _____)
- $\log 17 =$ _____ (Check: _____)
- $\log 1945 =$ _____(Check: _____)

Since our number system is _____ , the common log can be approximated by counting the _____ of the number. If a number has n digits to the left of the decimal point, its common log will be between _____ and _____ . If its first nonzero digit is n places to the right of the decimal point, its common log will be between _____ and _____ .

Example. Estimate each common log by counting the digits. Then find the log on your calculator, rounded to three decimal places.

- $\log 19271$
  - Counting: _____ $< \log 19271 <$ _____
  - Calculator: $\log 19271 \approx$ _____
- $\log 0.0083$
  - Counting: _____ $< \log 0.0083 <$ _____
  - Calculator: $\log 0.0083 \approx$ _____

4.3 Properties of Logarithms

The logarithm has many properties that allow us to manipulate it in convenient ways. In fact, historically, these rules came *first*; it only turned out later that the function with these properties is the inverse of an exponential. However, we're very familiar with exponents by the time we get to algebra, so introducing the logarithm as an inverse comes more naturally.

These properties seem like a came at this point, but we make significant use of them in solving equations. They will be much more important in calculus, where they can be used to turn incredibly tedious problems in to incredibly easy ones.

Colloquially, we call these properties "Log Rules."

Exponent Properties for Logarithms

The familiar rules for exponents can be rewritten as a rule for logarithms. In particular, we will be interested in the rules involving powers of the same _____ :

Theorem: Rules of Exponents

For any real numbers b, m, n, and p where $b > 0$:

- $b^m \cdot b^n = b^{m+n}$ (product of powers rule)
- $\dfrac{b^m}{b^n} = b^{m-n}$ (quotient of powers rule)
- $(b^m)^p = b^{mp}$ (power of a power rule)

Proof. These rules are easy to show for _____ exponents. Then we defined all other exponents (negative integers, fractions, real numbers*) to make the rules continue to hold. □

To get the same rules for logarithms, we will define:

$$x = b^m \qquad y = b^n$$

This means that $\log_b(x) =$ _____ and $\log_b(y) =$ _____.

* You might be thinking we didn't really define irrational exponents... and you're right. We use the fact that we can approximate any irrational number by a terminating decimal—which is rational—as closely as we'd like.

Log of a Product

Theorem: Log of a Product Rule

For any positive real numbers x, y, and b, $b \neq 1$:

$$\log_b(xy) = \log_b x + \log_b y$$

Proof. x, y, and b are positive real numbers, so $x = b^m$ and $y = b^n$ for some real numbers m and n.

- $\log_b(x) =$ _____
- $\log_b(y) =$ _____
- $xy =$ (_____)(_____) $=$ _____
- So $\log_b(xy) =$ _____ $=$ _____ $+$ _____

This proves the theorem. □

How do we use this?

Example. Rewrite $\log_4(5x^2)$ as separate logs.

Example. Rewrite $\log_3(x) + \log_3(y + 1)$ as a single log.

We refer to this rule as allowing us to "split up" or "combine" the logs.

Log of a Quotient

Theorem: Log of a Quotient Rule

For any positive real numbers x, y, and b, $b \neq 1$:

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

Proof. x, y, and b are positive real numbers, so $x = b^m$ and $y = b^n$ for some real numbers m and n.

- $\log_b(x) =$ _____
- $\log_b(y) =$ _____
- $\dfrac{x}{y} = \dfrac{(_____)}{(_____)} =$ _____
- So $\log_b\left(\dfrac{x}{y}\right) =$ _____ $=$ _____ $-$ _____

This proves the theorem. □

How do we use this?

Example. Rewrite $\log_5\left(\dfrac{28}{q}\right)$ as separate logs.

Example. Rewrite $\log_7(x+1) - \log_7(y^2)$ as a single log.

This rule also lets us "split up" or "combine" logs.

Log of a Power

Theorem: Log of a Product Rule

For any positive real numbers x and b, $b \neq 1$, and any real number p:

$$\log_b(x^p) = p \log_b x$$

Proof. x and b are positive real numbers, so $x = b^m$ for some real number m.

- $\log_b(x) =$ _____
- $x^p =$ _____ $=$ _____
- So $\log_b(x^p) =$ _____ $=$ _____

This proves the theorem. □

How do we use this?

Example. Rewrite $\log_3(x^5)$ as a simpler log.

Example. Rewrite $\log_3\left(\sqrt[5]{z}\right)$ as a simpler log.

Example. Rewrite $3\log(x+1)$ with everything inside the log.

We refer to "pulling the power out of" or "pushing the power into" the log.

Log Rules to Separate

One use for the log rules is to rewrite the log of a complex expression as a combination of logs of much simpler expressions.

Note. In the log rules, we assume that every expression whose log we take is positive, since the log is only defined for positive numbers. Sometimes, we will include an absolute value to make this less of a concern. In our work with log rules in this class, we will assume the values of all variables are such that there is no concern about domain.

Note. We have no log rules to split up _____ . If we have _____ (or _____) inside a log, we have to leave it alone.

Example. Rewrite $\log_7\left(\dfrac{2x^4\sqrt{x-4}}{y^5z}\right)$ using the simplest possible logs.

$=$

$=$

$=$

$=$

We notice the following, which gives us a shortcut.

- Factors in the _____ give us positive terms
- Factors in the _____ give negative terms

Let's see another example, using that shortcut.

Example. Rewrite $\log_4\left(\dfrac{3(x-1)^5\sqrt[3]{y}}{z^4\sqrt{y+1}}\right)$ using the simplest possible logs.

$=$

$=$

Log Rules to Combine

We will make extensive use of log rules to combine when solving equations.

Example. Use log rules to rewrite this expression as a single logarithm.

$$2\log_5 x - 3\log_5(y+7) - \tfrac{1}{3}\log_5 z + 4\log_5(x+1)$$

We follow the same pattern as before:

- Negative terms go in the _____
- Positive terms go in the _____

$=$

$=$

We can only do this if all the logarithms have the same _____ .

Example. Use log rules to combine these logarithms as much as possible.

$$3\log_3 x - 5\log_2 y + 4\log_3 z + 3\log_2 x$$

$=$

$=$

$=$

Using the Log of a Power to Solve Equations

If we know that both sides of an equation are _____ , we can take the log of both sides. This works especially well if one or both sides contains an _____ expression.

Example. Solve $3^x = 5$.

- We cannot easily write 3 and 5 as _____ of the same number.
- Therefore, we cannot solve this by _____ .
- We'll take the _____ of both sides, so we can use a calculator.

We could also solve this by writing the equation in log form:

_____

This suggests the last property of logarithms.

Change of Base for Logarithms

Theorem: Change of Base for Logarithms

For any positive real numbers b, c, and x, with $b, c \neq 1$, we have:

$$\log_b(x) = \frac{\log_c(x)}{\log_c(b)}$$

Proof. Write $y = \log_b(x)$.

- Then $x =$ _____
- Take _____ of both sides: _____ $=$ _____
- Using the _____ rule, _____ $=$ _____
- So _____ $=$ _____
- Divide both sides by _____ to get $y = \dfrac{(_____)}{(_____)}$
- Since both are equal to _____ , $\log_b(x) = \frac{\log_c(x)}{\log_c(b)}$, as desired.

$\square$

At the moment, our primary use for this fact will be to calculate logs on the calculator. *This is not its main purpose—you will need this property in calculus.*

Example. Use the change of base rule to find $\log_7 212$ on your calculator. Round your answer to three decimal places. Check your work by rewriting in exponential form.

4.4 Exponential Equations and Inequalities

We have already solved simple equations involving exponential expressions. We extend these techniques to work with more complicated equations and with inequalities involving exponential expressions.

Solving Equations with One Exponential

We now have a second method of solving exponential equations: by rewriting in log form. (We can think of this as taking the log of both sides!)

Example. Solve $4^{x+1} = 8$

Method 1:

Method 2:

We can still use the new method if we *can't* rewrite each side using the same base.

Example. Solve $3^{x-5} = 29$

There are two standard ways to give this answer.

- Give the exact solution in terms of the common log: _____
- Give a decimal approximation: _____

We may also need to _____ the exponential first.

Example. Solve $\frac{1}{2} \cdot 3^{2x-1} + 6 = 24$

Solving Equations with Two Exponentials

We have already seen a method for solving an equation where two exponential expressions have the same base. If the bases are different, we can use the logarithm in any base. (We usually choose the _____ log.)

Example. Solve $3^{x+5} = 7^{2x-1}$

Note that exponential expressions are always _____ , so we can easily have an equation with no solutions:

Example. Solve $3^{x+4} + 11^{x-3} = 0$

- _____ > 0 for every value of x
- _____ > 0 for every value of x, too.
- So _____ $+$ _____ > 0 for every value of x.

Since the expression on the the left is always _____ than 0, it can never _____ 0, so the equation has no solution.

We also can't deal with addition (in general).

Example. Solve $4^x = 3^x + 2$

We're stuck, because we can't split up a log of a _____ . For an equation like this, we will usually need to resort to _____ techniques.

Usually, an exponential equation with more than _____ terms cannot be solved.

Pretend Polynomials with Exponential Terms

There are a few cases in which we can solve an equation containing exponential expressions and more than two terms by pretending we have a polynomial.

Example. Solve $4^x + 4 = 5 \cdot 2^x$.

- We notice that $4 =$ _____
- So $4^x =$ _____ $=$ _____ $=$ _____

Sometimes, we have to work a little harder for this.

Example. Solve $3^{2x+3} - 3^{x+1} - 4 = 0$

Moves for Inequalities

We have the following moves for solving equations:

1. Basic moves (add, subtract, multiply, divide)
2. Treat as a polynomial and factor
3. Rewrite using an inverse

We've seen how the first two apply to inequalities, with some warnings.

1. Basic moves—multiplying by a _____ number reverses the inequality.
2. Factoring polynomials—detailed sign analysis needed
3. Rewrite using an inverse—?

When we solved _____ inequalities, we had to think about the _____, because the _____ function can be *increasing* or *decreasing*.

Definition
A function f is **increasing** if $f(a) > f(b)$ if and only if $a > b$. A function is **decreasing** if $f(a) < f(b)$ whenever $a > b$.

Example. $f(x) = 3x$ is an increasing function, because if $a > b$, then $3a > 3b$ and vice versa.

Example. $g(x) = -2x$ is a decreasing function, because if $a > b$, then $-2a < -2b$ and vice versa.

This gives us the third move for solving inequalities.

- We can apply any function to both sides of an inequality.
- If the function is _____, the inequality sign _____
- If the function is _____, the inequality sign _____
- If the function does both, we must _____ as we did the absolute value.

The $\log_b$ is an increasing function if b_____, and decreasing if b_____.

Solving an Exponential Inequality

We might solve an exponential inequality without taking a log explicitly...

Example. Solve $3^x < \frac{1}{9}$

...or we might need to actually write the log.

Example. Solve $1500(0.87)^x > 750$

4.5 Exponential Models

Although the math involved in working with exponential expressions may be unfamiliar, exponential models—that is, exponential functions used to describe things in the real word—are very natural.

Basic Exponential Models

In a *linear* model, a constant change in one variable leads to a constant change in another. Linear functions show a _____ rate of change.

Example. You have \$20, and you get \$5 every day. We can write the amount of money you have, m, as a function of the number of days that have passed, t:

$$m(t) = 20 + 5t$$

In an *exponential* model, a constant change in one variable leads to a constant multiplication in another. In exponential models, the rate of change is _____ when the value of the function is _____ , and _____ when the value of the function is _____ .

Example. You have \$5, and your money doubles every day. We can write the amount of money you have, m, as a function of the number of days that have passed, t:

$$m(t) = 5 \cdot 2^t$$

Exponential Model (Basic Form)
An exponential model is a function of the form $$f(x) = f_0 \cdot m^x$$ where f_0 is the starting value (the value of $f(x)$ when $x = 0$) and m is the multiplier (what $f(x)$ is multiplied by each time x increases by 1).

Example. A new web site is growing quickly in popularity: every week, the number of visitors triples. If the site has 10 visitors during is first week, write a function $v(t)$ showing the number of visitors t weeks after the web site launches.

- The word "triple" means multiply by _____ , so _____
- The starting value is _____ ("the first week" means _____ full weeks have passed) so _____

$$v(t) = _____$$

Exponential Models Using Percents

Often, we describe multiplication in terms of *percents* or *percent change*.

Example. Each year, approximately 65% of all aluminum used is recycled. If 50 kg (50,000 grams) of aluminum enter circulation this year, write an exponential function describing the amount A of this aluminum (in grams) that remains in circulation after t years.

- To find 65% of some amount, we multiply by _____ , so $m =$ _____
- We start with _____ grams of aluminum, so $A_0 =$ _____

$$A(t) = \underline{\hspace{6cm}}$$

Example. The population of a small town decreases by 4% each year. This year, the population of the town is 13,850. Write an exponential function describing the population p of the town t years from now.

- If the population decreases by 4%, then _____ remains. So each year we multiply by _____ . So $m =$ _____.
- The starting population is _____ , so $p_0 =$ _____.

$$p(t) = \underline{\hspace{6cm}}$$

Example. The value of a certain stock increases by 12% each year. This year, the stock is worth \$350 per share. Write an exponential function describing the value v of the stock t years from now.

- If the value increases by 12%, it ends up with _____ of its original value. So each year we multiply by _____ . So $m =$ _____.
- The starting value is _____ , so $v_0 =$ _____

$$v(t) = \underline{\hspace{6cm}}$$

When we have a percent change:

- If we have a percent *increase* we _____ .
- If we have a percent *decrease* we _____ .

Exponential Models and Change in Time

Another natural way to describe an exponential model is to choose a nice multiplier, and describe the amount of time it takes. We may also be given a value other than the starting value.

Example. A scientist culturing bacteria in a petri dish observes that the population doubles every 2.3 hours. After 10 hours, there are 10,000 bacteria in the dish. Write a function describing the population p of bacteria in the dish after t hours.

Time (hours)	Population	# of times doubled

$p(t) =$ _____

Exponential Models (General Form)

An exponential model is a function in the form

$$f(x) = f_1 \cdot m^{\frac{x - x_1}{\Delta x}}$$

where

- f_1 is the value of f when $x = x_1$.
- Δx ("delta x") is the change in x which results in the value of f being multiplied by m.

The multiplier may not be explicitly given, but this is easy to find.

Example. A scientist culturing bacteria in a petri dish observes that, after 5 hours, there are 3000 bacteria in the dish, and after 12 hours, there are 30,000 bacteria. Write a function describing the population p of bacteria in the dish after t hours.

Time (hours)	Population

$p(t) =$ _____

Compound Interest

One common place to find an exponential model is *compound interest*, in particular, when we start to charge interest *on the interest*.

Example. Suppose you borrow \$1000 at 10% interest. What happens each year?

Year	Amount owed at start	Interest	Amount owed at end
1			
2			
3			

- Each year, we multiply by _____
- This is an exponential model: $P(t) =$ _____

We could also charge interest more than once a year.

Example. What happens if we borrow \$1000 at 10% interest, but we "compound" the interest every six months?

Year		Amount owed at start	Interest	Amount owed at end
1	1st half			
	2nd half			
2	1st half			
	2nd half			

- Now we multiply by _____ , but we do it _____ times!
- This is another exponential model: $P(t) =$ _____

Definition: Compound Interest

If amount P_0 is invested at an annual interest rate r, compounded n times per year, then the balance after t years is given by

$$P(t) = P_0 \left(1 + \frac{r}{n}\right)^{nt}$$

Half-Life

A common concept in exponential decay (that is, decrease), especially common in science, is the *half-life*

Definition: Half-Life
The **half-life** of a substance is the amount of time required for half of the substance to remain.

Example. A certain radioactive isotope has a half-life of 183 days. A 50 gram sample is obtained. Write an exponential function describing the mass m of the isotope that will remain after t days.

- The starting value is _____ at _____
- The multiplier is _____
- Δt, the time it takes to multiply once, is _____

$$m(t) = _____$$

We can also answer various questions.

Example. A certain radioactive isotope has a half-life of 183 days. A 50 gram sample is obtained. How much will remain after one year (365 days)? Round your answer to two decimal places.

We might also be asked to compute a half-life, after an exponential model is described in some other way.

Example. A drug is metabolized by the body in such a way that the amount of drug in the patient's system dereases by 22% every hour. What is the half-life of the drug? Round to two decimal places.

Doubling Time

When we're concerned with exponential *growth*, that is, increase, we will often consider the doubling time.

Definition: Doubling Time
The doubling time of a quantity is the amount of time it takes for the quantity to double.

Example. The value of an investment doubles every 27 months (2.25 years). If \$50,000 is initially invested, write an exponential function describing the value, v, of the investment after t years.

- The starting value is _____ at _____
- The multiplier is _____
- Δt, the time it takes to multiply once, is _____

$$v(t) = _____$$

We can also answer various questions.

Example. The value of an investment doubles every 27 months (2.25 years). If \$50,000 is initially invested, what is the investment worth after 5 years?

Example. An investment pays 3.1% interest, compounded monthly. What is the doubling time for this investment?

Finding the Exponent

One of the things we most often want to do with an exponential model is to answer a question about the exponent.

Example. The population of a small town increases by 3% every year. If its population now is 12,000, how long will it take its population to reach 18,000? Round your answer to two decimal places.

Sometimes—especially when only integers make sense—we want to understand our question as an inequality.

Example. Alice deposits $1500 in an account which pays 5.5% interest, compounded annually. How long does it take for the value to be at least $2500?

- Since the interest is compounded _____ , interest is only added after a _____ passes.
- The value "is at least" $2500 means _____ .

Finding Percent Change

We might also compute the percent change for an exponential model described in some other terms.

Example. 10,000 copies of a limited edition collectible card are initially sold. After 15 years, only 10 cards in good condition remain in circulation. Wear and tear on collectibles follows an exponential decay model. What percent of the cards are destroyed each year? (Round to one decimal place.)

- In this problem, we must find the multiplier first.
- The multiplier is _____ .
- The time required to multiply by this is number is _____ years.
- There are _____ cards in circulation after _____ years.
- We can write the number of cards in circulation after t years as $n(t) =$ _____
- We want to know the percent lost *each year*. We rewrite the exponential part of the expression as _____ .
- The multiplier each year is _____ .
- What do we need to subtract from 100% to get _____ %? _____
- _____ % of the cards are destroyed each year.

Let's see an example of the work we would typically write out in a problem like this.

Example. After an economic boom, the population of a small town doubles over the course of 5 years. By what percent did the population increase each year? (Round to one decimal place.)

4.6 Eulers Number

One of the must unnecessarily confusing things in working with exponential functions is the number e and the notion of "continuous growth," which often seems to come out of left field. While the full development of these ideas will have to wait for calculus, we can preview them here at an intuitive level.

Interest Compounded Continuously

The more often we compound interest, the more interest we get.

Example. Suppose we deposit \$5000 at 5% interest for 10 years. How much is our account worth at various compounding frequencies?

Frequency	n	Expression	$P(10)$
Annual	$n = 1$	$5000\left(1+\frac{0.05}{1}\right)^{1\cdot 10}$	
Quarterly	$n = 4$	$5000\left(1+\frac{0.05}{4}\right)^{4\cdot 10}$	
Monthly	$n = 12$	$5000\left(1+\frac{0.05}{12}\right)^{12\cdot 10}$	
Weekly	$n = 52$	$5000\left(1+\frac{0.05}{52}\right)^{52\cdot 10}$	
Daily	$n = 360$	$5000\left(1+\frac{0.05}{360}\right)^{360\cdot 10}$	
Hourly	$n = 8640$	$5000\left(1+\frac{0.05}{8640}\right)^{8640\cdot 10}$	

As the interest is compounded more often, we reach a point that we can describe as "compounded continuously."

Interest Compounded Continuously

If an amount P_0 is invested at an interest rate r, *compounded continuously*, the balance after t years is given by

$$P(t) = P_0 e^{rt}$$

where e is *Euler's Number*.

Example. Suppose we deposit \$5000 at 5% interest for 10 years. How much is our account worth if the interest is compounded continuously?

Euler's Number

What is this number e? What's it doing in that formula? How does the interest rate get into the exponent?

- e is an _____ number.
- Like another irrational number, π, e is defined to be the number that works in certain formulas.
- π is defined to be the number that works in the _____ formula, $C = 2\pi r$
- e is defined to be the number that works in the _____ formula.

Shouldn't that depend on r? How do we get r into the exponent?

- Start with the compound interest formula, $P(t) = P_0 \left(1 + \frac{r}{n}\right)^{nt}$
- The P_0 part will stay the same—we'll look at just the $\left(1 + \frac{r}{n}\right)^{nt}$ part.
- Introduce a new variable, $x =$ _____
- Then $\frac{r}{n} =$ _____ and $n =$ _____
- So $\left(1 + \frac{r}{n}\right)^{nt} =$ _____ $=$ _____
- If n is _____ , x is _____ .
- What happens to _____ when x is large?

x	$\left(1 + \frac{1}{x}\right)^x$
10	_____
100	_____
1000	_____
10000	_____
100000	_____

- e is defined to be the number this gets close to.

The Natural Logarithm

Because we often want to consider exponential equations base e, and because of another way we could have defined logarithms using calculus, the logarithm base e has a special name and notation.

Definition: Natural Logarithm
The log base e is called the **natural logarithm** and denoted ln.

Example. Find $\ln(e^3)$.

- $\ln(e^3)$ means _____
- So by definition $\ln(e^3) =$ _____

The calculator has a button for the natural logarithm.

Example. Find $\ln(31)$ on the calculator. Round your answer to two decimal places.

- $\ln(31) \approx$ _____

The calculator also has an option to calculate powers of e. On TI calculators, it is the _____ function of the _____ button.

Example. Find e^3 on the calculator. Round your answer to three decimal places. Check by taking the natural log.

- On the calculator, $e^3 \approx$ _____
- To check, we take $\ln($ _____ $) \approx$ _____ $\approx$ _____

Change of Base for Exponentials

Sometimes, for various reasons, we want to rewrite an exponential expression in a different base, especially base e.

Example. A certain drug has a half-life in the body of 11 hours. A 200mg dose is injected. Write an exponential function $d(t)$ showing the amount of the drug in the patient's system after t hours. Then, rewrite the function in the form $d(t) = d_0 e^{kt}$. Round k to two decimal places.

- The half-life is _____ hours, so our exponent will be _____ .
- So our function is $d(t) =$ _____

Now, to rewrite base e:

- We can write our current exponential function as $d(t) =$ _____
- In our goal function, we will have $d(t) =$ _____
- So we want to find k so that _____

Our function is $d(t) =$ _____

Equivalent Interest Rates

One reason we might do this is to think of interest as compounded continuously.

Example. Suppose we take out a loan with 6.5% interest, compounded quarterly. What would be an equivalent interest rate compounded continuously? Give your answer as a percent, rounded to two decimal places.

The given interest rate is equivalent to _____ % interest compounded continuously.

We can also work in the other direction on compound interest, solving for r.

Example. Suppose we take out a loan with 8.2% interest, compounded continuously. What would be an equivalent interest rate compounded annually? Give your answer as a percent, rounded to two decimal places.

The given interest rate is equivalent to _____ % interest compounded annually.

Percent Rate of Change

This is a piece of subtle wording that is very confusing.

- We said "A value decreases by 4% each year" to mean that each year, the value is multiplied by _____ .
- If we say "A value decreases at a rate of 4% per year," this *should* mean the same thing. However, for complex reasons (which will become clear in calculus), this actually describes a "continuous change," like continuous interest.

Continuous Growth/Decay
Continuous growth or decay can be modeled by a function of the form $$f(t) = f_0 e^{rt}$$ where r is the rate of growth or decay, and f_0 is the value of $f(0)$. Note that this is considered *decay* if the value of f is decreasing; in this case, r is negative.

Example. The cost of goods increases at a rate of 3% per year due to inflation. If an item costs \$800 today, write a function describing its cost $c(t)$ after t years.

- $r =$ _____ $=$ _____
- $c_0 =$ _____
- So $c(t) =$ _____

Example. The amount of a drug in a patient's system decreases at a rate of 12% per hour. If a dose of 50mg is initially administered, write a function describing the amount $d(t)$ of the drug in the patient's system after t hours.

- $r =$ _____ $=$ _____ (_____ because the amount of the drug is _____)
- $d_0 =$ _____
- So $d(t) =$ _____

4.7 Advanced Logarithm Equations

Isolating a Logarithm

We have already solved basic equations including logarithms by rewriting the equation in _____ form.

Example. Solve $\log_5(2x+1) = 2$

Like with other functions, we must _____ the logarithm before applying an _____ .

Example. Solve $5\log_3(x-1) - 6 = 4$.

Isolate the logarithm:

Rewrite as an exponential:

Finish solving:

Equations with Multiple Logarithms

When we have equations with multiple logarithms in the same _____ , we use log rules to combine them.

Example. Solve $\log_3(x+2) = \log_3(x-6) + 2$.

Collect the log terms:

Use log rules to combine:

Rewrite in exponential form:

We now have a _____ equation. We need to recall our techiques for solving those.

Check:

Extraneous Solutions

The domain of the logarithm function contains only _____ numbers. If we multiply or divide two _____ numbers, the result is _____ . So when we _____ logarithms, we risk introducing extraneous solutions.

Example. Solve $\log_4(x+7) = 3 - \log_4(x-5)$. Discard any extraneous solutions.

Collect the log terms:

Combine the logs:

Rewrite in exponential form:

Finish solving:

To identify extraneous solutions, we plug in:

Notice that the extraneous solution was not in the _____ of one of the log functions. We could also note the restrictions on x up front:

We could have gone in knowing that only solutions such that _____ would be true solutions.

Another Example

Let's see an example with the full process of solving a logarithm equation. This one also gives rise to a slightly more challenging equation after we remove the exponents.

Example. Solve $\log_3(2x+17) - 2\log_3(x+6) + 5 = 6 - \log_3(x+8)$

Unit 5

Advanced Polynomial Topics

5.1 Polynomial Division

In this section, we will study the last operation on polynomials: polynomial division. Like division of whole numbers, division of polynomials presents some complications, but is useful in many ways. In particular, we will use division in order to identify factors of polynomials when they cannot be guessed easily using our basic techniques.

We begin with a review of the long division of whole numbers before meeting the basic algorithm for polynomial division. We then discuss the various difficulties that arise in polynomial division that do not come up in long division of numbers. Finally, we see an example which brings together the entire process.

An alternate method, more popular outside the US, is also discussed. Some students find this method easier. You may use whichever method you prefer.

Review of Long Division

We divide polynomials using a process very similar to the one we use to divide numbers. How do we divide $485 \div 21$?

Write: Think:

The process of divison long goes:

D
M
S
B

Each of the numbers in a division problem has a name. In the example above:

- The number 21 (that we're dividing by) is called the _____
- The number 485 (that we're dividing into) is called the _____
- The number 23 (the result of the division) is called the _____
- The number 2 (what was left over) is called the _____

We can check the results of our division by multiplying and then adding:

$$____ \times ____ + ____ = ____$$

Dividing Polynomials

Polynomials are like numbers, with the terms taking the place of digits. So if we want to divide

$$\left(4x^4 - 4x^3 - x^2 + 13x - 12\right) \div \left(2x^2 + x - 3\right)$$

we follow the same steps as for long division.

D

M

S

B

Write: | Think:

$$2x^2 + x - 3 \,\overline{\big)\, 4x^4 \quad - 4x^3 \quad - x^2 \quad + 13x \quad - 12}$$

Like with numbers, we can check this result by multiplying:

Remainders

When we divide numbers, we sometimes get a remainder.

- With numbers, the remainder must be less than the divisor.
- With polynomials, the _____ of the remainder must be less than the _____ of the divisor.

Example. Consider the division problem:

$$\left(2x^3 - 9x^2 + 14x - 5\right) \div \left(x^2 - 3x + 4\right)$$

The divisor has degree _____ , so the remainder has degree at most _____ .

Write: Think:

$$x^2 - 3x + 4 \overline{\big)\, 2x^3 \quad - 9x^2 \quad + 14x \qquad - 5}$$

The remainder is _____ which has degree _____ .
We write this result as:

$$\left(2x^3 - 9x^2 + 14x - 5\right) \div \left(x^2 - 3x + 4\right) = \underline{\qquad\qquad} + \underline{\qquad\qquad}$$

This is just like a mixed number! Like we did with numbers, we can also write our result in terms of multiplication and addition.

$$2x^3 - 9x^2 + 14x - 5 = \left(\underline{\qquad\qquad}\right)\left(\underline{\qquad\qquad\qquad}\right) + \left(\underline{\qquad\qquad}\right)$$

We can think of all these polynomials as functions, and write

$$P(x) = Q(x)D(x) + R(x)$$

- _____ is the dividend, _____
- _____ is the divisor, _____
- _____ is the quotient, _____
- _____ is the remainder, _____

Remember, the degree of the _____ is always _____ than the degree of the _____ .

Structural Issue: Holes

When we divide numbers, there's a digit in every place. Compare:

$$2304 \qquad \text{vs.} \qquad 2x^3 + 3x^2 + 4$$

In order to make the long division algorithm work, we must fill in missing terms with zeros.

Example. To divide

$$\left(8x^3 - 28x + 15\right) \div (2x - 3)$$

we must first fill in the gaps in the dividend:

Write: | Think:

$2x - 3 \,\big)\overline{\qquad\qquad\qquad\qquad\qquad\qquad}$

Example. We might also need to fill in missing terms in the divisor, as in this example.

$$\left(2x^4 - 6x^3 + 11x^2 - 9x + 12\right) \div \left(2x^2 + 3\right)$$

Write: | Think:

$\big)\overline{\;2x^4 \quad - 6x^3 + 11x^2 \quad - 9x \quad + 12\;}$

Structural Issue: Order

In numbers, the digits must come in order. In order to use the long division algorithm, we must write our polynomials (both divisor and dividend) in descending order.

Example. Divide:

$$\left(12x^2 - 21x - 2x^3 + 20\right) \div (4 - x)$$

$12x^2 - 21x - 2x^3 + 16 =$

$4 - x =$

Arithmetic Issue: Fractions

When we divide numbers, we might need to borrow and end up with numbers with too many digits. This cannot happen in dividing polynomials—we must use up an entire term at each round. As a result, some of our coefficients might be fractions.

Example. Divide:

$$\left(6x^3 + 13x^2 - 12x + 15\right) \div (2x + 6)$$

Write:

$$2x + 6 \overline{\big)\, 6x^3 + 13x^2 \quad - 12x \quad + 15}$$

Think:

A Long Example

Now let's see a longer example, which uses all of the skills we have learned (and one extra bit of weirdness).

Example. Divide:

$$\left(18x^2 - 11x^4 + 6x - 15 + 3x^5\right) \div (3x - 9)$$

An Alternate Method: Reverse Multiplication

This alternate method reverses the area method for multiplication.

Example. We begin with an example.

$$(2x^5 - 9x^4 + 19x^3 - 20x^2 + 7x + 3) \div (2x^2 - 5x + 1)$$

1. First, guess the degree of the result. In this example, The degree of the dividend is _____ and the degree of the divisor is _____ . The degree of the quotient will be _____ .
2. The number of columns we need is two more than the degree of the quotient. In this case we need _____ columns.
3. We need a row for each term of the divisor (including zeros). In this example we need _____ rows.
4. Label the columns with descending degrees. The second-last column is the constant. The last column is the remainder.
5. Write the divisor down the left, in descending order.
6. Write the first term of the dividend in the upper left corner.
7. Reverse multiplication to figure out what term goes above it.
8. Multiply to fill in the column.
9. Each diagonal must add up to the corresponding term. Figure out what number needs to go in the next empty box in the top row to get the correct sum.
10. Repeat steps 7 – 9 until you have filled in the top number in the remainder column.
11. Fill in the rest of the remainder column to get the appropriate totals.

5.2 Factors and Synthetic Division

Why would anyone want to divide polynomials?

It isn't just to complete the system we started to explore in Unit Unit 2 when we learned to add, subtract, and multiply polynomials. Ultimately, polynomial division will be an important tool in allowing us to solve problems involving polynomials. Remember that factoring polynomials is an important tool in our arsenal of methods for solving equations!

This section is one of the first times in the course that we do what mathematicians tend to think of as "real math." We will meet a number of *theorems* in this section, and we will *prove* some of them. In the videos which explain proofs, your main job is to understand the logic. After you watch these videos, you should try to put away your notes and re-write the proof on your own. Don't try to memorize it; rather, try to understand the underlying reasoning.

The Remainder Theorem

We begin with a theorem that will help us to get to grips with polynomials.

Theorem: Remainder Theorem
For any polynomial $P(x)$ and number k, there is a polynomial $Q(x)$ and a number R such that $$P(x) = Q(x)(x-k) + R$$ and $R = P(k)$.

Proof. We can find $Q(x)$ and R by dividing $P(x) \div$ _____. $Q(x)$ is the quotient, and R is the remainder.

Remember that the degree of the remainder must be less than the degree of the divisor. _____ has degree _____, so when we perform the division the degree of the remainder must be _____ —that is, the remainder must be a _____!

To see that $R = P(k)$, we just plug in $x = k$:

$$\begin{aligned} P(k) &= Q(k)(k-k) + R \\ &= Q(k) \cdot 0 + R \\ &= 0 + R \\ &= R \end{aligned}$$

So we can find $Q(x)$ and R, as described by the theorem, for any polynomial. $\square$

... so what?

Synthetic Division: Introduction

Because of the remainder theorem (among other things), we often want to divide by polynomials in the form $(x - k)$, that is, just the variable minus a number.

Example. Divide:

$$\left(2x^4 - 13x^3 + 26x^2 - 27x + 7\right) \div (x - 4)$$

$$x - 4 \overline{\big)\, 2x^4 - 13x^3 + 26x^2 - 27x + 7}$$

There are several problems with this process:

- It's *long*
- It takes up a lot of space
- We are writing the same number over and over
- It's easy to lose track of signs in subtraction

What we need is a *shortcut*. This shortcut is called _____ .

Synthetic Division: Method

Example. Use synthetic division to divide:

$$\left(2x^3 + x^2 - 13x - 1\right) \div (x + 3)$$

1. The divisor must be in the form $x - k$. $x + 3 = x -$ _____, so $k =$ _____.
2. In order to avoid subtraction, we _____ of the number that appears in the divisor. We write _____ in the upper left corner and put a bracket around it.
3. Note: the number we write in synthetic division is equal to _____
4. Now, write the coefficients of the _____ , in order, to the right of that number.
5. Make a line underneath that row of numbers, with room for another row of numbers between.
6. Copy the _____ below the line. In our example, we copy the number _____ .
7. Multiply the number you just wrote by the number in the bracket. Write the result above the line in the next column.
8. Add the numbers in that column, and write the result below the line.
9. Repeat steps 7 and 8 until you have filled all the columns.
10. Make a vertical line before the last number.

After we do synthetic division, we have to read the answer.

1. The degree of the quotient is one less than the degree of the _____
2. In our example, the degree of the quotient is _____ $- 1 =$ _____.
3. So we start with _____ and fill in the variables in the terms in descending order. The last number before the vertical line is the constant.
4. The number after the vertical line is the _____ .

We have found that

$$\left(2x^3 + x^2 - 13x - 1\right) \div (x + 3) =$$ _____

Synthetic Division: Complications

There are two things to be aware of with synthetic division.

- This form of synthetic division only works if the _____ is in the form $x - k$.

Which of these division problems are appropriate for synthetic division?

$$\left(4x^3 - 5x^2 + 7x - 3\right) \div (2x - 4)$$

_____

$$\left(3x^5 + 8x^2 - 7x - 2\right) \div \left(x^2 - 3\right)$$

_____

$$\left(4x^3 - 5x^2 - 6x + 1\right) \div (x + 7)$$

_____

- Just like with polynomial division, we must put the terms of the dividend in order and fill in holes.

Example. Divide using synthetic division:

$$\left(9x^3 + 16 - 21x - 2x^4\right) \div (x - 3)$$

$9x^3 + 16 - 21x - 2x^4 =$

$k =$

$\left(9x^3 - 16 - 21x + 2x^4\right) \div (x - 3) =$ _____

Using Synthetic Division to Evaluate

We can use synthetic division and the remainder theorem to evaluate a polynomial function.

- The remainder theorem says that if we divide $P(x)$ by _____ , the _____ is $P(k)$.
- Synthetic division makes it easy to divide by _____

Example. Use synthetic division to evaluate $P(-4)$ where

$$P(x) = 2x^4 + 8x^3 - 7x^2 - 23x + 20$$

We will divide $\Big($ _____ $\Big) \div \Big($ _____ $\Big)$

						2		8		−7		−23		20					

The remainder is _____ , so $P(-4) =$ _____

Polynomial Notation

At this point, we need a nice way to write a polynomial. This will be useful, but we need to meet some new notation.

Definition: Polynomial

A **polynomial** in **variable** x of **degree** n is an expression in the form

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

where each a_n is a number and $a_n \neq 0$.

The numbers a_n are called the **coefficients**. The polynomial is said to have **integer coefficients** if each $a_0, \ldots, a_n$ is an integer, **rational coefficients** if each $a_0, \ldots, a_n$ is a rational number, and **real coefficients** if each $a_0, \ldots, a_n$ is a real number.

This definition doesn't tell us anything we didn't know, but the notation might be hard to read. Have a look at this polynomial:

$$5x^3 - 11x^2 + \frac{2}{3}x - 7$$

- In this polynomial, $n =$ _____, so the degree is _____ .
- The coefficients are:

  $a_3 =$ _____ $a_2 =$ _____ $a_1 =$ _____ $a_0 =$ _____

- These coefficients are all _____ numbers, so this polynomial has _____ coefficients.

We can also write a polynomial in a more compact notation.

Definition: Polynomial

A polynomial expression in the form

$$\sum_{k=0}^{n} a_k x^k$$

where the numbers $a_0, \ldots, a_n$ are numbers.

The symbol $\sum$ is the Greek letter _____ , and it means to add up whatever comes after it by plugging in various numbers for k. This definition means the same thing as the first one.

Roots, Zeros, and Factors

There is a little bit of vocabulary that we're going to need in the rest of this unit.

Definition: Root

For a polynomial

$$\sum_{k=0}^{n} a_k x^k$$

a **root** r of the polynomial is a number such that

$$\sum_{k=0}^{n} a_k r^k = 0$$

Definition: Zero

For a function f, a **zero** z of the function is a number such that

$$f(z) = 0$$

Notice, if we have a polynomial function $P(x)$, its roots and its zeros are the same. The difference is:

- A _____ has roots
- A _____ has zeros

When we're dealing with polynomial functions, we sometimes get lazy with these terms and use them interchangeably.

This last term we already know, but now we are going to be explicit about it.

Definition: Factor of a Polynomial

A **factor** of a polynomial P is any polynomial F for which there is a polynomial Q such that

$$P = Q \cdot F$$

In other words, if a polynomial is written as a product of two other polynomials, those other polynomials are called factors.

The Factor Theorem

This theorem *seems* like something we already knew, but it actually says more.

Theorem: Factor Theorem
For any polynomial, a number r is a root of the polynomial if and only if $(x - r)$ is a factor of the polynomial.

We already knew that we can use the factors to find the roots, using our method of solving a polynomial equation by factoring. This theorem tells us that we can also use the roots to find factors. The phrase "if and only if" means that this theorem goes in both directions.

Proof. Call our polynomial $P(x)$. First, we'll show that if $(x - r)$ is a factor, then r is a root.

- If $(x - r)$ is a factor, that means that there's some other polynomial such that

  _____

- But then $P(r) =$ _____ $=$ _____ $=$ _____.

Now we'll show that if r is a root, $(x - r)$ is a factor.

- If r is a root of the polynomial, $P(r) =$ _____.
- According to the _____ theorem, $P(x) = Q(x)(x - r) + R$, where $R =$ _____
- So $P(x) = Q(x)(x - r) +$ _____ $=$ _____
- _____ $\square$

The point of this theorem is:

- If we know the _____ we can find the _____
- If we know the _____ we can find the _____

We can use this to write down a polynomial given its roots.

Applying the Factor Theorem

Example. Write a polynomial with roots 5, -3, 0 and 4.

- We can only add other factors if we're sure they're never _____
- We can be sure that a _____ is never _____

Example. Write a polynomial with roots -3, 4, and $\sqrt{5}$ and leading coefficient 7.

- We can leave these polynomials in _____ form unless we need another form.

Example. Write a polynomial with roots -4, $\sqrt{3}$, and $-\sqrt{3}$ and leading coefficint 1 in standard form.

The polynomial is

- We had to multiply this out because _____

Multiplicity

What if we have a power of a factor, as in this polynomial?

$$(x-4)(x+2)^2(x-7)^3(x+1)$$

We saw in Unit 2 that the equation

$$(x-4)(x+2)^2(x-7)^3(x+1)=0$$

only has four solutions:

But this is definitely a different polynomial from

$$(x-4)(x+2)(x-7)(x+1)$$

We can think of the factors with exponents—and therefore the roots that go with them—as appearing "more than once."

Definition: Multiplicity
Recall for a polynomial P and a number r, if r is a root of P, then $(x-r)$ is a factor of P. If $(x-r)^m$ is a factor of P (and m is the largest number this applies to), we say the root r has **multiplicity** m.

In our example

$$(x-4)(x+2)^2(x-7)^3(x+1)$$

- The root 4 has multiplicity _____
- The root -2 has multiplicity _____
- The root 7 has multiplicity _____
- The root -1 has multiplicity _____

Example. Write a polynomial with leading coefficient 2 whose roots are 0 with multiplicity 2, -1 with multiplicity 3, 4 with multiplicity 1, and 3 with multiplicity 2.

The polynomial is

A Note on Fractions

What if we have a factional root? Do we have to have fractions in our factors?

The number $\frac{2}{3}$ is a root of the polynomial $6x^3 + 23x^2 - 12$. What factors does that give us?

$\frac{2}{3}$	6	23	0	-12

But the notice that in the other factor, we have _____ .

So _____ is a factor. This will always happen if the coefficients are all integers.

Corollary: Factor Theorem with Integer Coefficients
For a polynomial P with integer coefficients, if a fraction $\frac{p}{q}$ (in lowest terms) is a root of P, then $(qx - p)$ is a factor of P, and the other factor has integer coefficients as well.

(A *corollary* is a theorem that follows easily from another theorem. This theorem is a corollary of the Factor Theorem. We won't prove this, though, because the proof is messy.)

5.3 Rational Roots Theorem

We now know that if we can factor a polynomial, we can find the roots, and if we can guess the roots, we can find the factors. In practice, we combine these techniques: we guess some of the factors, until we get to a polynomial "small" enough to use our basic techniques from Unit 2.

We could guess the roots by randomly plugging in numbers, but we would have to get lucky, and we could never know when we were done looking. In this section, we will learn how to come up with a list of all possible *rational* roots of a polynomial with rational coefficients, and to use this to find all possible factors with rational coefficients.

Multiplying Polynomials Quickly

In general, multiplying three or more polynomials takes a long time. For example, if we want to multiply:

$$(2x-3)(3x^2+4x+5)(x-1)$$

- The *middle* terms in this product are hard to find, because they require combining like terms.
- But the *first* and *last* terms are *easy*.
- The first term is called the _____ .
- The last term is called the _____ .
- To find the leading term, multiply the _____ of the factors.
- To find the constant, multiply the _____ of the factors.

Example. Find the leading term and constant term of this product:

$$(2x-5)(3x+4)(x-3)(x^2+x+5)$$

Leading Term:	Constant:
The leading term is	The constant is

Rational Roots Theorem

Theorem: Rational Roots Theorem

If a fraction $\frac{p}{q}$ (in lowest terms) is a root of a polynomial with *integer* coefficients

$$P = \sum_{k=0}^{n} a_n x^n$$

then p is a factor of a_0 (the constant term), and q is a factor of a_n (the leading coefficient).

Proof. If $\frac{p}{q}$ is a root of P, then the factor theorem tells us that _____ is a factor of P.

- So $P =$ _____ Q for some polynomial Q.
- The _____ of P is _____ times the leading coefficinet of Q
- So _____ is a factor of _____
- Also, the _____ of P is _____ times the constant term of Q
- So _____ is a factor of _____

Looking at the coefficients give us the result we want. □

Example. List all possible rational roots of the polynomial

$$6x^5 - 11x^4 + 8x^3 - 9x^2 + 15x - 15$$

In this polynomial:

- $a_n =$ _____, so the possible values of q are _____ .
- $a_0 =$ _____, so the possible values of p are _____ .

To find all possible rational roots:

	$p =$	$p =$	$p =$	$p =$
$q =$				
$q =$				
$q =$				
$q =$				

The possible rational roots are _____ . There are _____ possibilities.

Factoring with Rational Roots

Example. Use the rational roots theorem to find at least one factor of the polynomial

$$4x^4 - 8x^3 - 7x^2 + 11x + 6$$

First, we find all possible roots.

- $a_n =$ _____, so the possible values of q are _____
- $a_0 =$ _____, so the possible values of p are _____

	$p = \pm 1$	$p = \pm 2$	$p = \pm 3$	$p = \pm 6$
$q = \pm 1$				
$q = \pm 2$				
$q = \pm 4$				

The possible rational roots are _____ . We will try numbers from this list until we find one that is a factor. The easiest way is to use synthetic division.

___	4	-8	-7	11	6

___	4	-8	-7	11	6

We got lucky and found a factor in only two tries.

$$4x^4 - 8x^3 - 7x^2 + 11x + 6 = \Big(_____\Big)\Big(_____\Big)$$

Notice that the degree of each factor is _____ than the degree of the original. The new polynomials are "_____."

Note: We could have ended up with different factors. See what happens if you try $r = -1$ instead of $r = 2$.

Descartes's Rule of Signs

How can we predict how many zeros a polynomial will have? We have a theorem to help us that's a little tricky to understand, but easy to apply.

Theorem: Descartes's Rule of Signs

For a polynomial $P(x)$ with real coefficients written in descending order, the number of positive real roots is either equal to the number of *sign changes* of the coefficients, or less than that by an even number.

To find number of negative real roots, apply the same rule to $P(-x)$.

The proof of this theorem requires techniques we don't have yet, unfortunately, so we won't see it here.

The most important question about this theorem is, what does it mean by a "sign change"? Consider the polynomial

$$P(x) = x^3 + 3x^2 - 10x - 24$$

- The coefficients are: _____ _____ _____ _____
- _____ and _____ are both positive. This is not a sign change.
- _____ is positive, but then _____ is negative. This is a sign change: the _____ goes from positive to negative.
- _____ and _____ are both negative. This is not a sign change.
- The polynomial has _____ sign change, so according to the rule it must have _____ positive real root.

What about the negative real roots? For that, we have to work out $P(-x)$. For that, we need to remember two things we know about exponents:

$$(-x)^n = x^n \text{ if } n \text{ is even} \qquad (-x)^n = -x^n \text{ if } n \text{ is odd}$$

$P(-x) =$

- The coefficients are: _____ _____ _____ _____
- _____ is negative, but then _____ is positive. This is a sign change.
- _____ and _____ are both positive. This is not a sign change.
- _____ is positive, but then _____ is negative. This is a sign change.
- The polynomial has _____ sign changes. The polynomial has _____ or _____ negative real roots.

In fact the roots of this polynomial are -4, -2, and 3.

Upper Bounds

Let's try to factor the polynomial

$$6x^3 + 47x^2 - 9x - 8$$

Its possible rational roots are

	$p = \pm 1$	$p = \pm 2$	$p = \pm 4$	$p = \pm 8$
$q = \pm 1$				
$q = \pm 2$				
$q = \pm 3$				
$q = \pm 6$				

The polynomial has _____ sign change, so we know it has _____ positive real root. We might decide to try all the positive integers first.

1	6	47	−9	−8	2	6	47	−9	−8
		6	53	44			12	118	218
	6	53	44	36		6	59	109	210
4	6	47	−9	−8	8	6	47	−9	−8
		24	284	1100			48	760	6008
	6	71	275	1092		6	95	751	6000

None of them worked! It turns out that we could have known that without trying all of them.

Theorem

If we have a positive number k, and when performing synthetic division by $(x-k)$, every term in the quotient and the remainder have the *same* sign (positive or negative), k is an *upper bound* on the roots of the polynomial, that is, every root of the polynomial is less than or equal to k.

In the example above, _____ is an upper bound on the roots. So we didn't need to check _____ .

We should try some positive fractions (less than 1) next, or think about the negative roots.

Lower Bounds

Let's try to factor the polynomial

$$3x^3 + 4x^2 + 13x + 4$$

- This polynomial has _____ sign changes, so it has _____ positive real roots!
- Plugging in $-x$ for x gives us _____
- This has _____ sign changes, so our original polynomial has _____ or _____ negative real roots.

Its possible rational roots are:

	$p = \pm 1$	$p = \pm 2$	$p = \pm 4$
$q = \pm 1$			
$q = \pm 3$			

We might decide to try all the negative integers first.

-1	3	4	13	4	-2	3	4	13	4
		-3	-1	-12			-6	4	-34
	3	1	12	-8		3	-2	17	-30
-4	3	4	13	4					
		-12	32	-180					
	3	-8	45	-176					

We might have worried that -1 was an upper bound, but it's not, for two reasons:

- Our rule for upper bounds only works for _____ numbers
- The _____ is negative

However, there is another rule that helps us here.

Theorem

If we have a negative number k, and when performing synthetic division by $(x-k)$, the terms in the quotient and the remainder *alternate* signs, k is a *lower bound* on the roots of the polynomial, that is, every root of the polynomial is greater than or equal to k.

In this example, _____ was a lower bound, so we didn't need to try _____ . The numbers _____ are still possibilities.

Factoring Fully with Rational Roots

We now know enough to factor a lot of polynomials that we couldn't in Unit 2. Let's see the whole process, for an example that has only rational roots.

Example. Factor this polynomial as a product of linear factors.

$$2x^4 - 17x^3 + 68x - 32$$

This polynomial has _____ terms, so we should try _____

Grouping failed–but now we have more advanced techniques!

First, we want to know how many zeros we're looking for.

- The polynomial has _____ sign changes, so it should have _____ positive roots.
- Substituting $-x$ gives _____
- This has _____ sign changes, so the polynomial should have _____ negative roots.

We'll try to find at least one rational root.

- The factors of _____ are _____
- The factors of _____ are _____
- The possible rational roots are _____

	2	−17	0	68	−32

	2	−17	0	68	−32

We've found one factor:

$$2x^4 - 17x^3 + 68x - 32 = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}$$

Now we need to continue to factor _____ . This has _____ terms, so we should try _____ again–but that still doesn't work.

Let's look at what happens to the possible rational roots.

- _____ is no longer a factor of the constant, so _____ are no longer possible rational roots.
- We have also already eliminated _____ : _____ wasn't a factor of the original polynomial, so it can't be a factor of a factor.
- But _____ could happen again, so we have to check.

We are trying to factor _____ . Possible rational roots are _____ .

	2	−13	−26	16			2	−13	−26	16	
	2	−13	−26	16			2	−13	−26	16	

That gives us two factors!

$$2x^4 - 17x^3 + 68x - 32 = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}$$

The remaining factor has _____ terms, and it's _____ , so we can use the _____ method.

We now have linear factors, and we can't factor any further.

$$2x^4 - 17x^3 + 68x - 32 = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}$$

The roots of this polynomial are $x =$ _____

5.4 Fundamental Theorem of Algebra

Now we know how to find all the rational roots of a polynomial. What about roots that are not negative? How can we know that we're done looking for roots?

In this section, we will learn methods for dealing with irrational and complex roots of polynomials, along with how many roots we can expect a polynomial to have.

Factoring a Prime Polynomial?

Back in Unit 2, we had we called a polynomial like this one *prime*:

$$x^2 - 4x + 1$$

In order to factor this, the _____ method says we need to find two numbers whose product is _____ and whose sum is _____ , and _____ , so we _____ .

On the other hand, we can use the _____ or _____ to find two perfectly good solutions to the equation

$$x^2 - 4x + 1 = _____$$

So according to the factor theorem, we can factor the polynomial as

$$x^2 - 4x + 1 = (_____)(_____)$$

What gives? We need to make a distinction.

Definition: Prime Polynomial
We call a polynomial with rational coefficients **prime** if it cannot be factored as a product of polynomials with rational coefficients.

Definition: Irreducible Polynomial
We call a polynomial with real coefficients **irreducible** if it cannot be factored as a product of polynomials with real coefficients.

So $x^2 - 4x + 1$ is _____ but not _____ .

Identifying Irreducible Quadratics

We already have a tool to determine whether a quadratic is factorable (with rational factors), prime, or irreducible.

Definition

Discriminant The **discriminant** of a quadratic equation $ax^2 + bx + c = 0$ is the number

$$D = b^2 - 4ac$$

whose square root appears in the quadratic formula. The quadratic equation has:

- Two real solutions if $D > 0$
- One real solution if $D = 0$
- No real solutions if $D < 0$

If D is a perfect square, the solutions are rational. Otherwise, they are irrational.

Putting this in terms of factorability:

- If D is a _____, the quadratic has factors with _____ coefficients.
- Otherwise, if D is _____, the quadratic is prime (but has factors with _____ coefficients).
- If D is _____, the quadratic is irreducible.

Example. Identify each quadratic expression as factorable (with rational factors), prime, or irreducible.

a. $x^2 - 6x + 4$

_____

b. $x^2 - 6x + 8$

_____

c. $x^2 - 6x + 13$

_____

Fundamental Theorem of Algebra

Theorem: Fundamental Theorem of Algebra (Real Number Version)

Any polynomial with real coefficients can be factored uniquely as a product of linear binomials and irreducible quadratic factors. Each linear binomial corresponds to a real root of the polynomial. In notation:

For any polynomial

$$\sum_{k=0}^{n} a_k x^k$$

with *real* coefficients, there is a unique list of real numbers $r_1, r_2, \ldots, r_j$ and irreducible quadratics $Q_1, Q_2, \ldots, Q_k$ such that

$$\sum_{k=0}^{n} a_k x^k = a(x - r_1)(x - r_2)\cdots(x - r_j)(Q_1)(Q_2)\cdots(Q_k)$$

Further, by matching the degrees, $j + 2k = n$

We're not going to prove this theorem—it takes a *lot* of tools we don't have yet!—but we do need to understand what it means.

- A polynomial with degree n has at most _____ real roots.
- Each irreducible quadratic factor decreases the number of real roots by _____ .
- These two facts together mean that a polynomial with an _____ degree must have at least _____ real root.
- A polynomial with degree n and *no* irreducible quadratic factors has exactly _____ roots ("up to multiplicity", which means that roots with multiplicity _____ get counted _____ times.)
- Like numbers, polynomials factor *uniquely*. That means if we want to factor a polynomial, it doesn't matter _____ .

Irreducible Quadratic Factors

There are two ways we can discover an irreducible quadratic factor of a polynomial:

1. We may factor out enough linear factors that the remaining factor is quadratic, and discover that it is irreducible.
2. We may _____ or be _____ an irreducible quadratic factor. (In real life, this would typically be based on how we got the polynomial.)

Suppose that we've discovered an irreducible quadratic factor. How can we use it?

Example. The polynomial $x^4 - 4x^3 + 3x^2 + 2x - 6$ has $x^2 - 2x + 2$ as an irreducible quadratic factor. Use this fact to factor it fully.

First, we divide by

We can factor using basic techniques.

$x^4 - 4x^3 + 3x^2 + 2x - 6 =$

Factoring Fully

We now have all of our tools for factoring.

1. Try the basic techniques first—if they work, they're easier.
2. Sometimes we start out knowing an irreducible quadratic factor. (This might come from something about how the polynomial was set up, or some other source.) If so, divide by it.
3. Use Descartes's Rule of Signs to analyze the real roots.
4. Try rational roots until you get down to a quadratic factor, or until you have eliminated all possibilities.
5. If you get down to a quadratic factor, use the *ac* method, completing the square, or the quadratic formula to find the last two roots, if possible.

Example. Find all real roots of the polynomial $x^5 - 14x^3 - 26x^2 + 105x - 26$ and factor it fully, given that $x^2 + 6x + 13$ is a factor of the polynomial.

5.5 Polynomial Inequalities

We now know everything we need to know in order to find, at least the rational solutions to polynomial equations. What about inequalities?

We haven't discussed inequalities in a long time—not since Unit 1! That's because, in general, non-linear inequalities have some complexities that we don't want to get too tangled with. But the techniques we use with polynomial inequalities will be very useful in general, when we work with more complicated functions in the future.

Introduction to Polynomial Inequalities

- To solve a polynomial equation, we start with a factored polynomial equal to _____ .
- We do this because if a _____ is 0, then one of the _____ is also 0.
- There are also facts about how products to compare to zero.
- If a number is _____ 0, it is _____ .
- If a number is _____ 0, it is _____ .
- We can predict the _____ of a product from the _____ of the factors.

Example. Solve the polynomial inequality

$$(x-3)(x+1) \geq 0$$

Note that the equation $(x-3)(x+1) = 0$ has two solutions: $x =$ _____ and $x =$ _____. These divide the number line into parts:

$x < -1$

$x = -1$

$-1 < x < 3$

$x = 3$

$x > 3$

The solution set of the inequality is:

or in interval notation:

Solving Polynomial Inequalities in Factored Form

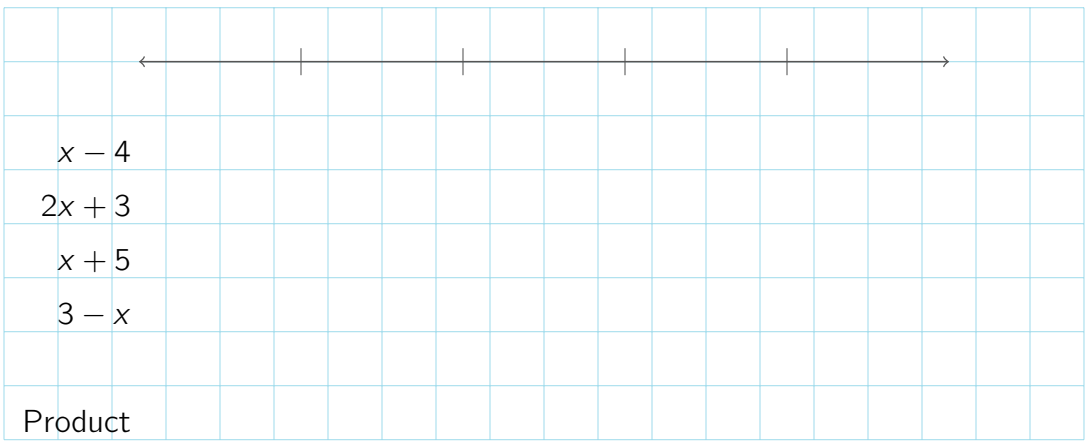
The method of solving a polynomial inequality is based on analyzing the signs.

Example. Solve the polynomial inequality

$$(x-4)(2x+3)(x+5)(3-x) < 0$$

First, we set each factor equal to zero to find the cut-off points.

Mark these points (in order, but spacing doesn't matter) on a number line, and analyze each factor's sign. This is commonly done with a diagram like this one.



We are looking for the set where the polynomial is _____ 0, so where the product is _____ . So the solution set is

$$\left\{ x \;\middle|\; _____ \text{ or } _____ \text{ or } _____ \right\}$$

or in interval notation

$$_____ \cup _____ \cup _____$$

Handling Multiplicity

How do we handle a polynomial inequality in which some factors have multiplicity greater than one?

- If the exponent is _____ , that factor _____ .
- If the exponent is _____ , that factor _____ .

Example. Solve the polynomial inequality

$$x(x-3)^2(x+4)^3 \geq 0$$

x

$(x-3)^2$

$(x+4)^3$

Product

Solution set:

Handling Irreducible Quadratics

What if a polynomial has irreducible quadratic factor?

- An irreducible quadratic factor is never _____ for real numbers x—that's the point.
- So they're either always _____ or always _____ .
- The sign matches the _____ .

Example. Solve the polynomial inequality

$$(x-4)(x^2-4x+7)(2x+1) > 0$$

When $x = 0$, $x^2 - 4x + 7 =$

So $x^2 - 4x + 7$ for all real numbers x.

$x - 4$

$x^2 - 4x + 7$

$2x + 1$

Product

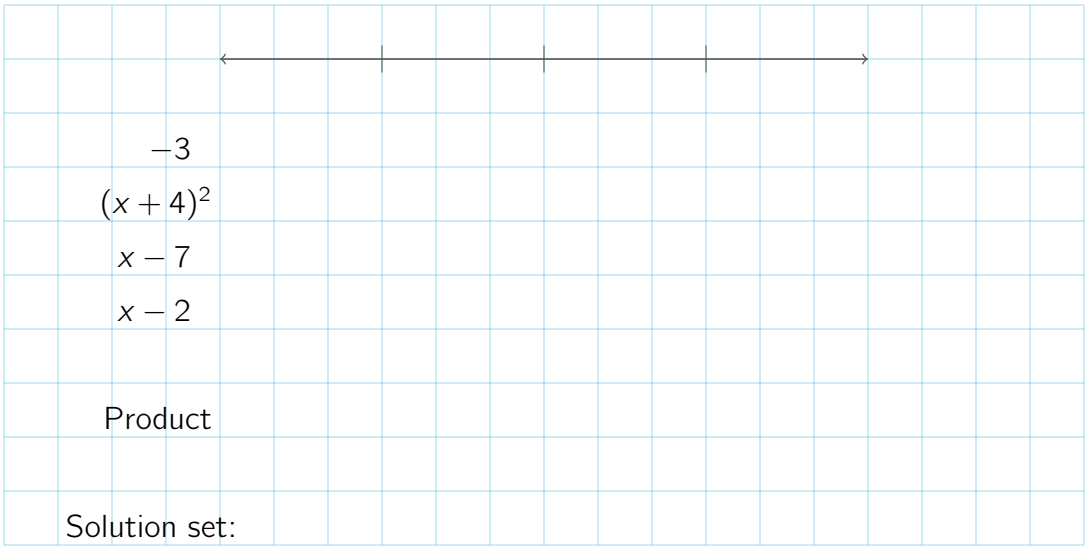
Solution set:

Handling Leading Coefficients

Leading coefficients are easy to deal with, since the _____ of a number doesn't vary.

Example. Solve the polynomial inequality

$$-3(x+4)^2(x-7)(x-2) < 0$$



A Polynomial Inequality in Standard Form

The standard form of a polynomial equation is

$$P(x) = 0$$

Similarly, the standard form of a polynomial inequality is one of

$$P(x) < 0 \quad P(x) > 0 \quad P(x) \leq 0 \quad P(x) \geq 0$$

Given a polynomial inequality in standard form, we must _____ the polynomial first.

Example. Solve

$$-2x^3 - 4x^2 + 18x - 4 \geq 0$$

Polynomial Inequalities in Other Forms

Like equations, if a polynomial inequality comes to us in something other than Factored or Standard form, we must first put it in standard form, then factor.

Example. Solve the inequality

$$x^3 + 7x^2 \leq 36$$

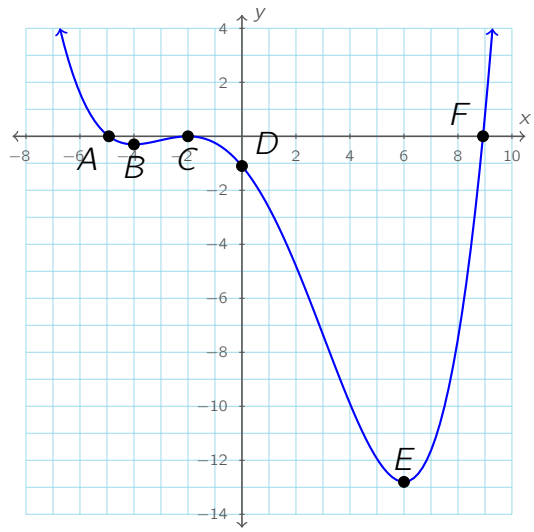
5.6 Graphs of Polynomials

In this section, we will discuss the basic properties of the graphs of polynomials. We will not be able to fully graph polynomials—determining the exact coordinates of a point requires either more advanced techniques, or more involved algebra than we want to be involved with—but we will be able to recognize the shape of a graph of a polynomial, and an expression for a polynomial function from its graph.

The Graph of a Polynomial

Consider the graph below, which is the graph of

$$f(x) = \frac{1}{160}\left(x^4 - 28x^3 - 96x^2 - 88\right)$$



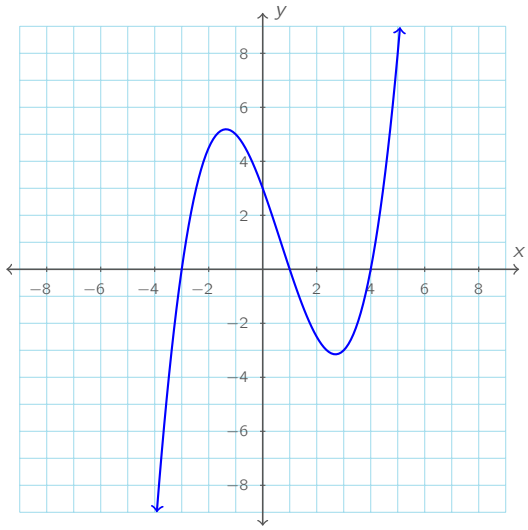
There are several features here that are typical of a polynomial graph.

- The function is _____ —we could trace the entire graph without lifting our pencil.
- Points _____ are the _____ -intercepts. Their x-coordinates are the _____ of the polynomial.
- Points _____ are called "local minima"—their y-coordinate is _____ than that of nearby points. (Singular would be a "minimum.")
- Point _____ is called a "local maximum"—its y-coordinate is _____ than that of nearby points. (Plural would be "maxima".)
- Point _____ is the _____ -intercept.
- We also see the function's "end behavior": what _____ does for very large (positive or negative) values of _____ . For a polynomial, when _____ is very large, _____ is also very large.

Zeros on a Graph

Consider the graph below of

$$f(x) = \frac{1}{4}\left(x^3 - 2x^2 - 11x + 12\right)$$



This graph has x-intercepts at $(___, ___)$, $(___, ___)$, and $(___, ___)$. This is where _____ .

- $f(___) =$ _____
- $f(___) =$ _____
- $f(___) =$ _____

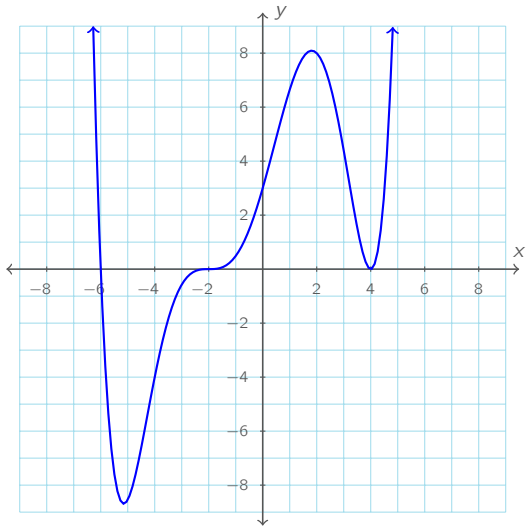
These are the zeros of $\frac{1}{4}\left(x^3 - 2x^2 - 11x + 12\right)$, as we can see if we factor:

... but what if the roots have multiplcity?

Multiplicity on a Graph

Consider the graph below of

$$f(x) = \frac{1}{256}(x-4)^2(x+2)^3(x+6)$$



- The zero _____ has multiplicity _____ . Near (___, ___), the graph looks like the graph of _____ (it's almost a _____)
- The zero _____ has multiplicity _____ . Near (___, ___), the graph looks like the graph of _____
- The zero _____ has multiplicity _____ . Near (___, ___), the graph looks like the graph of _____ (it's almost a _____)

If a root r has multiplicity m, then near that point, the graph looks like _____

End Behavior

We can also discuss the "end behavior" of a polynomial. This refers to what happens to the value of f when x is a very large _____ number or a very large _____ number.

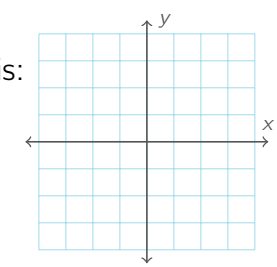
- To say, "x gets very large," we write _____
- To say, "x gets very large negative," we write _____

When x is very large (positive or negative), the _____ of the polynomial is the most important.

Example. Describe the end behavior of the polynomial

$$f(x) = -3x(x-5)(2x+3)(4-x)^2$$

- The leading term is _____
- The _____ is _____
- The _____ is _____
- The graph of $y =$ _____ looks like this:



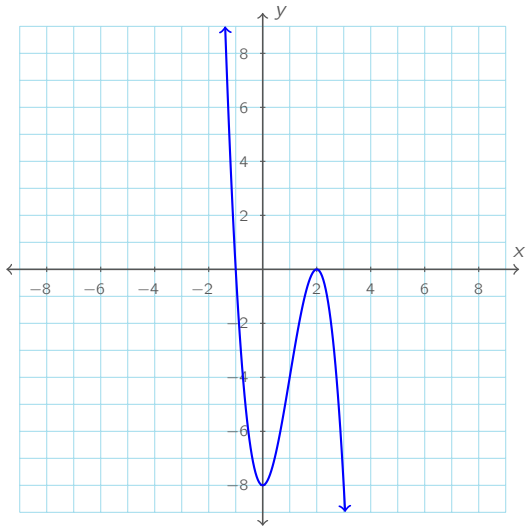
- We describe this end behavior as:
  - $f(x) \to$ _____ when $x \to +\infty$
  - $f(x) \to$ _____ when $x \to -\infty$

This is the easiest way to find the end behavior: find the leading term and sketch its graph.

Writing a Polynomial Function From a Graph

Given a graph of a polynomial, we can now guess the expression for the polynomial.

Example. The graph of a polynomial function $f(x)$ is shown. Give an expression for $f(x)$.



zeros:

multiplicity: (guess)

To find a, choose a point on the graph.

To check, choose another point.

We find $f(x) =$ _____

Unit 6

Additional Topics

6.1 Coordinate Geometry

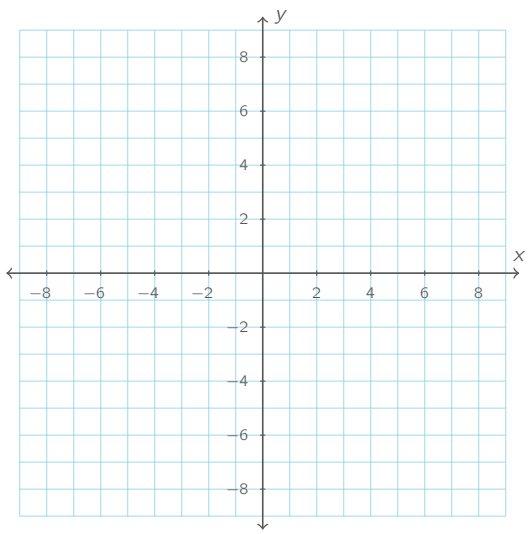
This section serves as our introduction to the first topic in this unit: *coordinate geometry*, that is, geometry done in the coordinate plane. This is an important part of building the relationship between geometry and algebra.

Points in the Plane

In geometry, a *point* is one of the key undefined concepts.

- It has _____ but no _____ .
- We denote points with _____ letters.
- In the plane, we will refer to a point by its _____ .

Example. In the coordinate plane below, plot the points $A = (-2, 5)$, $B = (3, -2)$, and $C = (-6, -4)$

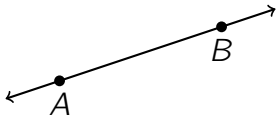
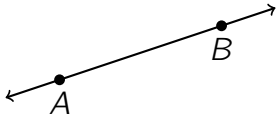
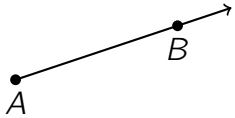
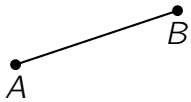
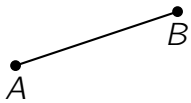


Lines, Segments, and Rays

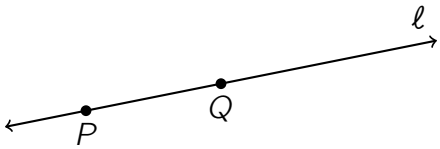
If we have two points, we can draw a line, ray, or segment.

- A line _____ in both directions.
- A ray _____ in one direction, and _____ in the other.
- A segment _____ in both directions.

We name these figures using the two points and a marking over them.

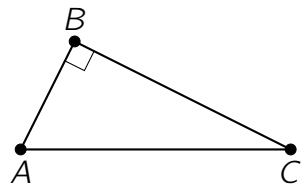
Symbol	Read	Image
$\overleftrightarrow{AB}$	_____	
$\overleftrightarrow{BA}$	_____	
$\overrightarrow{AB}$	_____	
$\overrightarrow{BA}$	_____	
$\overline{AB}$	_____	
$\overline{BA}$	_____	

We can also denote a line with a _____ . In the figure below, _____ is also called _____ (read " _____ ").



Right Triangles

If we have a triangle with corners at A, B, and C, we write $\triangle ABC$.

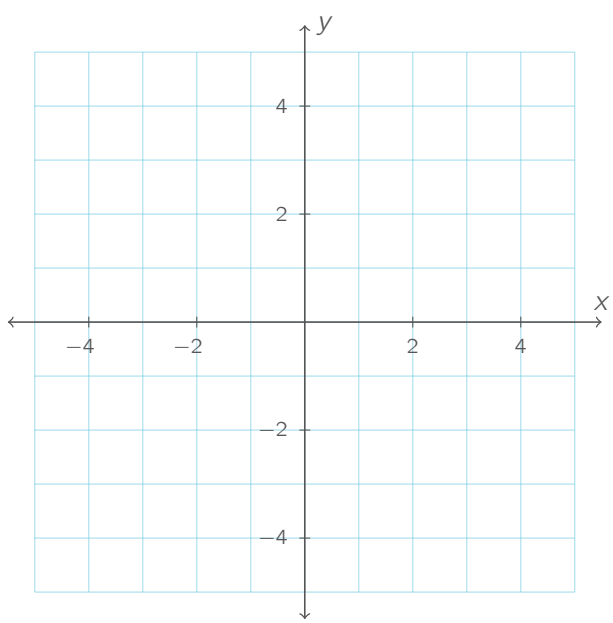


In this diagram:

- We see a small square at $\angle B$ ("_____").
- This marks a _____ angle.
- So $\triangle ABC$ is a _____ triangle.
- $\overline{AC}$ is called the _____ of the triangle.
- $\overline{AB}$ and $\overline{BC}$ are called the _____ of the triangle.

If we have a diagonal line segment in a coordinate plane (one that isn't _____ or _____) there is a natural way to draw a right triangle with that segment as the hypotenuse.

Example. Plot $A = (-2, 3)$ and $B = (3, -1)$. Choose a point C so that $\triangle ABC$ is a right triangle with hypotenuse $\overline{AB}$.



- We could put C at (___, ___)or (___, ___)
- If $A = (x_A, y_A)$ and $B = (x_B, y_B)$, we could put C at (___, ___)or (___, ___).

Distance in the Plane

The *distance* between A and B, written $d(A, B)$, is the length of $\overline{AB}$, written AB. It is easy to find for _____ and _____ segments.

Example. Plot the points $A = (-3, 2)$, $B = (1, 2)$, $C = (2, 2)$, $D = (2, -1)$. Then find AB and CD.

In general:

- If $A = (x, y_A)$ and $B = (x, y_B)$ (same _____ coordinate), $AB =$ _____
- If $A = (x_A, y)$ and $B = (x_B, y)$ (same _____ coordinate), $AB =$ _____

For other segments, we use the _____ theorem.

Theorem: Pythagorean Theorem
If the legs of a right triangle have lengths a and b, and the hypotenuse has length c, then $$a^2 + b^2 = c^2$$

Example. Plot the points $A = (-3, -2)$ and $B = (1, 1)$. Then find AB.

Distance Formula
If $A = (x_A, y_A)$ and $B = (x_B, y_B)$, then $$d(A, B) = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$$

Midpoint of a Segment

The *midpoint* M of a segment AB is the point on $\overline{AB}$ such that $AM = MB$. This is easy to find if $\overline{AB}$ is _____ or _____ .

Example. Let $A = (-3, 2)$, $B = (-3, 0)$, $C = (-1, 1)$, and $D = (3, 1)$. Plot these points, and find the midpoints of $\overline{AB}$ and $\overline{CD}$.

$\overline{AB}$ has midpoint

$\overline{CD}$ has midpoint

In general:

- If $A = (x, y_A)$ and $B = (x, y_B)$ (same _____ coordinate), the midpoint of $\overline{AB}$ is $M = ($_____, _____$)$
- If $A = (x_A, y)$ and $B = (x_B, y)$ (same _____ coordinate), the midpoint of $\overline{AB}$ is $M = ($_____, _____$)$

For other segments, we use the properties of congruent triangles.

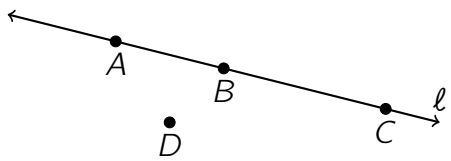
Example. Let $A = (-2, -3)$ and $B = (4, 1)$. Plot these points, and find the midpoint of $\overline{AB}$.

Midpoint Formula
If $A = (x_A, y_A)$ and $B = (x_B, y_B)$, then the midpoint of $\overline{AB}$ is $$M = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}\right)$$

Collinear Points

Three (or more) points are *collinear* if they lie on the same line.

Example. In this figure, _____ , _____ , and _____ are collinear points (they lie on line _____), but _____ is not collinear with any pair of them.



Note. We only talk about sets of _____ or more points being collinear or not, since we can draw a line through any _____ points.

If we have three collinear points, the following property of the Euclidean plane holds.

Segment Addition Postulate
Given three collinear points, we can label them A, B, and C so that B is on $\overline{AC}$. In that case $$AB + BC = AC$$

We can use this, along with the distance formula, to determine whether three points in the plane are collinear.

Example. Let $A = (-2, 5)$, $B = (4, 1)$, and $C = (2, \frac{7}{3})$. Plot these points. Are they collinear?

$AB =$

$BC =$

$AC =$

_____ , the points _____ collinear, because _____ + _____ = _____ .

Triangle Inequality

If three points are *not* _____ , they form a triangle, which gives us the name of the following property of the Euclidean plane.

Triangle Inequality
Given any three points A, B, and C: $$AC \leq AB + BC$$ with equality if and only if the three points are colinear with B on $\overline{AC}$.

We sometimes phrase this as "_____ ."

Note. This is primarily interesting when $\overline{AC}$ is the longest of the three segments.

To practice our understanding of this inequality, we may ask whether a list of three numbers are possible *sides of a triangle*, that is, possible lengths of three segments AB, BC, and AC.

Example. Which of these are possible lists of lengths of the sides of a triangle?

1. $AB = 5$, $BC = 6$, $AC = 10$: _____ , so _____
2. $AB = 1$, $BC = 6$, $AC = 3$: _____ , so _____

   Note. This _____ for any three _____ points—this is _____ .

3. $AB = 5$, $BC = 7$, $AC = 8$: _____ , so _____
4. $AB = 2$, $BC = 4$, $AC = 6$: _____ , so _____

   Note. This is _____ a triangle because the points are _____ . We might call this a "degenerate triangle."

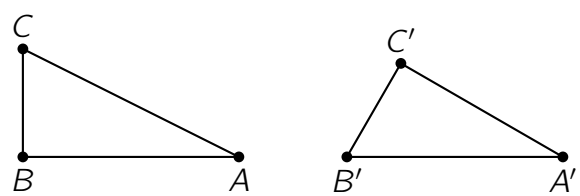
Distance and Angles

We classify triangles based on their largest angle.

Definition: Classification of Triangles

- An **obtuse triangle** is one in which the largest angle is _____, that is, larger than a right angle (measures _____ 90°).
- A **right triangle** is one in which the largest angle is a _____ angle (measures _____ 90°).
- An **acute triangle** is one in which the largest angle is _____, that is, smaller than a right angle (measures _____ 90°).

The largest side of a triangle is always opposite the largest angle. Further, for two triangles $\triangle ABC$ and $\triangle A'B'C'$, if all sides are the same length except $\overline{AC}$ is _____ than $\overline{A'C'}$, then $\angle B$ is _____ than $\angle B'$ (as shown in this figure):



Using this fact, we can determine the type of triangle by looking at the side lengths and comparing with the pythagorean theorem.

Let a, b, and c be the side lengths of a triangle, labeled so that $a \leq b \leq c$ (c is the largest side length). Then:

- If $c^2 < a^2 + b^2$ the triangle is _____
- If $c^2 = a^2 + b^2$ the triangle is _____
- If $c^2 > a^2 + b^2$ the triangle is _____

Example. Let $A = (-2, -3)$, $B = (1, 1)$, and $C = (13, -4)$. Is this triangle right, acute, or obtuse?

$AB =$

$BC =$

$AC =$

6.2 Lines

Equation of a Line

Lines in a plane are described by a _____ equation in two variables. This equation can be given in several forms.

Definition: Slope-Intercept Form

The **slope-intercept form** of the equation of a line is

$$y = mx + b$$

where m is the **slope** of the line and b ys the ***y*-intercept** of the line.

Definition: Standard Form

The **standard form** of the equation of a line is

$$Ax + By = C$$

where A, B, and C are integers with no common factor, and $A > 0$

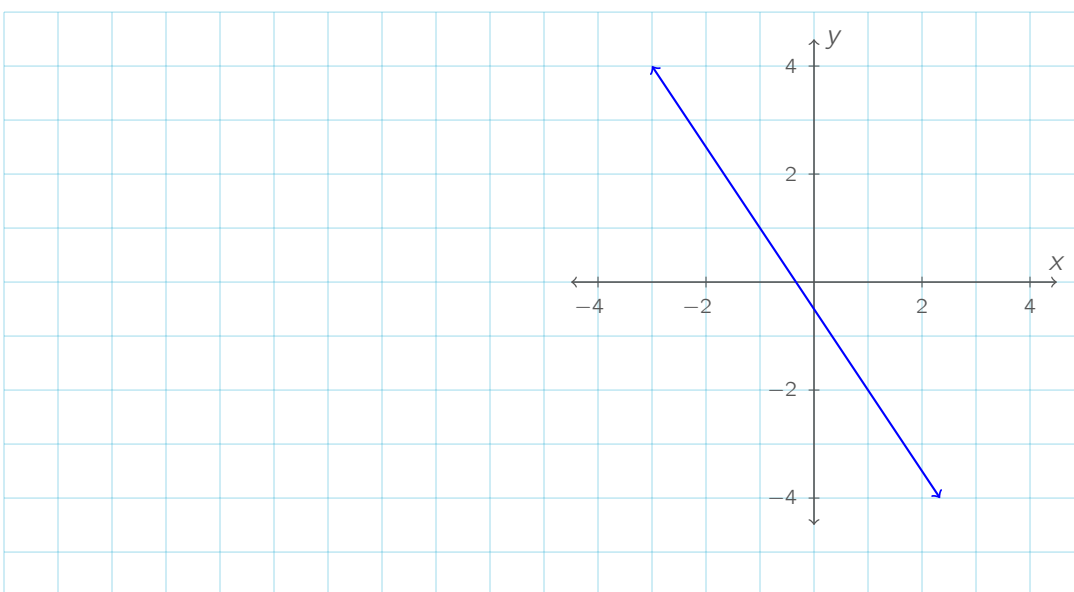
Definition: Point-Slope Form

A **point-slope form** equation of a line is in the form

$$y - y_0 = m(x - x_0)$$

where m is the slope and (x_0, y_0) is any point on the line.

Example. Give an equation of the line graphed below in point-slope form. Then rewrite the equation in slope-intercept form and standard form.



Parallel Lines

Definition: Parallel Lines
Two lines are **parallel** if they do not intersect.

In coordinate terms, parallel lines have the same slope.

Example. Give an equation of the line parallel to the graph of $y = \frac{2}{3}x - 5$ through the point $(-4, 2)$.

- The given line has slope _____
- A line parallel to the given one has slope _____
- So the line we want (in point-slope form) is _____
- Rewrite this as _____

If a line is horizontal or vertical, parallel lines are of the same type.

Example. Give an equation of the line parallel to the graph of $y = 7$ through $(-2, 4)$.

- The given line is _____
- So the desired line is _____
- The line we want is _____

What if a line is given in standard form?

Example. Give an equation of the line parallel to $2x - 3y = 5$ through $(-2, 1)$.

Given a line in standard form, we can find a parallel line by _____ coefficients, then _____ to find the constant.

Perpendicular Lines

Definition: Perpendicular Lines
Two lines are **perpendicular** if they meet at a right angle.

In coordinate terms, the slopes of parallel lines are *opposite reciprocals*.

Example. Give an equation of the line perpendicular to the graph of $y = \frac{3}{4}x + 2$ through the point $(6, -3)$.

- The given line has slope _____
- A line perpendicular to the given one has slope _____
- So the line we want (in point-slope form) is _____
- Rewrite this as _____

If a line is horizontal, perpendicular lines are vertical. If a line is vertical, perpendicular lines are horizontal.

Example. Give an equation of the line perpendicular to the graph of $y = 3$ through $(5, -2)$.

- The given line is _____
- So the desired line is _____
- The line we want is _____

What if a line is given in standard form?

Example. Give an equation of the line perpendicular to $3x - 5y = 10$ through $(-2, 1)$?

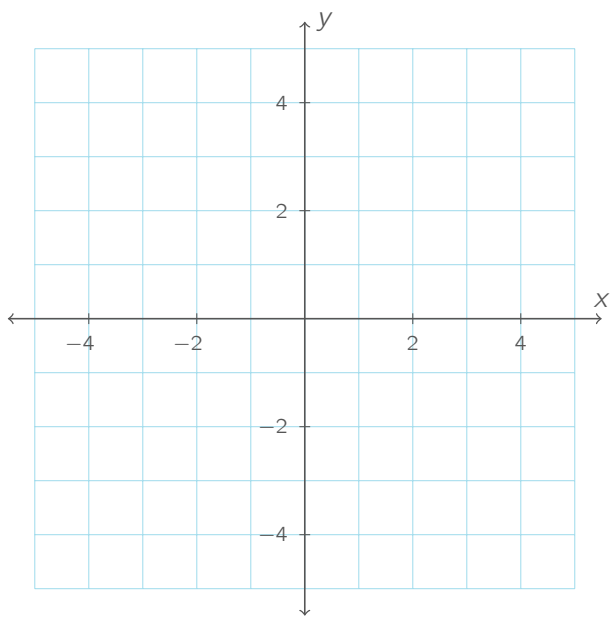
Given a line in standard form, we can find a perpendicular line by _____ the coefficients and _____, then _____ to find the constant.

Point of Intersection of Lines

If we have two different lines that are not parallel, they _____ at exactly one point.

Example. Graph the lines described below, and find their point of intersection.

$$\begin{cases} y = -\frac{2}{3}x + 3 \\ y = x - 2 \end{cases}$$



The lines intersect at (___, ___)

Notice:

- _____ $= -\frac{2}{3}($_____$) + 3$, so (___, ___)is a solution to the first equation.
- _____ = _____ $- 2$, so (___, ___)is a solution to the second equation.

This is called a *system of equations*.

Definition: System of Equations
A **system of equations** is a set of two or more equations that are to be solved simultaneously, that is, equations that should all be true at the same time.

If we _____ a system of equations, the solution to the system is the point where the lines _____ .

Solving a System – Substitution Method

Rather than drawing a graph, we can solve a system algebraically. We have two methods for this. The first is the *substitution method*.

Example. Use substitution to solve the system

$$\begin{cases} y = 3x + 5 \\ 3x + 2y = 1 \end{cases}$$

The solution is (___, ___)

1. Solve one equation for one variable. (We'll usually choose substitution when this is _____ .)
2. Substitute that into the second equation.
3. You now have an equation in _____ variable. Solve it.
4. You now have the _____ of one variable.
5. Plug into one of the equations to find the value of the _____ variable.
6. Check by plugging into the other equation.

Solving a System – Elimination Method

The other common method for solving a linear system is the *elimination* method. This method is harder to get our head around, but easier in most cases.

Example. Use elimination to solve

$$\begin{cases} 3x - 5y = 21 \\ 4x + 3y = -1 \end{cases}$$

The solution is (___, ___)

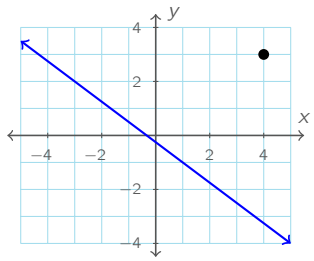
1. Put both equations in standard form. (We'll usually choose elimination when this is _____ , and we'll _____ the restrictions on A, B, and C.)
2. Multiply both sides of each equation by numbers chosen so that one variable will have opposite coefficients. (Swapping coefficients and switching one sign will always work.)
3. Add together the left-hand sides and the right-hand sides.
4. You now have an equation in _____ variable. Solve it.
5. You now have the _____ of one variable.
6. Plug into one of the equations to find the value of the _____ variable.
7. Check by plugging into the other equation.

Distance From a Line

We know how to find the distance between two _____ . How do we find the distance between a point and a line?

- The distance depends on what _____ on the line we choose.
- We choose the _____ possible _____ .
- This is on a _____ line.

Example. How far is the point $(4, 3)$ from the line $y = -\frac{3}{4}x - \frac{1}{4}$?



To find the distance without the picture:

1. Find a perpendicular line through the given point.
2. Find where this intersects the given line.
3. Find the distance between them.

Example. Find the distance between $(-3, 1)$ and $y = -3x + 12$.

Distance from a Horizontal or Vertical Line

Finding a distance from a horizontal or vertical line is significantly easier.

Example. Find the distance from $(-3, 1)$ to the line $x = 7$.

- The line perpendicular to $x = 7$ through $(-3, 1)$ is _____
- This crosses $x = 7$ at $(___, ___)$.
- The distance from $(-3, 1)$ to $(___, ___)$is _____

We can do this in general, without going through the whole process.

- The distance from (x_1, y_1) to the _____ line $x = x_0$ is _____
- The distance from (x_1, y_1) to the _____ line $y = y_0$ is _____

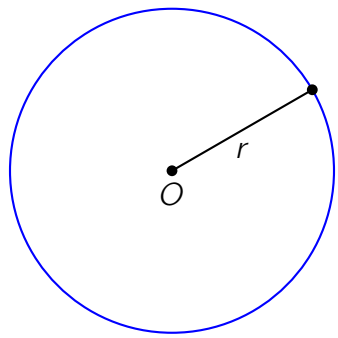
6.3 Circles

What is a Circle?

As a geometric figure, how do we define a circle?

Definition: Circle
Given a point O in the plane and a real number $r > 0$, the **circle of radius r with center O** is the set of all points C with $d(O, C) = r$.

So a circle consists of all points at some given distance from the center.



Note. The number r is called *the* radius of the circle. A segment running from the center to the circle is called *a* radius. Each radius has length r.

We can use this definition to come up with an equation for a circle.

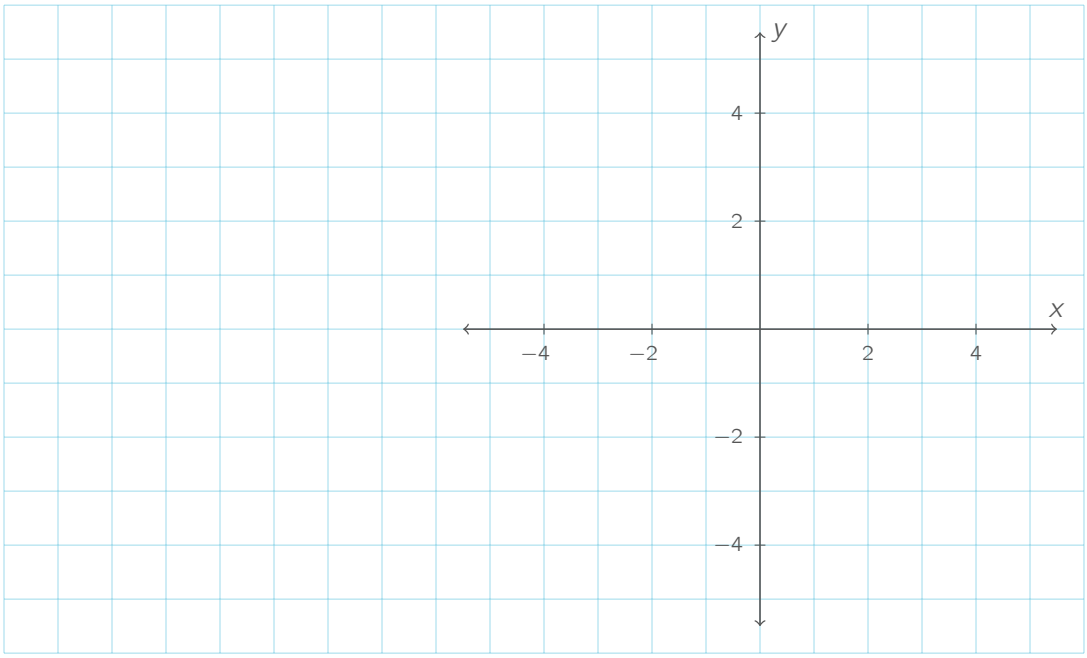
- We call the _____ $O = (h, k)$.
- We're writing an equation for a shape in the plane, so we call the point on the circle $P = (x, y)$
- Then $d(O, P) =$ _____
- Our basic equation is _____ $= r$
- But square roots are hard to work with, so we _____ both sides.
- This gives us _____

Equation of a Circle (center-radius form)
The circle centered at (h, k) with radius r is the graph of the equation $$(x - h)^2 + (y - k)^2 = r^2$$

Working With the Circle Equation

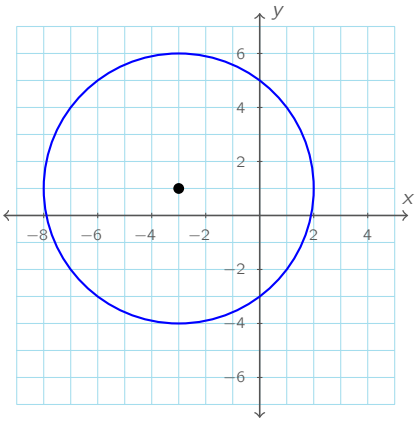
Given the equation of a circle, we can sketch it.

Example. Sketch the graph of $(x-2)^2 + (y+1)^2 = 9$.



We can also give the equation of a circle we see.

Example. Give the equation of the circle graphed below.



- $(h, k) = (___, ___)$
- $r =$ _____
- So the equation of the circle is _____

Equation of a Circle Given Center and Point

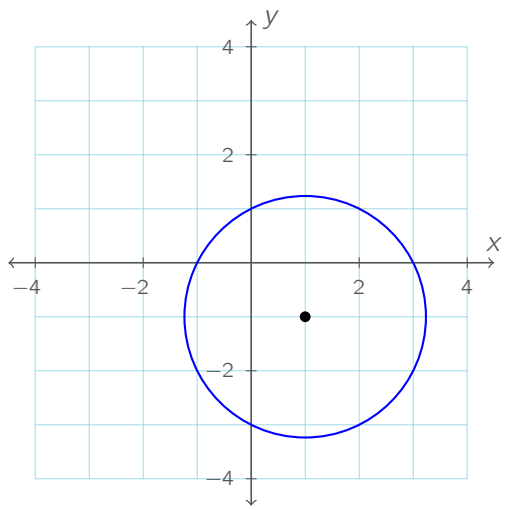
We can give the equation of a circle given the center and a point on the circle.

Example. Give the equation of the circle with center $(-2, 4)$ and passing through $(2, -1)$.

- $(h, k) = (___, ___)$
- If the circle passes through $(2, -1)$, the radius must be the _____ from _____ to $(2, -1)$.
- $r =$ _____ $=$ _____
- Note $r^2 =$ _____, so we don't need to worry about the _____ .
- The equation is _____

We can also use this to write the equation of a graphed circle if the radius isn't easy to read.

Example. Write the equation of the circle graphed below.

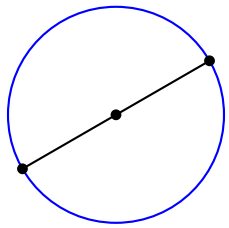


- $(h, k) = (___, ___)$
- $(___, ___)$ is on the circle
- $r =$ _____ $=$ _____
- So the equation of the circle is _____

Equation of a Circle Given a Diameter

Definition: Diameter

A **diameter** of a circle is a line segment passing through the center of the circle with end points on the edge. *The* diameter of a circle is the length of such a segment.



Note. The segment on each side of the center is a _____ , so each must be the same length. So the center is the _____ of any diameter.

Note. The diameter d of a circle is _____ its radius r. In other words, the radius is _____ the diameter.

We can write the equation of a circle given the endpoints of a diameter.

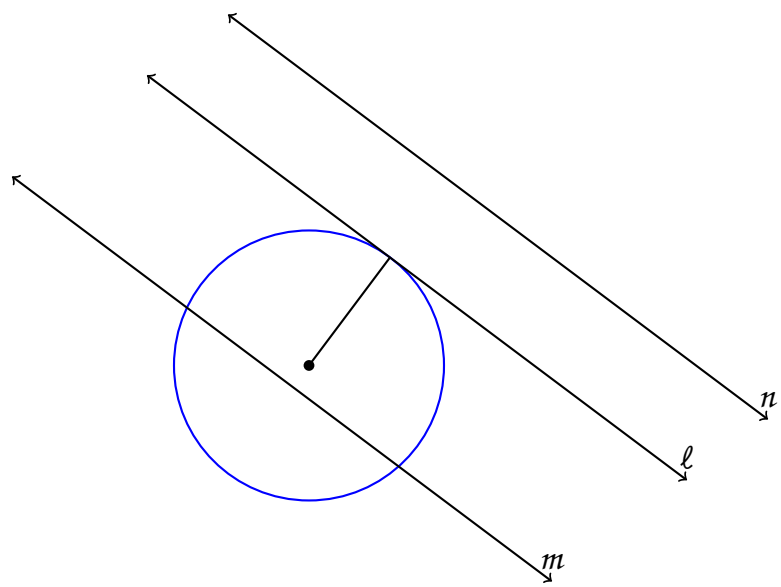
Example. Give the equation of a circle whose diameter has endpoints $A = (-3, 1)$ and $B = (5, 2)$.

- The center is at the midpoint of _____ .
- $h =$ _____ $=$ _____
- $k =$ _____ $=$ _____
- The diameter is the length of _____ .
- $d =$ _____ $=$ _____
- $r = \frac{d}{2}$, so $r^2 =$ _____ $=$ _____
- In this example, $r^2 =$ _____
- The equation is _____ .

Tangent Lines

Definition: Tangent Line
A line is **tangent** to a circle if it touches the circle at exactly one point.

In the diagram below, line ℓ is tangent to the circle, and lines m and n are not.



- Line m isn't tangent to the circle because it _____ .
- Line n isn't tangent to the circle because it _____ .
- Line ℓ is tangent to the circle. Notice that it is _____ to the radius.
- The _____ from the center to the tangent line is equal to the _____ of the circle.

Equation of a Circle Given a Tangent Line

We can find the equation of a circle given a center and a tangent line.

Example. Give an equation of the circle with center $(2, -5)$ and with tangent line $y = \frac{2}{3}x + 11$.

- $(h, k) = (___, ___)$
- The radius is the distance from the center to the line.

Perpendicular line:

Find the point of intersection

Point of intersection is

$r^2 =$

- So the equation of the circle is _____ .

This is easier if the line is _____ or _____ .

Example. Give an equation of the circle with center $(-3, 1)$ and tangent to the line $x = 2$.

- $(h, k) = (___, ___)$
- r is the distance from $(___, ___)$to _____
- So $r =$ _____ $=$ _____
- The equation of the circle is _____ .

The x- and y-axis are special horizontal and vertical lines.

Example. Give an equation of the circle with center $(-1, 4)$ and tangent to the x-axis.

- The x-axis is the line _____
- $(h, k) = (___, ___)$
- r is the distance from $(___, ___)$to _____
- So $r =$ _____ $=$ _____
- The equation of the circle is _____ .

Conic Sections

Definition: Conic Sections

Circles are one of a family of curves called **conic sections**. The equation of any conic section has a *standard form*:

$$Ax^2 + By^2 + Cx + Dy + Exy + F = 0$$

where all coefficients are integers (if possible) and $A \geq 0$.

The family of conic sections includes some curves we've met before, or will meet soon.

- If $A \neq 0$, $B = 0$, and $E = 0$, we have

  $$Ax^2 + Cx + Dy + F = 0$$

  If we solve this for y, we get _____, which is the equation of a _____ .

- If $A = B = E = 0$, we have

  $$Cx + Dy + F = 0$$

  This is equation is _____, so its graph is a _____

- If $A = B = 0$, then we have

  $$Cx + Dy + Exy + F = 0$$

  We can solve this for y to get _____, a _____ equation. We will learn to graph this soon.

In a circle, we must have _____ = _____, _____ $\neq 0$, and _____ $= 0$.

Circles in Standard Form

Equation of a Circle (standard form)
An equation of a circle in *standard* form is of the form $$Ax^2 + Ay^2 + Cx + Dy + F = 0$$ with A, C, D, and F integers, $A > 0$

Example. Write the equation of the circle centered at $(-2, 5)$ with radius 7 in standard form.

Given the equation a circle in standard form, we can write it in center-radius form by _____ .

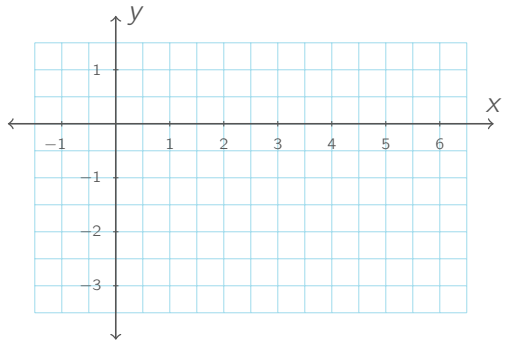
Example. Write the equation of the circle $4x^2 + 4y^2 - 24x + 8y + 15 = 0$. Then, sketch the graph.

Divide by the coefficient of x^2:

Rearrange terms to complete the square:

Complete the square (twice):

The center is (___, ___), and the radius is _____ . So our graph is:



Degenerate and Impossible Circles

Two things might go wrong when we try to rewrite a circle in center-radius form.

Example. Rewrite $x^2 + y^2 - 4x + 2y + 5 = 0$ in center-radius form.

In this equation, we have $r^2 =$ _____. That's not a circle... is it?

This equation is true only if $(x, y) = (___, ___)$. The graph of this equation is just _____. We call this a _____ circle.

Example. Rewrite $x^2 + y^2 - 8x + 10y + 45 = 0$

In this equation, we have $r^2 =$ _____. But r must be a _____.

If r^2 is a _____ number, the equation has _____. We can say there is _____.

6.4 Linear Systems

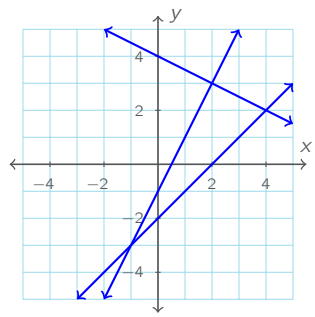
Larger Linear Systems

We have already studied systems of _____ linear equations with _____ variables. We can add more equations and more variables.

- If a system has more equations than variables, the system is **overdetermined**, and will typically have _____ solutions. An overdetermined system has _____ information.

Example. Consider the system

$$\begin{cases} y = x - 2 & (I) \\ y = -\frac{1}{2}x + 1 & (II) \\ y = 2x - 1 & (III) \end{cases}$$

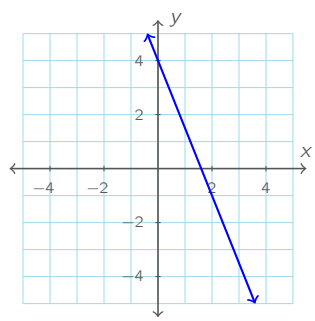


Every _____ of equations has a shared solution, but there is no solution shared by _____ .

- If a system has fewer equations than variables, the system is **underdetermined**, and will typically have _____ solutions. An underdetermined system has _____ information.

Example. Consider the system

$$\{\ 5x + 2y = 8 \qquad (I)$$



Every point on the _____ is a solution to this "system."

- To have a unique solution, we must typically have a **balanced** system, in which the number of equations is _____ the number of variables.

In the rest of this section, we will concern ourselves with systems of _____ linear equations in _____ variables.

Note. The "typically" refers to the fact that well chosen systems of each type may behave differently. If we write down a system at random, it will almost certainly have the typical behavior.

Basic Strategy for Larger Systems

Let's consider our basic strategy for solving a larger system.

- When we had a system of 2 equations with 2 variables we _____ them to get _____ equation with _____ variable.
- We could then _____ .
- In general, when we have 3 equations with 3 variables, we will combine them to give _____ equations with _____ variables.
- We then solve this by _____ .
- Note that if we have 3 equations, there are _____ possible combinations. We will choose _____ of them.
- If we have _____ equations, there are _____ possible combinations! We will only choose _____ of them.

The hardest part of solving a larger system is _____ of what we're doing. In order to solve such a system, we will _____ the equations and _____ what we're doing at each step.

A System of Three Equations by Elimination

In general, we'll use elimination to solve a system of more than two equations, unless the system is set up to make substitution exceptionally easy.

Example. Solve

$$\begin{cases} 2x + 3y - z = -2 & (I) \\ 3x - 5y + 2z = 23 & (II) \\ x + 2y - 3z = -7 & (III) \end{cases}$$

The solution is $(x, y, z) = (___, ___, ___)$

Note. The work shown here is the actual work to solve this system—including the notes in words.

A System of Three Equations by Substitution

Occasionally, we encounter a system that is set up to make substitution especially easy. These arise in some common applications.

Example. Alice, Bob, and Carol are splitting a dinner bill. Bob ordered an expensive meal, so he will pay \$5 more than Alice. Carol is paying for Dave's meal as well, so she will pay twice as much as Alice. The total bill is \$65. How much will each person pay?

- Start by defining some variables:

$$\begin{aligned} A &= \text{the amount Alice pays} \\ B &= \text{the amount Bob pays} \\ C &= \text{the amount Carol pays} \end{aligned}$$

- Our equations are:

$$\begin{cases} \underline{\qquad\qquad\qquad\qquad} & (I) \\ \underline{\qquad\qquad\qquad\qquad} & (II) \\ \underline{\qquad\qquad\qquad\qquad} & (III) \end{cases}$$

- This is set up to make substitution easy, because the first two equations express _____ and _____ in terms of _____ .

Alice pays:

Bob pays:

Carol pays:

Complications in Substitution

This system is also set up to make substitution easy, but we must be careful.

Example. Solve

$$\begin{cases} x = 2z & (I) \\ y = x - 5 & (II) \\ 2x - 3y + z = 9 & (III) \end{cases}$$

- All three equations include _____
- Since it appears in all three equations, we'll replace _____ *last*.

The solution is $(x, y, z) = (___, ___, ___)$

The next example is more confusing than difficult.

Example. Solve

$$\begin{cases} y = 1 - 2x & (I) \\ z = 2y - 8 & (II) \\ x = \frac{1}{3}z - 5 & (III) \end{cases}$$

Note that, while each equation contains _____ variables, the entire system contains _____ variables.

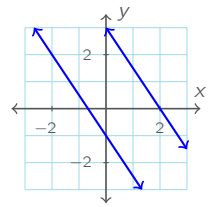
The solution is $(x, y, z) = (___, ___, ___)$

Inconsistent and Dependent Systems

The solution to a system of two equations in two variables represents the point of intersection of two lines. What can go wrong?

Example. Consider the system:

$$\begin{cases} 3x + 2y = 6 \\ y = -\frac{3}{2}x - 1 \end{cases}$$



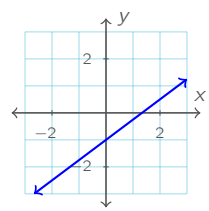
These lines are _____ so they _____ . This system has _____ . If we try to solve it:

Substitution:	Elimination:

These equations contain no _____ and are _____ . A system like this is called _____ . Notice that in standard form, these are obviously _____ lines.

Example. Consider the system:

$$\begin{cases} 3x - 4y = 4 \\ y = \frac{3}{4}x - 1 \end{cases}$$



These two equations' graphs are _____ . This system has _____ . If we try to solve it:

Substitution:	Elimination:

These equations contain no _____ and are _____ . A system like this is called _____ . Notice that in standard form, these are obviously _____ line.

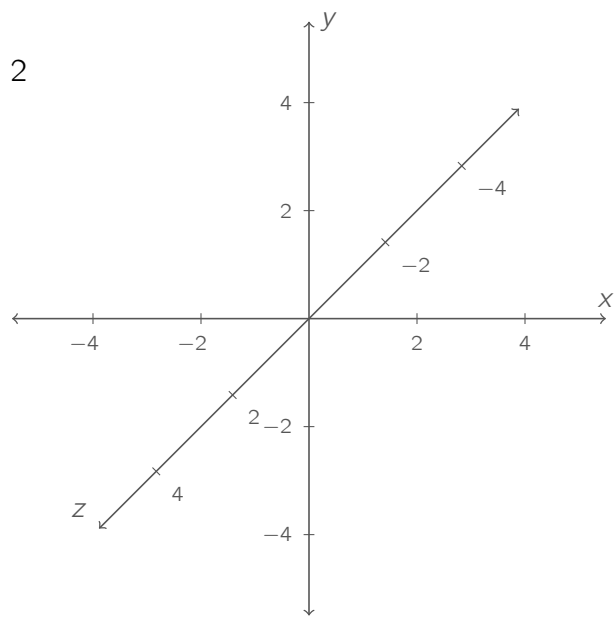
Linear Equations in Three Variables

What does an equation in three variables "look like"?

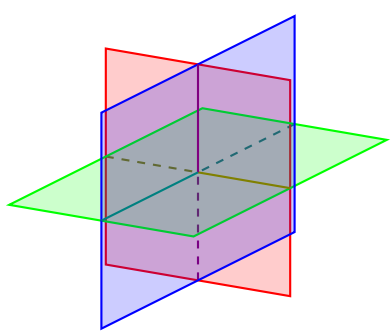
Example. Sketch the graph of $x + y + z = 2$

- Remember, the graph of an equation is a picture of all the _____ to the equation.
- Since there are _____ variables in this equation, its solution is an ordered *triple*, that is, _____ values (x, y, z).
- To graph this, we need _____ axes. We draw the z-axis on a _____ , but imagine that it "comes out of the screen" toward you.
- It is easy to see that (___ , ___ , ___), (___ , ___ , ___), and (___ , ___ , ___) are solutions to this equation.

$x + y + z = 2$



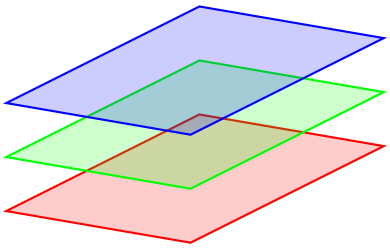
The graph of a linear equation in three variables is a _____ in space. When a system of three equations has a unique solution, we have three _____ intersecting at a _____ :

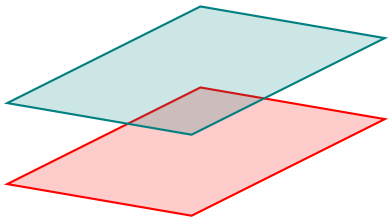


What could go wrong?

Inconsistent Systems in Three Variables

There are three basic ways that a system in three variables can be inconsistent.

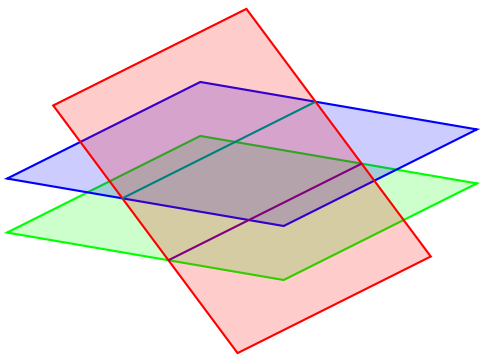




In this figure, all of the planes are _____ . If all of the equations are in _____ form:

$$\begin{cases} 2x + 3y - 4z = 5 \\ 2x + 3y - 4z = 2 \\ 2x + 3y - 4z = -1 \end{cases} \qquad \begin{cases} 2x + 3y - 4z = 2 \\ 2x + 3y - 4z = 2 \\ 2x + 3y - 4z = -1 \end{cases}$$

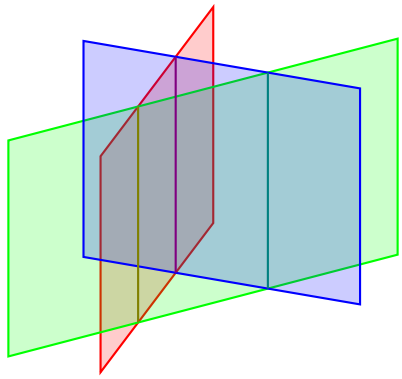
If we combine any _____ of these equations, we will _____ and the resulting equations will be _____ .



Here *two* of the planes are _____ , so they intersect the third plane at _____ lines. An example of planes like this, written in _____ form:

$$\begin{cases} 3x - 5y + 2z = 7 \\ 3x - 5y + 2z = 9 \\ 2x + 8y - 3z = 4 \end{cases}$$

We see _____ equations that, if we combine them, we will _____ and the resulting equations will be _____ .



Here, each _____ of planes intersect at a _____ , but the lines are _____ .

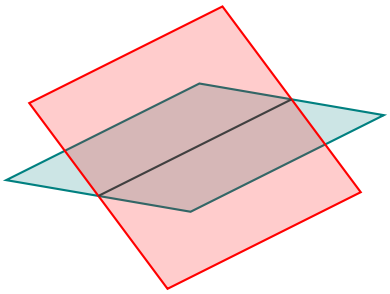
$$\begin{cases} 8x + y - 3z = -3 & (I) \\ 10x + 4y - z = 10 & (II) \\ 2x + 3y + 2z = 2 & (III) \end{cases}$$

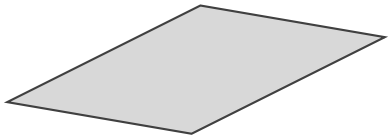
Written in _____ form, this looks fine. What happens when we try to solve?

When we get a system of _____ equations in _____ variables, the new system is _____ .

Dependent Systems in Three Variables

There are two basic ways to have a dependent system of three equations.

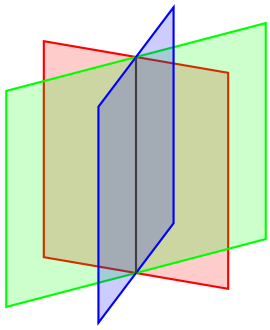




Two or all three planes are _____ .

$$\begin{cases} 3x + 1y - 2z = 6 \\ 3x + 1y - 2z = 6 \\ 5x - 2y + z = -1 \end{cases} \qquad \begin{cases} 7x - 2y + 5z = 8 \\ 7x - 2y + 5z = 8 \\ 7x - 2y + 5z = 8 \end{cases}$$

In standard form, two or all three equations are _____ .



Here, _____ planes meet at _____ .

$$\begin{cases} 4x - y - 3z = -3 & (I) \\ 2x + y - z = 1 & (II) \\ 5x + 4y - 2z = 5 & (III) \end{cases}$$

Written in standard form, this system looks fine. If we try to solve it, we will end up describing the _____ .

Describing the Solutions of a Dependent System

We saw that a dependent system in three variables has a _____ of solutions. Can we say what the solutions are, instead of just "infinitely many solutions"?

A line in three dimensions
We can describe a line in space as an ordered triple containing only one variable, in one of these forms: $$(x, y(x), z(x)) \qquad (x(y), y, z(y)) \qquad (x(z), y(z), z)$$ We can choose any variable we like to describe a line. We pick whichever makes the math easy.

Example. Solve

$$\begin{cases} 4x - y - 3z = -3 & (I) \\ 2x + y - z = 1 & (II) \\ 5x + 4y - 2z = 5 & (III) \end{cases}$$

This is a dependent system:

Our system of two variables contains _____ and _____ . We can choose either one write our line in terms of. We'll choose _____ .

The solution set is all points of the form (_____, _____, _____)

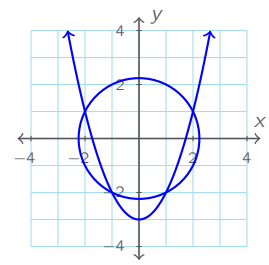
6.5 Non-Linear Systems

A Non-Linear System

The graphs of linear systems are _____ . This gives a major an advantage in solving these systems: two lines in a plane intersect _____ most of the time, or _____ if they're parallel. What about nonlinear equations?

Example. Consider the system:

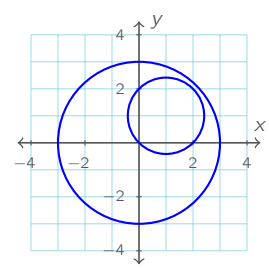
$$\begin{cases} x^2 + y^2 = 5 & (I) \\ y = x^2 - 3 & (II) \end{cases}$$



- The graph of (I) is a _____ and the graph of (II) is a _____
- The curves intersect at _____ points!
- Each point is a solution of the system:

Example. Consider the system:

$$\begin{cases} (x-1)^2 + (y-1)^2 = 2 & (I) \\ x^2 + y^2 = 9 & (II) \end{cases}$$



- The graphs of these equations are both _____ .
- These curves _____ .
- This system has _____ .

Systems of nonlinear systems might have:

- No solution because the curves go around each other
- Several solutions because the curves cross each other more than once
- All the complications involved in solving nonlinear equations

Solving a Non-Linear System by Substitution

Substitution will be our preferred method for solving a system of nonlinear equations. It will often be easy.

Example. Solve

$$\begin{cases} y = x^2 & (I) \\ xy = 8 & (II) \end{cases}$$

In this system, _____ is already solved for _____ .

This system has _____ solution: (___, ___).

Example. Solve

$$\begin{cases} x^2 + y^2 = 65 & (I) \\ y = 5x - 13 & (II) \end{cases}$$

Note that this *system* is nonlinear, even though _____ is linear.

We got _____ values of x. We must plug in _____ of these to find the corresponding value of y.

This system has _____ solutions: (___, ___)and (___, ___).

Solving a Non-Linear System by Elimination

It is usually not possible to solve a nonlinear system by elimination.

Example.

$$\begin{cases} x^2 + xy + y^2 = 7 & (I) \\ xy = 2 & (II) \end{cases}$$

We *cannot* solve this by elimination, because the second equation doesn't have _____ or _____ terms. We must solve this by _____ .

This system has _____ solutions!

But if we're lucky, we will be able to use elimination. We should use it if we can.

Example.

$$\begin{cases} x^2 + y^2 = 5 & (I) \\ 3x^2 + 5y^2 = 23 & (II) \end{cases}$$

This system has _____ solutions: _____

A Challenging Example

When we meet a nonlinear system, we should:

1. Use _____ if possible (it'll make our lives easier!)
2. Use _____ if one of the equations can be solved for one variable (sometimes it can't!)
3. Use _____ if both of those fail.

Example. Solve

$$\begin{cases} x^2 + xy + y^2 = 19 & (I) \\ y^2 - x^2 = 5 & (II) \end{cases}$$

We can't solve this by _____ because equation _____ doesn't have an _____ term. So we'll try _____

It looks like this system has _____ solutions! Are they all real?

Extraneous Solutions to a Non-Linear System

In the previous example, we solved the system

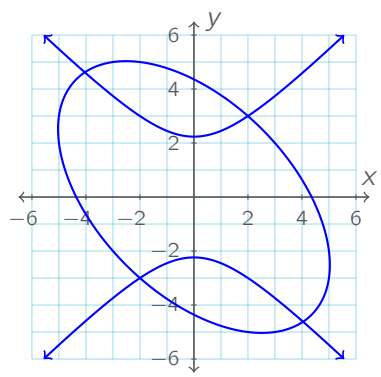
$$\begin{cases} x^2 + xy + y^2 = 19 & (I) \\ y^2 - x^2 = 5 & (II) \end{cases}$$

and got _____ (apparent) solutions:

$(x, y) = (2, 3)$ $\quad (x, y) = (2, -3)$ $\quad (x, y) = (-2, 3)$ $\quad (x, y) = (-2, -3)$

$(x, y) = \left(\frac{7}{\sqrt{3}}, \frac{8}{\sqrt{3}}\right)$ $\quad (x, y) = \left(\frac{7}{\sqrt{3}}, -\frac{8}{\sqrt{3}}\right)$ $\quad (x, y) = \left(-\frac{7}{\sqrt{3}}, \frac{8}{\sqrt{3}}\right)$ $\quad (x, y) = \left(-\frac{7}{\sqrt{3}}, -\frac{8}{\sqrt{3}}\right)$

But when we graph this system, we see:



These curves only cross _____ times! We must have _____ solutions. Steps that introduce extraneous solutions include:

- Multiplying both sides of an equation by a _____ expression
- Taking an _____ power of an _____ radical
- Combining _____ expressions

In our example, we _____ . So we must check for extraneous solutions. (When in doubt, check!)

Example (continued)**.** Check for extraneous solutions to

$$\begin{cases} x^2 + xy + y^2 = 19 & (I) \\ y^2 - x^2 = 5 & (II) \end{cases}$$

Note. We used (II) to find the possible solutions, using steps that did *not* risk introducing extraneous solutions. We need to check which solutions also work in (I).

$(2)^2 + \quad (2) \quad (3) + \quad (3)^2 =$ _____ $\qquad \left(\frac{7}{\sqrt{3}}\right)^2 + \left(\frac{7}{\sqrt{3}}\right)\left(\frac{8}{\sqrt{3}}\right) + \left(\frac{8}{\sqrt{3}}\right)^2 =$ _____

$(2)^2 + \quad (2)(-3) + (-3)^2 =$ _____ $\qquad \left(\frac{7}{\sqrt{3}}\right)^2 + \left(\frac{7}{\sqrt{3}}\right)\left(-\frac{8}{\sqrt{3}}\right) + \left(-\frac{8}{\sqrt{3}}\right)^2 =$ _____

$(-2)^2 + (-2) \quad (3) + \quad (3)^2 =$ _____ $\qquad \left(-\frac{7}{\sqrt{3}}\right)^2 + \left(-\frac{7}{\sqrt{3}}\right)\left(\frac{8}{\sqrt{3}}\right) + \left(\frac{8}{\sqrt{3}}\right)^2 =$ _____

$(-2)^2 + (-2)(-3) + (-3)^2 =$ _____ $\qquad \left(-\frac{7}{\sqrt{3}}\right)^2 + \left(-\frac{7}{\sqrt{3}}\right)\left(-\frac{8}{\sqrt{3}}\right) + \left(-\frac{8}{\sqrt{3}}\right)^2 =$ _____

6.6 Transforming the Graph of a Function

One of our most important ways of getting new functions from old functions is by transforming the graph of a function. In this section, we make explicit some tools we've been secretly using all semester. When we work with the more complex functions in trigonometry, we will need to use these tools explicitly.

Introduction

As we have studied each new function, we have added something like, "It's still a function of [type] if it's composed with something _____ ."

Example. These are all _____ functions:

$$f(x) = 2|3x-5|+4 \qquad g(x) = -\tfrac{1}{3}|x+2|-7 \qquad h(x) = \left|\tfrac{1}{3}x+4\right|$$

Example. These are all _____ functions:

$$f(x) = 3^{2x-4}-7 \qquad g(x) = 2\left(\tfrac{1}{3}\right)^{x-3}+5 \qquad h(x) = 2^{6-3x}$$

Example. These are all _____ functions:

$$f(x) = 3\sqrt{2x-4} \qquad g(x) = -\sqrt{5-x}+3 \qquad h(x) = \frac{1}{2}\sqrt{x-2}-5$$

These "families" of functions have a lot in common.

- We find their _____ and _____ the same way.
- Their _____ have the same basic shape.

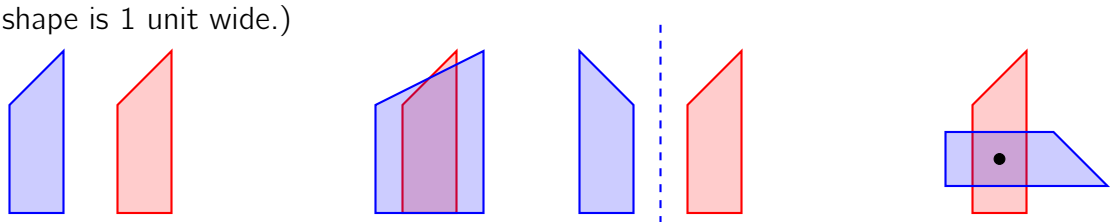
In this section, we see why: composing with a linear function has the effect of a geometric *transformation*.

Definition: Transformation

A **transformation** of a figure in the plane is any change to the figure that preserves its basic shape. We have four types of transformations:

- **Translation** moves the figure without changing its shape, or orientation.
- **Scaling** stretches or shrinks the figure while preserving its basic shape.
- **Reflecting** a figure across a line "flips" the figure without changing its size or shape.
- **Rotating** a figure around a point "turns" the figure without changing its size or shape.

In this figure, the blue shape is the result of transforming the red shape. (The original shape is 1 unit wide.)



Translate to the left by 2 units | Scale horizontally by a factor of 2 | Reflect across the dashed line | Rotate 90° to the right around the point.

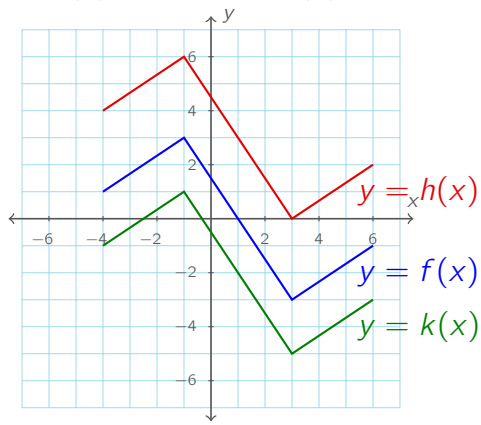
Vertical Translation

To translate a point upward by c, we add c to its y-coordinate. To translate it downward by c, we subtract c from its y-coordinate.

Example. In the figure below, the graph of:

- $h(x)$ is obtained by translating the graph of $f(x)$ three units upward
- $k(x)$ is obtained by translating the graph of $f(x)$ two units downward

Give expressions for $h(x)$ and $k(x)$ in terms of $f(x)$.



$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
	$h(x) =$ _____	$k(x) =$ _____

Vertical Translation

For any number $c > 0$:

- If $g(x) = f(x) + c$, the graph of g is the same as the graph of f, translated to the _____ by c units.
- If $g(x) = f(x) - c$, the graph of g is the same as the graph of f, translated to the _____ by c units.

Note. The direction of translation is the _____ as the sign of the number.

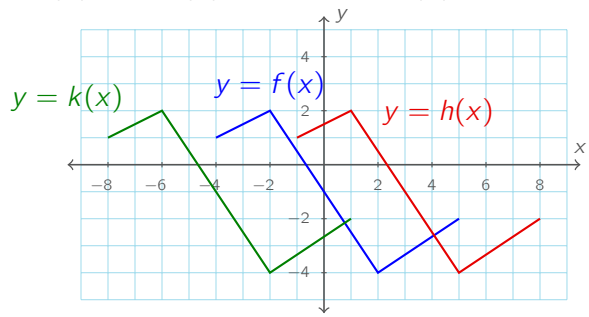
Horizontal Translation

To translate a point to the right by c, we add c to its x-coordinate. To translate it to the left by c, we subtract c from its x-coordinate.

Example. In the figure below, the graph of:

- $h(x)$ is obtained by translating the graph of $f(x)$ three units to the right
- $k(x)$ is obtained by translating the graph of $f(x)$ four units to the left

Give expressions for $h(x)$ and $k(x)$ in terms of $f(x)$.



$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
$f($_____$) =$ _____	$h($_____$) =$ _____	$k($_____$) =$ _____
	$h(x) =$ _____	$k(x) =$ _____

Horizontal Translation

For any number $c > 0$:

- If $g(x) = f(x - c)$, the graph of g is the same as the graph of f, translated to the _____ by c units.
- If $g(x) = f(x + c)$, the graph of g is the same as the graph of f, translated to the _____ by c units.

Note. The direction of translation is the _____ of the sign of the number.

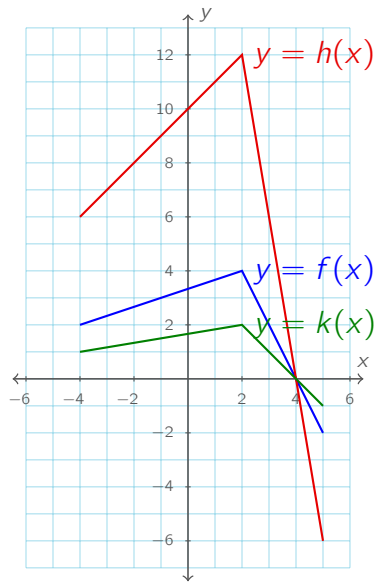
Vertical Scaling

To scale a point vertically by a factor of c, we multiply its y-coordinate by c.

Example. In the figure below, the graph of:

- $h(x)$ is obtained by scaling the graph of $f(x)$ vertically by a factor of 3
- $k(x)$ is obtained by scaling the graph of $f(x)$ vertically by a factor of $\frac{1}{2}$

Give expressions for $h(x)$ and $k(x)$ in terms of $f(x)$.



$f(_____) = _____$ $\quad$ $h(_____) = _____$ $\quad$ $k(_____) = _____$

$f(_____) = _____$ $\quad$ $h(_____) = _____$ $\quad$ $k(_____) = _____$

$f(_____) = _____$ $\quad$ $h(_____) = _____$ $\quad$ $k(_____) = _____$

$h(x) = _____$ $\quad$ $k(x) = _____$

Vertical Scaling
For any number $c > 0$: • If $g(x) = \frac{1}{c}f(x)$, the graph of g is the same as the graph of f, scaled vertically by a factor of _____ . • If $g(x) = cf(x)$, the graph of g is the same as the graph of f, scaled vertically by a factor of _____ .

Note. The scale factor is the _____ as the coefficient of $f(x)$.

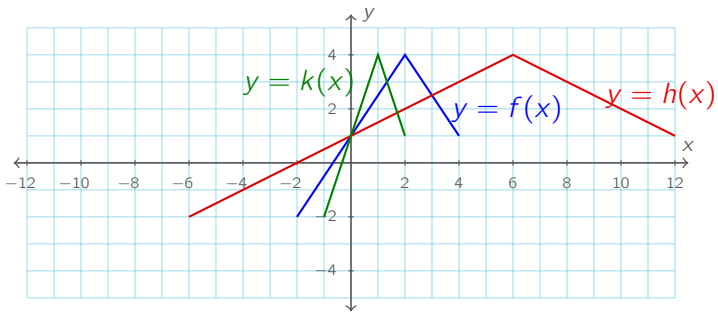
Horizontal Scaling

To scale a point horizontally by a factor of c, we multiply its x-coordinate by c.

Example. In the figure below, the graph of:

- $h(x)$ is obtained by scaling the graph of $f(x)$ horizontally by a factor of 3
- $k(x)$ is obtained scaling the graph of $f(x)$ horizontally by a factor of $\frac{1}{2}$

Give expressions for $h(x)$ and $k(x)$ in terms of $f(x)$.



$f(_____) = _____$ $\quad h(_____) = _____$ $\quad k(_____) = _____$

$f(_____) = _____$ $\quad h(_____) = _____$ $\quad k(_____) = _____$

$f(_____) = _____$ $\quad h(_____) = _____$ $\quad k(_____) = _____$

$h(x) = _____$ $\quad k(x) = _____$

Horizontal Scaling
For any number $c > 0$: • If $g(x) = f(\frac{1}{c}x)$, the graph of g is the same as the graph of f, scaled horizontally by a factor of _____ . • If $g(x) = f(cx)$, the graph of g is the same as the graph of f, scaled horizontally by a factor of _____ .

Note. The scale factor is the _____ of the coefficient of x.

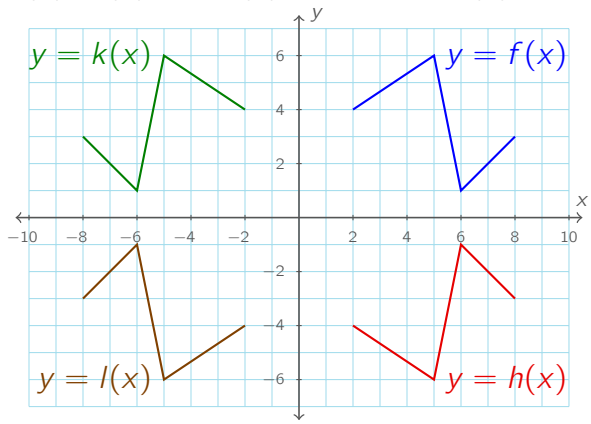
Reflection and Rotation

To reflect a point across the x-axis, we negate its y-coordinate. To reflect across the y-axis, we negate the x-coordinate. To rotate by 180°, we negate both coordinates.

Example. In the figure below, the graph of:

- $h(x)$ is obtained by reflecting the graph of $f(x)$ across the x-axis
- $k(x)$ is obtained by reflecting the graph of $f(x)$ across the y-axis
- $l(x)$ is obtained by rotating the graph of $f(x)$ 180° around the origin

Give expressions for $h(x)$, $k(x)$, and $l(x)$ in terms of $f(x)$.



$f(____) = ____$	$h(____) = ____$	$k(____) = ____$	$l(____) = ____$
$f(____) = ____$	$h(____) = ____$	$k(____) = ____$	$l(____) = ____$
$f(____) = ____$	$h(____) = ____$	$k(____) = ____$	$l(____) = ____$
$f(____) = ____$	$h(____) = ____$	$k(____) = ____$	$l(____) = ____$
	$h(x) = _____$	$k(x) = _____$	$l(x) = _____$

Reflection and Rotation

- If $g(x) = f(-x)$, the graph of g is the same as the graph of f, reflected across the _____ .
- If $g(x) = -f(x)$, the graph of g is the same as the graph of f, reflected across the _____ .
- If $g(x) = -f(-x)$, the graph of g is the same as the graph of f, *rotated* around _____ .

Note. The rotation is the _____ reflections.

Transforming a Graph: Order of Operations

Let's summarize what we've found.

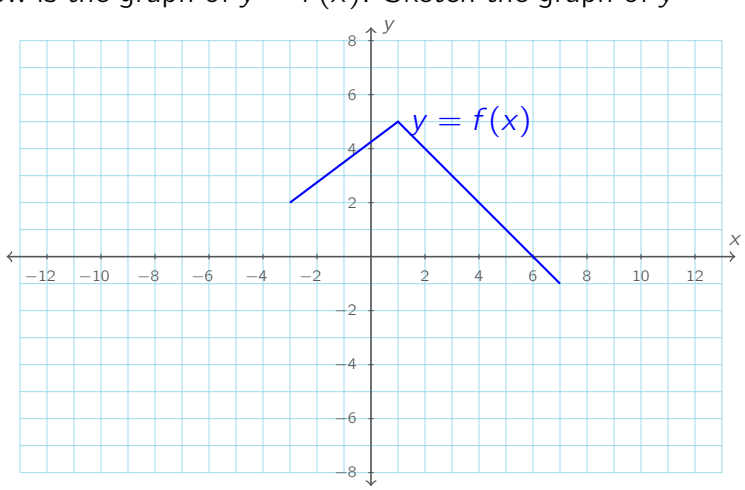
- Addition and subtraction correspond to _____ .
- Multiplication comes in two parts:
  - Multiplication by a _____ number corresponds to _____ .
  - If the number is _____ , there is also a _____ .
- Anything *inside* the function has a _____ effect.
- Anything *outside* the function has a _____ effect.
- _____ effects are _____ .

What about order?

- *Outside* the function, we follow order of operation: _____ before _____ .
- *Inside* the function, we *reverse* order of operations (remember: everything inside the function is _____ !), so _____ before _____ .

When we work, we will usually work on the _____ rather than on the picture directly.

Example. Below is the graph of $y = f(x)$. Sketch the graph of $y = -2f(\frac{2}{3}x+5)+3$

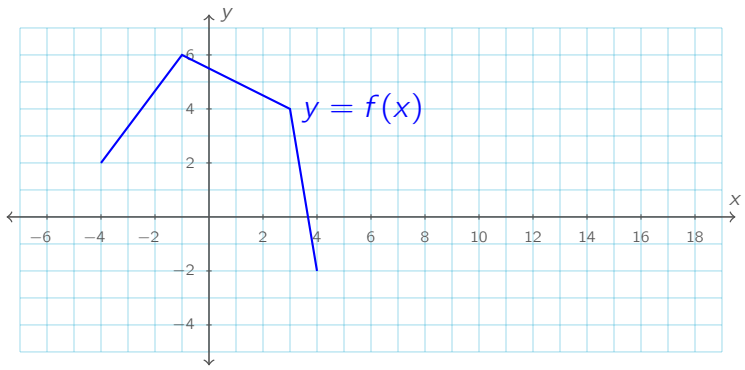


0. Starting Point		(___ , ___)	(___ , ___)	(___ , ___)
1. Horizontal translation	$x \to$ _____	(___ , ___)	(___ , ___)	(___ , ___)
2. Horiz. scaling/reflection	$x \to$ _____	(___ , ___)	(___ , ___)	(___ , ___)
3. Vert. scaling/reflection	$y \to$ _____	(___ , ___)	(___ , ___)	(___ , ___)
4. Vertical translation	$y \to$ _____	(___ , ___)	(___ , ___)	(___ , ___)

Transforming a Graph: Alternate Method

An alternate method of transforming a graph involves looking at the _____ and _____ of the original function. You might find this method helpful if you sometimes lose track.

Example. Below is the graph of $y = f(x)$. Sketch the graph of $y = \frac{1}{2}f\left(-\frac{1}{3}x + 2\right) - 3$



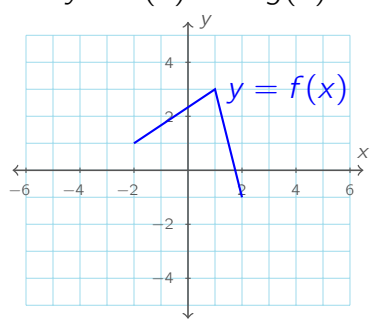
x	input	output	y

Transforming Domain and Range

We can easily find the domain and range of a transformed function. (In fact, we've been doing this all semester!)

- _____ transformations affect the _____
- _____ transformations affect the _____

Example. Below is the graph of $y = f(x)$. Let $g(x) = 2f(3 - x) - 1$.



- Dom f = _____ and Rng f = _____

Let's sketch the graph of $g(x)$ to see its domain and range.

- Dom g = _____ and Rng g = _____
- To find the domain, apply only the _____ transformations.
- To find the range, apply only the _____ transformations.

Dom f =

Rng f =

Dom g =

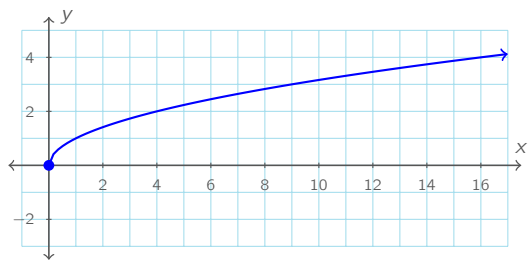
Rng g =

Why Transform Functions?

Understanding transformations of functions allows us to do two things.

- Easily understand the graphs of new functions as we encounter them.
- Transform the graphs of functions to be more easily studied.

Example. Below is the graph of $y = \sqrt{x}$. Sketch the graph of $y = \frac{3}{2}\sqrt{2x-6} - 2$.



Symmetry

Definition: Symmetry

A figure is **symmetric** under a transformation if applying the transformation leaves the figure the same.

Definition: Even Functions

A function $f(x)$ is called **even** if:

- its graph symmetric under reflection across the y-axis.
- $f(-x) = f(x)$ for all x.

Definition: Odd Functions

A function $f(x)$ is called **odd** if:

- its graph symmetric under rotation by $180°$ around $(0, 0)$.
- $f(-x) = -f(x)$ for all x.

Note. If a function were symmetric across the x-axis, we would have _____ = _____, which only happens if $f(x) =$ _____ for all x.

Example. Is $f(x) = x^4 - 3x^2 + |x|$ even, odd, or neither?

f is _____ .

Example. Is $f(x) = x^3 + 6x + \sqrt[3]{x}$ even, odd, or neither?

f is _____ .

Example. Is $f(x) = x^2 - 4x + 3$ even, odd, or neither?

f is _____ .

6.7 A Library of Functions

Now that we know how to transform a function, we need some baseline functions to transform. We think of this as our "library" of functions: basic function shapes that we recognize and know how to graph.

Constant and Linear Functions

Constant Function

Form:

Example:

Domain:

Range:

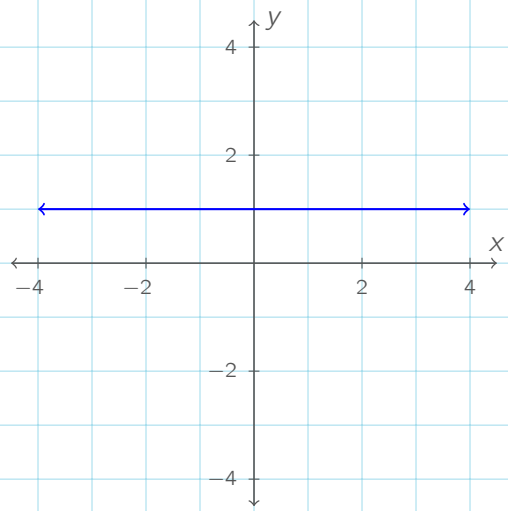
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Linear Function

Form:

Example:

Domain:

Range:

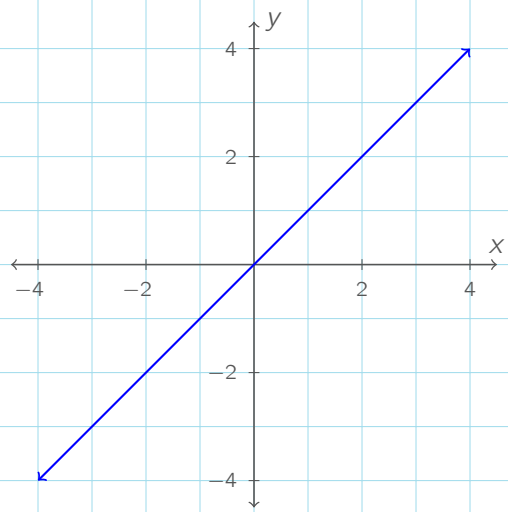
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Note. A constant function is a _____ line. A _____ line is not a function.

Power Functions: Whole Number Powers

Power Functions (even powers)

Form:

Example:

Domain:

Range:

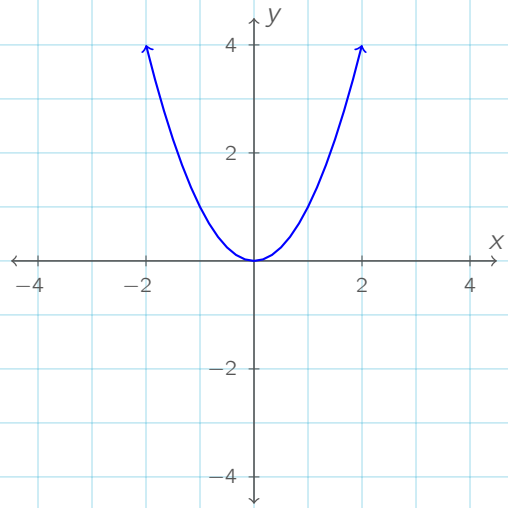
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Power Functions (odd powers)

Form:

Example:

Domain:

Range:

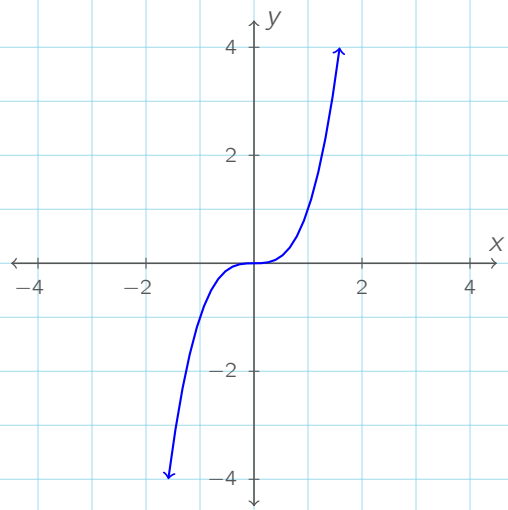
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Note. If the power is larger, the graph becomes _____ near zero and _____ far from zero.

Note. Because we can _____ , every quadratic function is a power function. This is not true of all _____ .

Note. Technically, a _____ function is a power function with exponent _____ .

Power Functions: Negative Integer Powers

Power Functions (odd powers)

Form:

Example:

Domain:

Range:

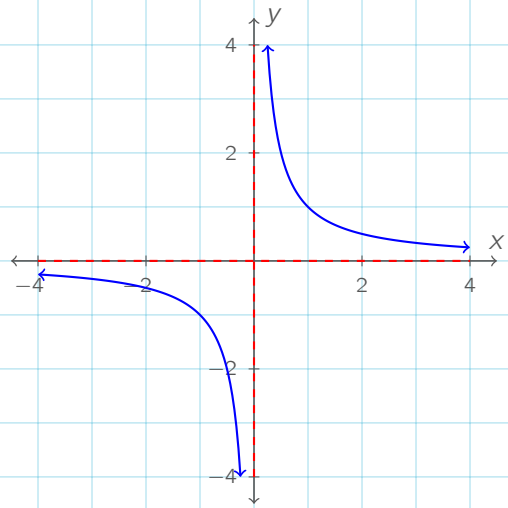
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Power Functions (even powers)

Form:

Example:

Domain:

Range:

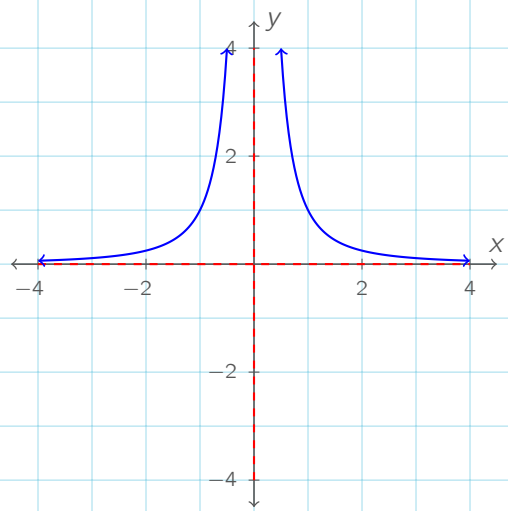
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Note. We can think of $F(x) = x^{-p}$ as $(f \circ g)(x)$, where $f(x) =$ _____ and $g(x) =$ _____. Most of these properties can be derived that way.

Power Functions: Radicals

Note. A radical function is a power function, because $\sqrt[p]{x} =$ _____.

Power Functions (even roots)

Form:

Example:

Domain:

Range:

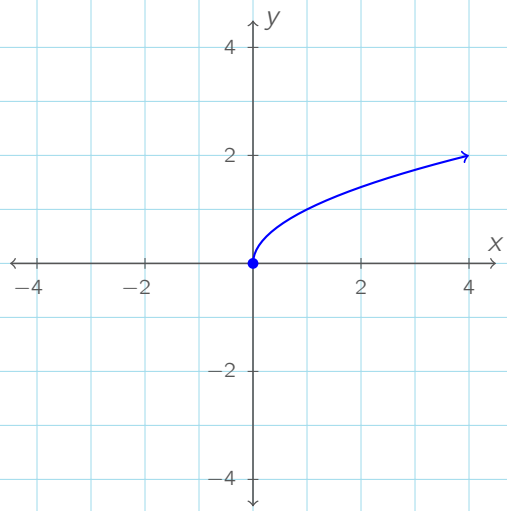
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Power Functions (odd roots)

Form:

Example:

Domain:

Range:

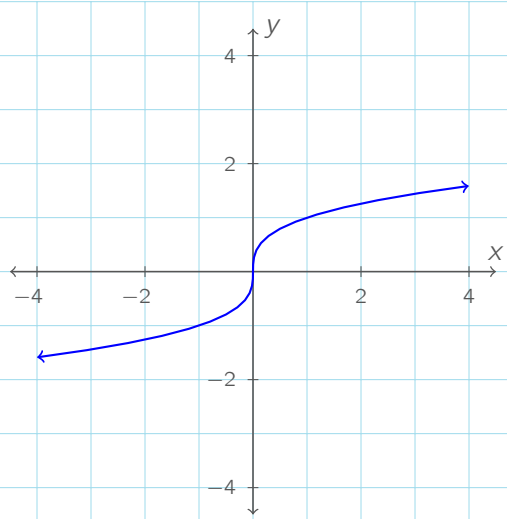
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Note. We can derive the rest of the rational power functions by $F(x) = x^{m/n}$ as $(f \circ g)(x)$, where $f(x) =$ _____ and $g(x) =$ _____.

Exponential and Logarithmic Functions

Exponential Functions

Form:

Example:

Domain:

Range:

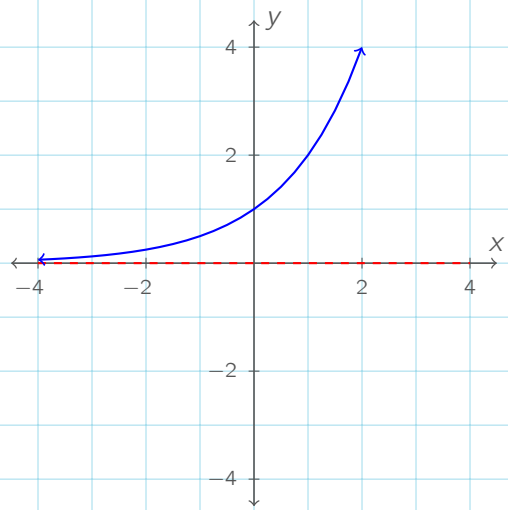
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Logarithmic Functions

Form:

Example:

Domain:

Range:

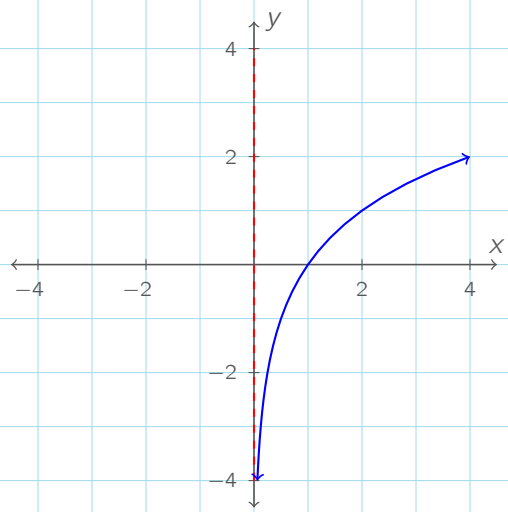
Inverse:

Symmetry:

Asymptotes:

Key Points:

Notes:



Note. In reality, we can transform everything from base e:

- $f(x) = \log_b x =$ _____, a _____ scale with factor _____
- $f(x) = b^x =$ _____, a _____ scale with factor _____

This isn't really useful in graphing (the points we end up with all have _____ in them), but it will be very useful in Calculus.

Absolute Value Functions

Exponential Functions

Form:

Example:

Domain:

Range:

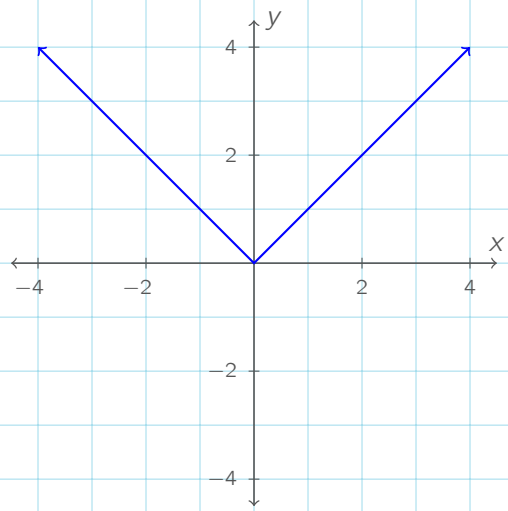
Inverse:

Symmetry:

Asymptotes:

Key Points:

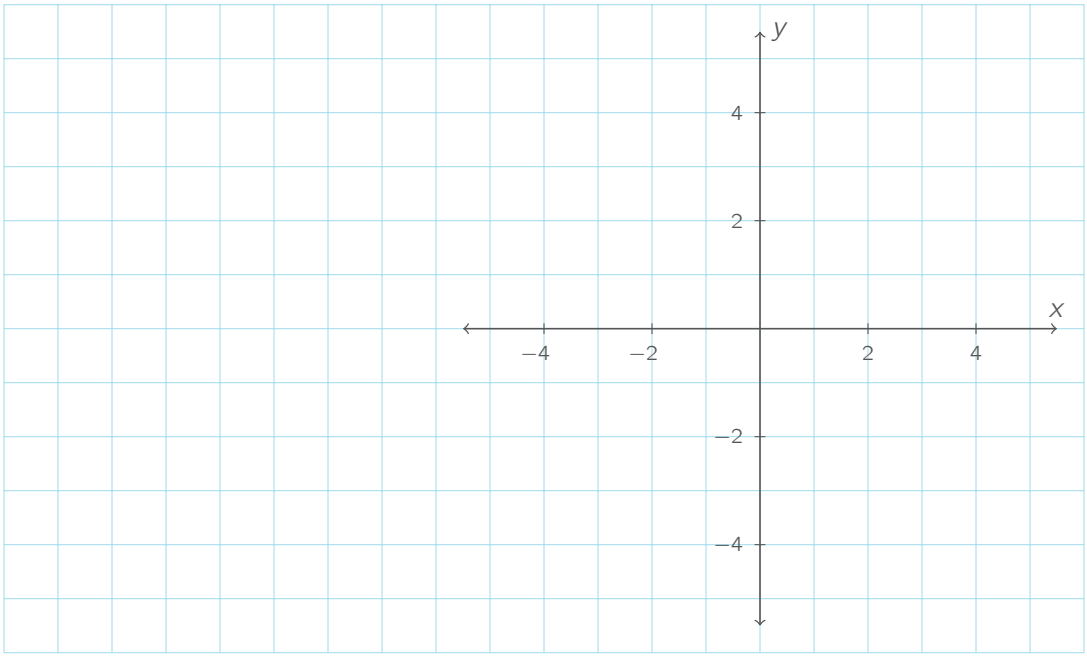
Notes:



Using the Library

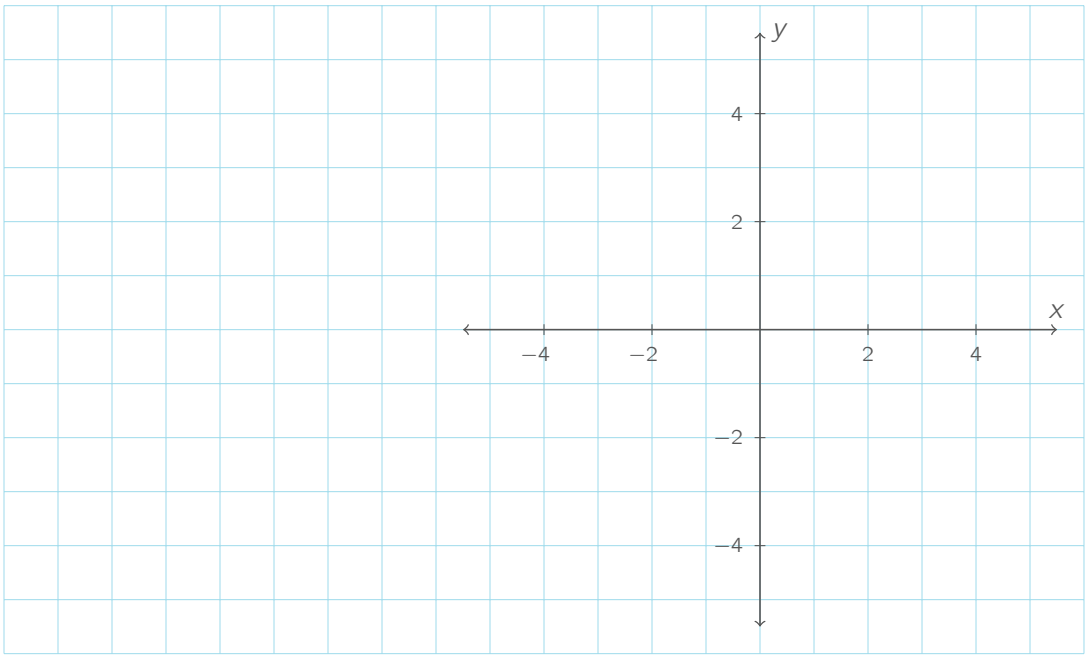
We use the library whenever we want to sketch the graph of a function.

Example. Let $f(x) = -3\sqrt[3]{x-1} + 2$. Sketch its graph, and give its domain and range.



Sometimes, it's not obvious where a function fits in the library.

Example. Let $g(x) = \dfrac{3x-10}{x-4}$. Sketch its graph, and give its domain and range.



6.8 Piecewise Defined Functions

Sometimes, we aren't able to find a single formula to describe a function. One of the ways we can handle this is to define the function *piecewise*, that is, with a different formula on different parts of its domain.

Piecewise Functions

Sometimes, a function can't be easily modeled by a single formula.

Example. A store sells flour in bulk for \$3 per pound. If a customer buys at least 5 pounds, the bulk discount kicks in, and the price drops to \$2.50 per pound. Write a function $P(x)$ describing the price of x pounds of flour.

To evaluate a piecewise-defined function:

- Figure out which range the input variable is in.
- Use the formula designated for that range.
- If none of the ranges contain the input value, the input value is not in the domain, so the function is undefined.

Example. Let $f(x) = \begin{cases} 3x & \text{if } -5 < x < -3 \\ -\frac{1}{2}x^2 & \text{if } -3 < x < 3 \\ 3 & \text{if } 3 \leq x < 4 \\ x & \text{if } x \geq 4 \end{cases}$. Find:

$f(-4)$

$f(2)$

$f(3.5)$

$f(5)$

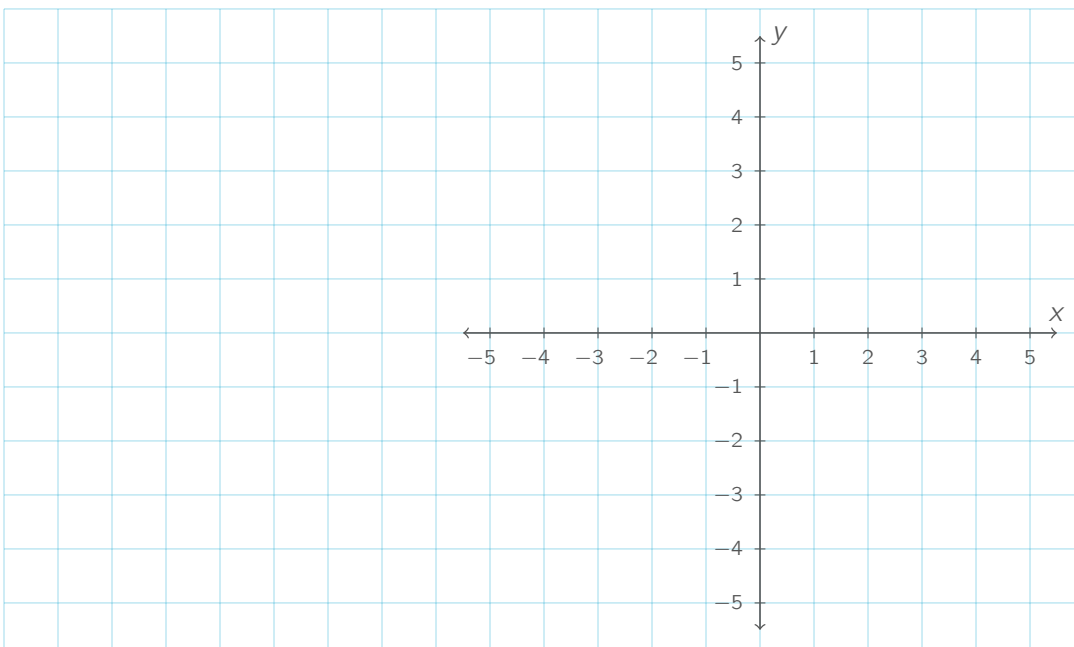
$f(-3)$

Graphing a Piecewise Linear Function

The simplest piecewise functions are *piecewise linear* functions–where each piece is linear.

Example. Sketch the graph of $f(x) = \begin{cases} 2x - 1 & \text{if} \quad -2 \leq x < -1 \\ 3 - x & \text{if} \quad -1 \leq x \leq \ \ 2 \\ \frac{1}{2}x + 1 & \text{if} \quad \ \ 2 < x \leq \ \ 4 \end{cases}$

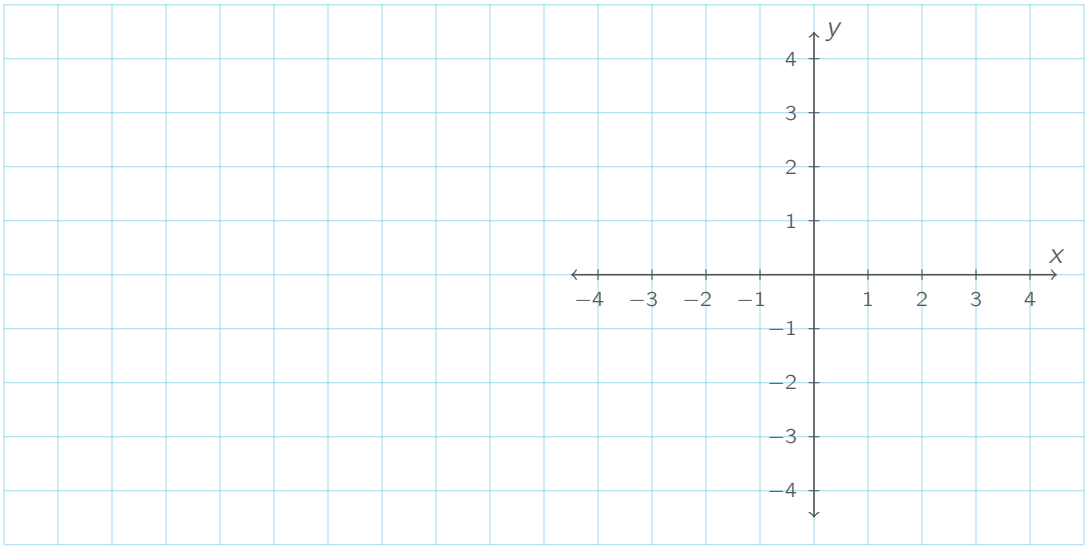
- We imagine cutting the graph into vertical "strips" corresponding to each _____ of the _____ .
- Each piece of this graph is a _____ , so we need _____ points to sketch it.
- We choose the edges of the strips.
- If the point is included in the graph, we draw a _____ circle there.
- If the point is not included, we draw an _____ circle there.



Other Functions

Any part of a function can make up part of a piecewise function.

Example. Sketch the graph of $f(x) = \begin{cases} x+3 & \text{if} \quad x < -1 \\ (x-1)^2 - 2 & \text{if} \quad -1 \leq x < 2 \\ -1 & \text{if} \quad x \geq 2 \end{cases}$

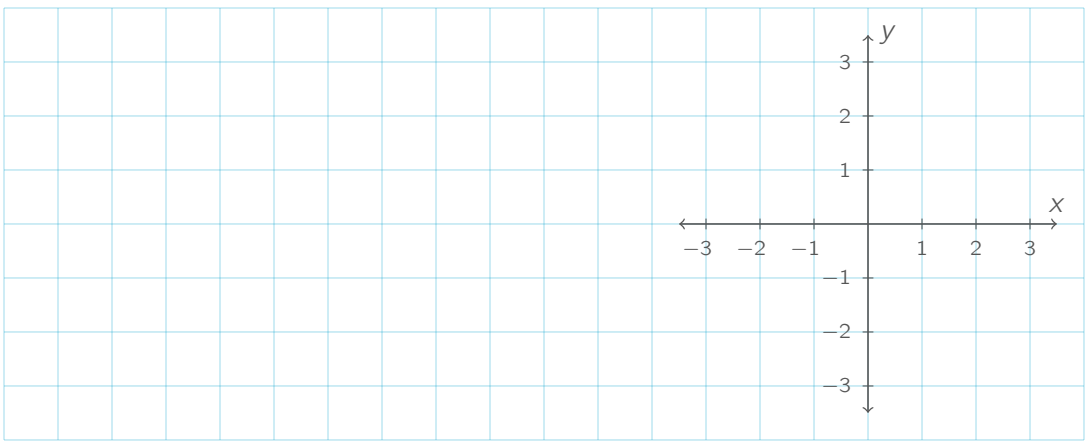


Holes in a Graph

Sometimes, a piecewise graph has just one special point.

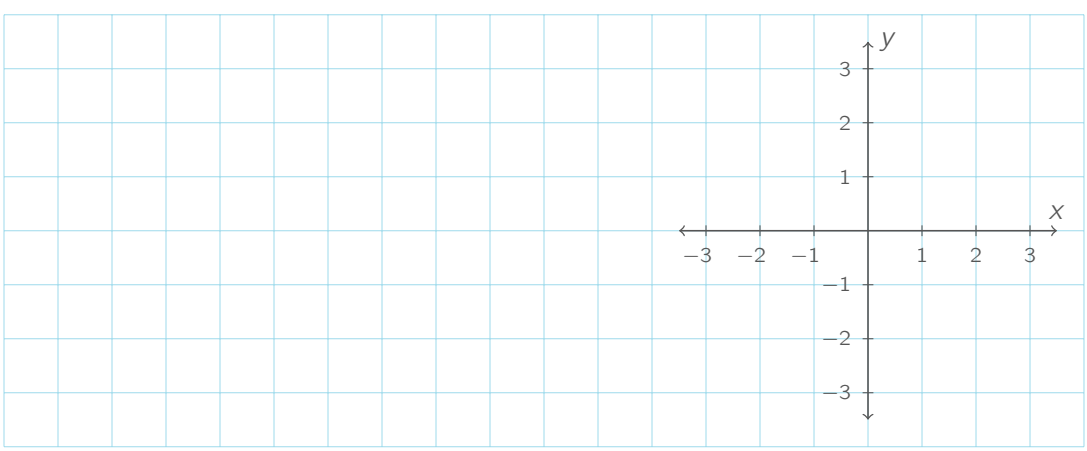
Example. Sketch the graph of $f(x) = \begin{cases} x^2 - 2 & \text{if} \quad x \neq 1 \\ 3 & \text{if} \quad x = 1 \end{cases}$

- First, we graph the main function (the one that applies for _____ values of x)
- Then, we make a "hole" in the graph at $x =$ _____.
- Finally, we plot the special point.



Sometimes, the special point comes between other pieces.

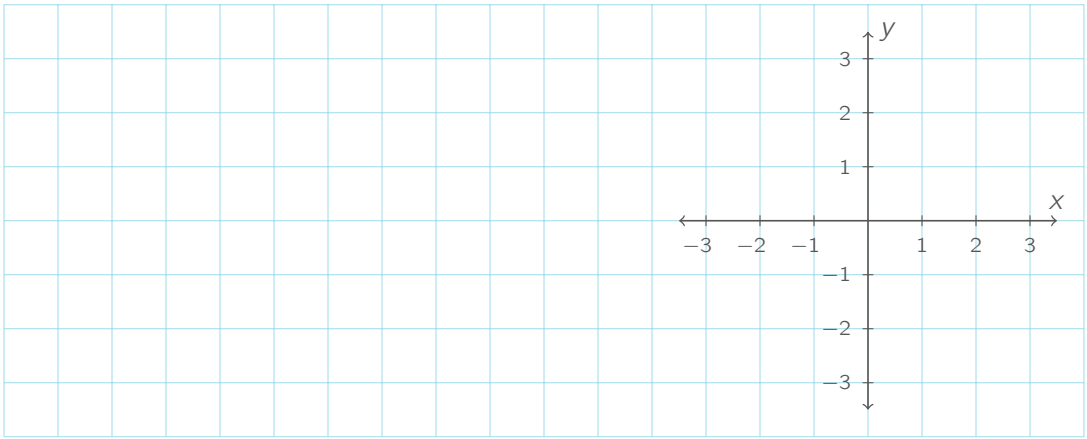
Example. Sketch the graph of $f(x) = \begin{cases} 3 & \text{if} \quad x < -2 \\ -1 & \text{if} \quad x = -2 \\ x + 3 & \text{if} \quad x > -2 \end{cases}$



Step Functions

A "step function" is a piecewise function whose pieces are all constant.

Example. Sketch the graph of $f(x) = \begin{cases} -2 & \text{if} \quad -3 \leq x < -1 \\ 3 & \text{if} \quad -1 \leq x \leq 1 \\ 1 & \text{if} \quad 1 < x \leq 3 \end{cases}$



Domain

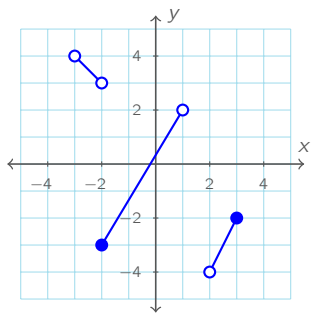
Finding the domain of a piecewise function is straightforward: we simply combine all of the domains of the pieces.

Example. Give the domain of $f(x) = \begin{cases} 3x & \text{if} \quad -2 \leq x < 1 \\ 4 - x & \text{if} \quad 1 < x < 5 \\ 6 & \text{if} \quad 5 \leq x \leq 6 \\ -2x + 4 & \text{if} \quad x > 8 \end{cases}$

Dom f = _____

If we are looking at the graph of a piecewise function, we go back to our most basic technique: _____ the graph into the _____ axis.

Example. Give the domain of the piecewise function graphed below.



Dom f = _____

Range

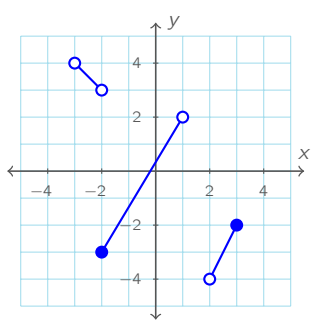
We find the range of a piecewise function in the same way we find the domain, but it can be more challenging, because the ranges of the pieces can _____ . They may also be harder to find from the expressions.

Example. Give the domain of $f(x) = \begin{cases} 3x & \text{if} \quad -2 \le x < 1 \\ 4 - x & \text{if} \quad 1 < x < 5 \\ 6 & \text{if} \quad 5 \le x \le 6 \\ -2x + 4 & \text{if} \quad x > 8 \end{cases}$

Rng f = _____

If we are looking at the graph of a piecewise function, we go back to our most basic technique: _____ the graph into the _____ axis.

Example. Give the domain of the piecewise function graphed below.



Rng f = _____